

Clinac Technical Reference Guide

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Clinac® iX, Clinac® Cx
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Clinac® 21 EX, 23 EX
Clinac® DHX, DMX
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Abstract	This document provides reference information and procedures for configuring Varian medical linear accelerators using C-Series software version 9.5. This publication is the English-language original.
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Chapter 1 Introduction

Introduction

The *Clinac Technical Reference Guide* describes how to configure, test, and maintain all models of Varian High Energy and UNIQUE Linear Accelerators.

Varian Medical Systems manufactures High Energy medical linear accelerators that feature different groups of X-ray and electron energies, as well as UNIQUE accelerators with a single X-ray energy. Some of the following procedures apply only to Varian High Energy Linear Accelerator models.



Note: *Your Varian Linear Accelerator is equipped with options and software that are installed and configured for your specific clinical setting. Depending on your configuration, options and software screens may differ from those shown in this guide.*

Intended Audience for This Document

The *Clinac Technical Reference Guide* is written for advanced users of Varian High Energy or Varian UNIQUE Linear Accelerators: physicists, dosimetrists, and authorized service personnel. It provides instructions for obtaining fast and comprehensive access to machine operations and conditions.

If you are new to Varian High Energy or Varian UNIQUE Linear Accelerators, you can find basic information on how to operate them in the appropriate *Instructions for Use*.



WARNING: Do not attempt to perform maintenance procedures on the accelerator. Maintenance procedures are restricted to service personnel who receive rigorous training and are authorized to work on the accelerator. Service personnel perform periodic maintenance inspections which include cleaning accelerator components, interlock verification, and measuring operational parameters. Performing maintenance procedures without proper training can result in serious injury or death.

Visual Cues

This publication uses the following visual cues to help you find information:



WARNING: A warning describes actions or conditions that can result in serious injury or death.



CAUTION: A caution describes hazardous actions or conditions that can result in minor or moderate injury.



NOTICE: A notice describes actions or conditions that can result in damage to equipment or loss of data.



Note: A note describes information that may pertain to only some conditions, readers, or sites.



Tip: A tip describes useful but optional information such as a shortcut, reminder, or suggestion, to help get optimal performance from the equipment or software.

Accelerator Specifications

Before operating the accelerator, you should be familiar with the following specifications, which apply to all Varian medical linear accelerators.

For imaging and system setup that routinely occur as part of delivering a treatment, refer to the appropriate *Instructions for Use*. For detailed safety information, refer to the *Clinac Safety Guide*.

Operating Specifications

- Type of protection against electric shock: Class I
- Degree of protection against electric shock: Type B,



- Operation: Varian Linear Accelerators are classified as being suitable for continuous connection to the supply main in the Standby state and for specified permissible loadings.
- Varian Linear Accelerators are not for use in the presence of flammable anesthetic mixtures.

- Varian Linear Accelerators are not intended for use in an oxygen-rich environment.
- Degree of protection against Ingress of Water: IPX0 [zero] (nonprotected)
- Method of Sterilization: not applicable. No sterile parts.
- Method of Disinfection: Use the disinfectant required by the employer (or, if none is specified, with a solution of 5.25% sodium hypochlorite [household bleach] diluted between 1:10 and 1:100 with water).



Note: The decision to decommission an installed linear accelerator is entirely at the discretion of the owner. Normally, a Varian Linear Accelerator remains operational until the owner decides to replace it.

Electrical Requirements

- Varian UNIQUE Linear Accelerator models, input voltage: 200 to 240 V_{AC} 50 or 60 Hz 42 A_{max} @208V; or 360 to 440 V_{AC} 50 or 60 Hz 22 A_{max} @ 400V.
- Varian High Energy Linear Accelerator models, input voltage: 200 to 240 V_{AC} 50 or 60 Hz 125 A_{max} @ 208V; or 360 to 440 V_{AC} 50 or 60 Hz 65 A_{max} @ 400V.

Environmental Operating Specifications

- Humidity range: 15–80% relative humidity, non-condensing
- Temperature range: 16–27°C (60–80°F)

Cooling Water Specifications

The accelerator uses an internal water cooling system to maintain a stable temperature for some of the machine's components.

Cooling external water flows from the external facility to the heat exchanger in the accelerator stand. Internal water and facility water never mix.

The internal water system uses a pump to move the internal water through the heat exchanger that extracts heat generated by power-dissipating components, and maintains them at 40° C.

The internal water is stored in a reservoir. The water pump runs at a constant speed to maintain the flow rate (in gallons per minute) of the internal water as it flows through components.

Sensors in the system measure the water flow rates and temperature of the internal water. The system monitors these sensors to regulate and control the cooling system.

The facility plumbing system discharges heated external facility water to a drain or recirculating system when the facility water valve is open. When a chiller is in use, the external facility water valve is closed.

Quality of the internal water is important for proper operation of the cooling system. For detailed information on internal water specifications, see *Customer Technical Bulletin CTB-WA-239B, Clinac Inhibited Water System*.

Symbols Used on Equipment Labels

Table 1 describes symbols found on the labels applied to the Varian medical linear accelerators, couch, and Multileaf Collimator (MLC):

Table 1 Symbols on Labels

Symbol	Description
	On (power: connection to the mains)
	Off (power: disconnection from the mains)
	Standby
	Direct current
	Alternating current
	Protected earth (ground)
	Type B applied part: This part complies with the specified requirements of EN 60601-1 to provide protection against electric shock, particularly regarding allowable leakage current.
	Equipment manufacturer. This symbol appears adjacent to the name and address of the equipment manufacturer.
	This equipment has been designated as electrical and electronic equipment (EEE) that is not to be disposed of at the end of its life as unsorted municipal waste. EEE contains substances that may present hazards to human health and to the environment. It must be recovered, reused, recycled, or otherwise treated, and properly disposed of.
	Reference number: This symbol appears adjacent to the manufacturer's reference number (catalog number).

Symbol	Description
	Serial number: This symbol appears adjacent to the manufacturer's serial number.
	Conforms to the official journal of the European Communities Council Directive 93/42/EEC.
	ETL listed. Conforms to UL STD 60601-1, EN STD 60601-2-1. Certified to CAN/CSA STD C22.2 No 601.1-M90.
	Indoor use only, use in a dry location.
	Equipment requirement: water. This equipment requires treated water to minimize damage. To avoid equipment damage, fill with de-mineralized water.
	People's Republic of China Electronic industry standard SJ/T11364-2006
	Heavy object. Can cause muscle strain or back injury. Use lifting aids and proper lifting techniques when removing or replacing.
	Maximum load
	Sitting prohibited
	Temperature range (limits)
	Humidity range (limits)
	Atmospheric pressure range (limits)
	Consult instructions for use. Follow operating instructions.
	Emergency Off: This symbol indicates the control for immediately shutting off the equipment, the beam, and stopping equipment movements.
	Caution: Indicates potential injury. Observe safety precautions. Consult accompanying documents.
	Laser radiation. Do not stare into beam; observe laser safety precautions.
	ESD (electrostatic discharge) susceptibility. Connections are ESD sensitive. Wear an ESD wrist strap when servicing this equipment.

Symbol	Description
	Radio frequency radiation: Radio frequency energy can cause injury or burns. Do not operate this equipment with the covers, output cable, or interlock cover removed.
	Hazardous voltage. Contact may cause electric shock or burn. Disconnect power before servicing.
	Pinch point: This symbol indicates places on the device that could pinch fingers or other body parts. To avoid injury, keep your hands away from these areas.
	Pinch point: This symbol indicates places on the device that could pinch toes or other body parts. Keep your foot away from these areas to avoid pinching toes. This symbol is only used on the EXaCT Targeting System.

Contact Varian Customer Support

Varian Customer Support is available on the internet, by e-mail, and by telephone. Support services are available without charge during the initial warranty period.

The MyVarian website provides contact information, product documentation, and other resources for all Varian products.

1. Go to www.MyVarian.com.
2. Choose an option:
 - If you have an account, enter your User login information (email and password).
 - If you do not have an account, click **Create New Account** and follow the instructions. Establishing an account may take up to two working days.
3. Click **Contact Us** at the top of the window to display customer support and training options, and international e-mail addresses and telephone numbers.
4. From the Contact Us page, choose an option:
 - Call Varian Medical Systems support using a phone support number for your geographic area.
 - Complete the form corresponding to your request for use on a call with a live Varian representative; then follow the instructions to complete the remote connection.

You can order documents by phone, request product or applications support, and report product-related issues. Links on the MyVarian website navigate to other support resources for products, services, and education.

5. To find documents, click **Product Documentation**.

Online documents in PDF format include customer technical bulletins (CTBs), manuals, and customer release notes (CRNs).

Chapter 2 PHYSICS Mode

Physics Mode

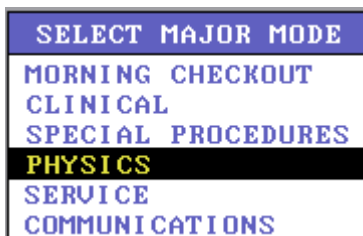
Physics mode allows medical physicists to customize morning checkout, dose rates, and other parameters to the specific clinic requirements. Entering and operating a Varian medical accelerator in Physics mode requires a password.



WARNING: Configuring the accelerator in Physics mode by unauthorized persons could cause a machine malfunction that may result in damage to equipment, bodily injury, or death. Only persons authorized by the owner should configure the accelerator in Physics mode. Do not divulge the password to unauthorized individuals.

Enter Physics Mode

After you finish daily start-up of the accelerator or when you exit from all major modes, the system displays the **SELECT MAJOR MODE** menu. Using the arrow keys on the dedicated keyboard, select **PHYSICS**.



If the system displays a message that this mode is currently unavailable, check that:

- PHYSICS is selected.
- Power keyswitch is in the **on** position (extreme clockwise).

To enter Physics mode, follow these steps:

1. From the **SELECT MAJOR MODE** menu, select **PHYSICS** and press **ENTER**.

The ENTER PASSWORD box appears.

2. In the password box, type the Physics password, and press **ENTER** (to exit, press **F1**).



Note: If you do not type the password correctly, an error message appears. Type the password again to continue. If you do not enter the correct password after four tries, the Physics mode entry program ends and the **SELECT MAJOR MODE** menu reappears.

After you type the password, the **SELECT OPTION** menu appears.

3. In the **SELECT OPTION** menu, make a selection:

- **CONFIGURATION**

The **CONFIGURATION** menu provides options that allow you to configure readouts, parameters, interfaces with some other devices, and other settings.

- **UTILITIES**

The **UTILITIES** menu selection displays a message stating utilities functions are available on the Windows workstation.

The menu you have selected appears.



Note: Depending on the configuration of your accelerator, you may not have all the options listed in the *SELECT CONFIGURATION* menu.

SELECT MAJOR MODE
MORNING CHECKOUT
CLINICAL
SPECIAL PROCEDURES
PHYSICS
SERVICE
COMMUNICATIONS

Select **PHYSICS** and press **ENTER**.

ENTER PASSWORD: ****

Enter the Physics password and press **ENTER** (press **F1** to Exit).

SELECT OPTION
CONFIGURATION
UTILITIES

SELECT CONFIGURATION
DEFAULT DOSE RATE
TIME FACTOR
PASSWORDS
MAXIMUM DOSES
MORNING CHECKOUT
CUSTOMER INTERLOCKS
DATE FORMAT
TIME FORMAT
ACCESSORIES
DYNAMIC TREATMENTS
REMOTE AUTO MOTIONS
BEAM DELIVERY TIME INTERLOCK
BEAM GATING ICON
RV INTERFACE
BEAMSTOPPER
LOW DOSE RATE
SMALL MOTIONS
SMART CONNECT

Table 2 Options in the SELECT CONFIGURATION Menu

Option	Use
DEFAULT DOSE RATE	Sets the default dose rate for fixed or the maximum allowed dose rate for dynamic treatments.
TIME FACTOR	Sets the time adjustment factor the system uses to calculate default backup time (maximum allowable time for the treatment), if the beam does not automatically terminate after the specified MU dose has been delivered. Trips the TIME interlock.  Note: <i>The default time does not take into consideration the leaf speed for IMRT and conformal arcs.</i>
PASSWORDS	Establishes the Dosimetry, Physics, Service, and HDTSe- passwords.
MAXIMUM DOSES	Sets the maximum dose warning level for each treatment type.
MORNING CHECKOUT	Establishes the default morning checkout treatments.
CUSTOMER INTERLOCKS	Configures a customer-defined interlock, and logging of the interlock.
DATE FORMAT	Sets the convention for displaying the date. The date is updated automatically.
TIME FORMAT	Specifies either a 24-hour or 12-hour clock for displaying the time. The time is updated automatically.
ACCESSORIES	■ Specifies the X and Y field sizes for special accessories. ■ Sets the accessory target-to-tray distance for accelerators without an MLC.
DYNAMIC TREATMENTS	■ Enables Enhanced Dynamic Wedge treatments. ■ Specifies whether pendant selection of Enhanced Dynamic Wedge orientation is required. ■ Enables and disables Dynamic MLC communication. ■ Enables and disables Dose Export (EXGI). ■ Enables the Extended Travel Range (ETR) limit of the couch.
REMOTE AUTO MOTIONS	■ Enables accelerator axes to self-position automatically to positions specified by the record-and-verify (RV) system. ■ Allows the positions of mechanical axes to be entered manually by the operator. ■ Allows for the gantry or couch rotation exclusion zone margin to be extended by up to 14 degrees. ■ Enables the option that allows the door to be opened during motions.

Option	Use
BEAM TIME DISPLAY	Changes the time indicator in Clinical and Special Procedures modes and changes the way the TIME interlock works. With this option enabled, both elapsed time and beam time are displayed and the system activates the TIME interlock if the beam time is exceeded. With this option disabled (default setting), only the elapsed time is displayed and the system activates the TIME interlock if the elapsed time is exceeded.
BEAM GATING ICON	The Beam Gating icon is enabled by default. The system allows 30 seconds for the moving axes to reach the target positions required by the treatment plan, and displays two indicators to let you know when the system is withholding the beam.
RV INTERFACE	Activates the record-and-verify (RV) system, enables auto mode-up, allows skipping verification, allows for repeated port films, and enables setting of RV interface choices.
BEAMSTOPPER	Specifies the modes in which the beamstopper (if installed) is required and the gantry angles between which it must be used. You can configure this option to enable or disable the beamstopper interlock.
LOW DOSE RATE	Enables the Low Dose Rate (LDR) range of dose rates for X energies.
SMALL MOTIONS	Enables small remote couch corrective motions.
SMART CONNECT	Enables a secure internet connection so Varian support representatives can view and troubleshoot your accelerator system online.

Tips for Navigating in Physics Mode

To navigate Physics mode:

- You must use the dedicated keyboard to perform tasks in Physics mode.
- To select items in the menus and data boxes, press the arrow keys or press **ENTER**.
- For most tasks, you can view settings without changing them and return to a higher-level menu by pressing **SETUP** or **F1 EXIT**.
- Depending on the task you are performing, keys on the dedicated keyboard may be assigned special functions. These soft key assignments are displayed at the bottom of the screen. For example, when the soft key assignment **F5 SAVE** appears at the bottom of the screen, you can press **F5** on the dedicated keyboard to save the configuration changes you have made.
- In most cases, numbers that you type in data boxes remain changed after you leave the data box (however, the values are not saved until you press **F5**). To

change the number back to the original value, you must type over the value, or press **CLEAR** and type the original value in the empty space.

Save Machine Configuration

When the **F5 SAVE** indicator appears at the bottom of the screen, you can save configuration data by pressing the **F5** key on the dedicated keyboard.

Using the **F5** function key saves all changes stored in memory since the last time changes were saved. You can either use **F5** once (after you make all changes) or you can use it multiple times (for example, after each change).

When you make configuration changes, the system retains the changes in memory only while you are in Physics mode. To retain the changes for the future, you must save them before you exit PHYSICS mode.

To save the machine configuration:

1. In the Physics mode **SELECT OPTION** menu, use the dedicated keyboard to select **CONFIGURATION** and press **ENTER**.
The **SELECT CONFIGURATION** menu appears.
2. Choose a location for the configuration file:
 - To save to the hard drive, proceed to the next step.
 - To save to external media, select **File > Change Root Directory**, and change the root directory path to the path to the external media.
3. Press **F5 SAVE**.
4. To save configuration changes, select **YES**
5. Press **ENTER**.

If you have selected YES, a message appears stating that the data is being saved.

The **SELECT CONFIGURATION** menu reappears.

6. To save your configuration settings, select **NO** at the **EXIT CONFIGURATION?** dialog box, or press the **SET UP** key.



Note: Reports of configuration data saved in PHYSICS mode can be viewed and printed in LogViewer.

If you do not save the changes made to Physics configuration data in the current session, the changes are lost.

If you have made any configuration changes and you attempt to exit Physics mode without saving them, a warning message appears. To save your configuration settings, select **NO** at the **EXIT CONFIGURATION?** dialog box, or press the **SET UP** key.

Save a New Password

If a new Physics password was specified, the system displays:

WARNING: Physics Password Has Been Modified!

- Make sure you know the new Physics password before you exit Physics mode.

Exit Physics Mode

To exit Physics mode:

1. On the dedicated keyboard, press **F1** (corresponding to the F1 EXIT function key label displayed at the bottom of the screen).
The system prompts you to confirm.
2. Select **YES** and press **ENTER** (to cancel, select **NO** and press **ENTER**).
The SELECT MAJOR MODE menu reappears.

Chapter 3 PHYSICS Mode: CONFIGURATION Menu

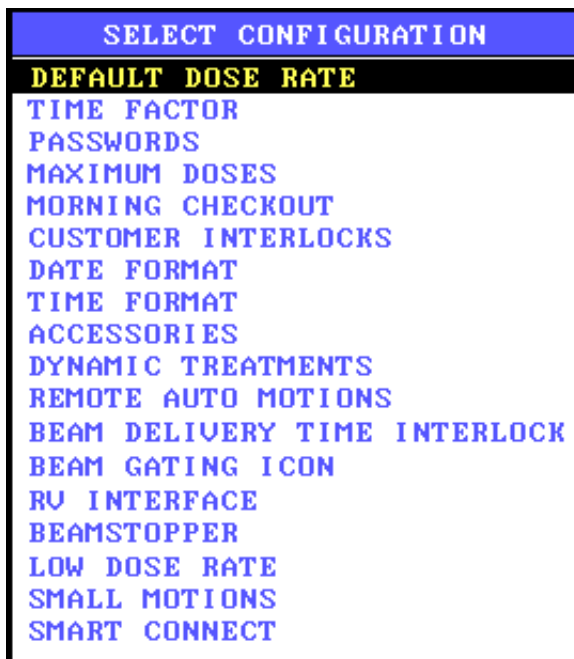
SELECT CONFIGURATION Menu

Each section in this chapter describes how to configure accelerator options that are available in the Physics mode **SELECT CONFIGURATION** menu. You can also view the current settings without changing them.



WARNING: Operation of the accelerator in Physics mode by unauthorized persons could cause a machine malfunction that may result in damage to equipment, bodily injury, or death. Only persons authorized by the owner should operate the accelerator in Physics mode. Do not divulge the password to unauthorized individuals.

The Physics mode **SELECT CONFIGURATION** menu allows you to configure many of the limits and settings on the accelerator; for example, the distance the treatment couch can be moved from outside the treatment room.





Note: Depending on the configuration of your accelerator, you may not have all the options listed in the **SELECT CONFIGURATION** menu.

DEFAULT DOSE RATE

The **DEFAULT DOSE RATE** option in the **SELECT CONFIGURATION** menu allows you to view and configure the default dose rate for treatments. Later on, at the time of treatment setup, you can change the dose rate by selecting a different dose rate from the list of predetermined monitor units/minute (MU/min).

To view and configure the default dose rates:

1. In the Physics mode **SELECT OPTION** menu, select **CONFIGURATION** and press **ENTER**.

The **SELECT CONFIGURATION** menu appears.

2. Select **DEFAULT DOSE RATE** and press **ENTER**.

The **DEFAULT DOSE RATE** data box appears, displaying the current dose rates.

DEFAULT DOSE RATE		
	STD	DYNAMIC
PORT FILM	100	
6 MV	300	600
6 MV	1000	1000
18 MV	300	600
6 MeV	300	1000
9 MeV	300	1000
12 MeV	300	1000
15 MeV	300	1000
18 MeV	300	1000
22 MeV	300	1000
6 HDTSe-	888	

3. To select the dose rate you want to change:
 - Use the arrow keys.
 - Press **ENTER**.
4. To change the dose rate:
 - Press **F5 DR DOWN** to decrease the dose rate.
 - Press **F6 DR UP** to increase the dose rate.
5. When you are finished, you can return to the **SELECT CONFIGURATION** menu by pressing **F2**.

6. To save your changes, press **F5**.
7. To exit the **SELECT CONFIGURATION** menu, press **F1**.

TIME FACTOR

You can use the **TIME FACTOR** option in the **SELECT CONFIGURATION** menu to view and configure the default time factor.

The default time factor is the number the system uses to calculate a default value for the treatment time. The default time factor is a tested limit that ensures appropriate treatment. When you specify treatment parameters, the system displays this default in the **TIME** area of the accelerator console screen.

The accelerator software calculates default time by:

1. Calculating the theoretical length of time a treatment should take. This figure is based on monitor units and dose rate, plus additional parameters for dynamic treatments (such as axis speeds, start and stop angles for arc treatments; and energy, wedge angle, and field size for Enhanced Dynamic Wedge treatments).
2. Multiplying the resulting value by the time factor to yield the default time.
3. For dynamic treatments, the software adds an additional 0.1 minute to the default time.

You must type a time factor that is either zero, or a number from 1.00 to 1.30:

- A time factor of zero disables the calculation of default treatment time.



Note: If the time factor is set to zero, the system uses the time supplied by the RV system. In stand-alone systems, if the time factor is set to zero, the system does not display a default and prompts you to type a time value manually.

- A time factor within the 1.00 to 1.30 range provides an excess from 0% to 30% over the theoretical treatment time.

To view and configure the default time factor:

1. In the Physics mode **SELECT OPTION** menu, select **CONFIGURATION** and press **ENTER**.

The **SELECT CONFIGURATION** menu appears.

2. Select **TIME FACTOR** and press **ENTER**.

The **ENTER FACTOR** data box appears, displaying the current time factor.

ENTER FACTOR	
Time Adjustment	1.20

3. Type in a new time factor, using the format #.##.

Valid entries are 0.00, or a number between 1.00 and 1.30.



Note: If you type an invalid value, an error message appears. You must type a new value to continue.

4. To confirm the new value, press **ENTER**.

The value you typed is confirmed, and the **SELECT CONFIGURATION** menu reappears.

5. To save your changes, press **F5**.

PASSWORDS

The accelerator software requires you to type a password to:

- Clear dosimetry interlocks
- Enter Physics mode
- Enter Service mode
- Set up HDTSe- treatments

The **PASSWORDS** option in the **SELECT CONFIGURATION** menu allows you to change any of these required passwords. *Each password must be a four-digit number, ranging from 0000 to 9999.*

Selecting the **PASSWORDS** option displays the **ENTER PASSWORDS** data box (see procedure [Change a Password](#)) and the current passwords for each mode.



Note: Varian recommends that you do not change the Service mode password set at the factory. If the Service mode password is modified from the factory default, the only way to restore the password is to re-install the software using the upgrade process.

Change a Password

To view and change a password:

1. In the Physics mode **SELECT OPTION** menu, select **CONFIGURATION** and press **ENTER**.

The **SELECT CONFIGURATION** menu appears.

2. Select **PASSWORDS**, and press **ENTER**.

The **ENTER PASSWORDS** data box appears, displaying the current passwords.

ENTER PASSWORDS	
Dosimetry	3210
Physics	8888
Service	TH86
HDTSe-	9876

3. Use the arrow keys to select the password you want to change, and type the new password over it.
4. To return to the **SELECT CONFIGURATION** menu, press **SET UP**.
5. To save changes, press **F5**.

Cancel a Password Change

To cancel a saved password change:

1. Return to the previous password by selecting the changed password and retyping the previous password.
Password changes are not saved until you save the configuration file.
2. To save changes, press **F5**.

MAXIMUM DOSES

Maximum dose levels are doses above which a user is not normally allowed to operate a stand-alone accelerator system in the Clinical and Special Procedures modes. A stand-alone system is a system that is not also using a record-and-verify (RV) system such as Varian 4DITC. Exceptions to maximum dose levels can be established as described in [Override Maximum Doses During Treatments](#) on page 29. These maximum dose levels set in Physics mode must fall below the maximum dose *limits* that the accelerator can deliver.



WARNING: Use of the system without 4DITC omits a component that is beneficial to the safe and effective use of the system. Treatment without 4DITC eliminates the safety enhancements provided by 4DITC. Varian strongly recommends that 4DITC be used for patient treatment.

If you are using a stand-alone accelerator, and you specify a dose higher than the maximum dose that is set using the **MAXIMUM DOSES** function, a warning message appears.



Note: The maximum dose warning message is only enabled for stand-alone accelerator operation, when the dose has been entered manually. The maximum dose warning does not appear if you are using an RV system and the treatment is sent to the accelerator by the RV system.

The **MAXIMUM DOSES** option in the **SELECT CONFIGURATION** menu allows you to view the current maximum dose levels displayed beside the factory-set maximum dose levels, and configure the maximum dose levels for your treatments.

Override Maximum Doses During Treatments

During a treatment, if you specify a dose higher than the maximum dose that is set using the **MAXIMUM DOSES** function, a warning message appears on the monitor. You are prompted to confirm that the higher dose is valid, in order to override the maximum dose level (up to the maximum dose limits set by Varian). The maximum dose limit values allowed are shown in [Table 3](#).

Table 3 Maximum Dose Limits Set by Factory

Type of Treatment	Monitor Units	Type of Treatment	Monitor Units
Port film	20	Total body X and e–	9000
Fixed X and e–	999	HDTS e–	9000
Arc X and e–	999	Enhanced Dynamic Wedge	999
SRS	6000	SRT	6000

To view and configure maximum dose levels:

1. In the Physics mode **SELECT OPTION** menu, select **CONFIGURATION** and press **ENTER**.

The **SELECT CONFIGURATION** menu appears.

2. Select **MAXIMUM DOSES** and press **ENTER**.

The **ENTER MAXIMUM DOSES** data box appears, displaying the current maximum dose levels for your accelerator in the **LEVEL** column. Maximum dose limits appear in the **LIMIT** column.



Note: The factory-set dose limits in the **LIMIT** column cannot be changed or exceeded.

ENTER MAXIMUM DOSES		
	LEVEL	LIMIT
PORT FILM	12	20
FIXED X	200	999
FIXED e ⁻	200	999
ARC X	300	999
ARC e ⁻	300	999
TBI X	3000	9000
TBI e ⁻	3000	9000
HDTSe ⁻	3000	9000
SRS X	3000	6000
SRS ARC X	3000	6000
E DWEDGE X	200	999
UMAT	3000	200
UMAT SRS	6000	9999

3. Use the arrow keys to select the maximum dose level you want to set.
4. Type the new maximum dose value.

If you type an invalid value for an exposure, the system displays an error message. To proceed, you must press **CLEAR** and type a new value.

To cancel a change:

- Manually return to the original entry (by pressing **CLEAR** and retyping it) before you exit the data box.
 - Press **F1 ABORT**.
5. After you have made all changes, press **F2 DONE**.
The **SELECT CONFIGURATION** menu reappears.
 6. To save your changes, press **F5**.

MORNING CHECKOUT

The **MORNING CHECKOUT** option in the **SELECT CONFIGURATION** menu allows you to configure the types of treatments that you want to verify in Morning Checkout mode. You can use the **MORNING CHECKOUT** option to:

- Add a treatment
- Modify a treatment
- Remove a treatment

Tips for Navigating Morning Checkout

To navigate Morning Checkout configuration:

- After you select an option in the Morning Checkout **SELECT OPTION** menu, a black box appears to the left of the first treatment in the list, indicating that the treatment is selected. To select another treatment, use the arrow keys to move the black box to the line you want to select.
- The treatment summary area lists up to eight treatment setups at a time. To page through all treatment setups, press **F3** (NEXT PAGE). To return to the first page, press **F3** again.
- To cancel an action such as adding or modifying a treatment and return to the **SELECT OPTION** menu, press **SET UP**. To return to the **SELECT CONFIGURATION** menu, press **F1**.
- The **ADD A TREATMENT** and **MODIFY A TREATMENT** options provide you with a series of menus for configuring treatments. These menus are similar to those you use to set up a treatment in the Clinical or Special Procedures mode, detailed in the appropriate *Instructions for Use*.
 - For arc treatments, the system displays an additional data box for you to type the **START ANGLE** and **STOP ANGLE**.
 - For Enhanced Dynamic Wedge treatments, the **FIELD Y** entry should be the same as the entry for the **FIELD SIZE**. When these values are not the same, an IPSN interlock prevents beam-on during the morning checkout of EDW.

Morning Checkout Sequence

Morning Checkout is a machine validation check that the radiation technologist should make each day before starting patient therapy. This mode verifies that the Clinac accelerator is operating properly and the system is ready to begin treatment.

For best performance of Morning Checkout, the energies should be programmed as follows:

1. Enter **Physics Mode**.
2. Modify the **Morning Checkout** screen to arrange the energy order as shown in the following table.



Note: A minimum of MU 100 MU can be used. However, a dose setting of 200 MU is preferred to ensure the beam remains on long enough to allow the Morning Checkout logs to fully populate.

Table 4 Morning Checkout Energy Order

Order	Energy	Treatment Type	MU
1	9e	Fixed	100 MU minimum (200 MU preferred)
2	12e	Fixed	100 MU minimum (200 MU preferred)
3	15e or 16e	Fixed	100 MU minimum (200 MU preferred)
4	18e or 20e	Fixed	100 MU minimum (200 MU preferred)
5	22e, if installed	Fixed	100 MU minimum (200 MU preferred)
6	4e, if installed	Fixed	100 MU minimum (200 MU preferred)
7	6e	Fixed	100 MU minimum (200 MU preferred)
8	Hi-X	Fixed	100 MU minimum (200 MU preferred)
9	Lo-X	Fixed	100 MU minimum (200 MU preferred)
10	6X	Fixed	250 MU (proceed if RPM QA is required)

RPM Respiratory Gating QA

If your system includes RPM Respiratory Gating or a third-party gating system, perform at least one patient QA treatment that tests gating.



WARNING: Regularly scheduled quality assurance and service must be performed on the equipment used to deliver treatment. Check all systems for operational safety to reduce the risk of personal injury, death or damage to equipment, according to departmental, local, and national guidelines. Varian does not make recommendations on the type or frequency of tests for a QA program. The frequency of checks must be based on the professional judgment of institution personnel and the requirements of the existing QA program.

ADD A TREATMENT to Morning Checkout

To add a treatment to morning checkout:

1. In the Physics mode **SELECT OPTION** menu, select **CONFIGURATION** and press **ENTER**.

The **SELECT CONFIGURATION** menu appears.

2. Select **MORNING CHECKOUT**, and press **ENTER**.

The **SELECT OPTION** menu appears. **ADD A TREATMENT** is selected.

PHYSICS							
	TREATMENT TYPE	ENERGY	MU	ORIG MU		TIME	ACCESSORY
	PORT FILM	6 MU	12			0.20	
	FIXED e-	6 MeV	100			0.50	10x10 APPL
	FIXED e-	9 MeV	100			0.50	10x10 APPL
	FIXED e-	12 MeV	100			0.50	10x10 APPL
	FIXED e-	15 MeV	100			0.50	10x10 APPL
	FIXED e-	18 MeV	100			0.50	10x10 APPL
	FIXED X	6 MU	200			1.00	NO ACCESSORY
	FIXED X	18 MU	100			0.50	NO ACCESSORY

SELECT OPTION

ADD A TREATMENT

MODIFY A TREATMENT

REMOVE A TREATMENT

F1 EXIT	F2	F3 NEXT PAGE	F4	F5	F6
---------	----	--------------	----	----	----



Note: The values represented in the figure are meant only as an example. They are not meant to represent a recommended setup.

3. Press ENTER.

You are prompted to select a row where a treatment should be added. A black box beside a table row indicates that the row is selected.

- Use the arrow keys move the black box, in order to select the treatment you want to replace with a new treatment.
- Press F3 to page through all the morning checkout treatments, and use the arrow keys to select a blank line.

4. Press ENTER.

An empty treatment line appears in the list.

The SELECT TREATMENT menu also appears.

SELECT TREATMENT	
X-RAYS FIXED	ENHANCED D-WEDGE X-RAYS
ELECTRONS FIXED	DYNAMIC ARC ELECTRONS
DYNAMIC ARC X-RAYS	TOTAL BODY X-RAYS
PORT FILM	TOTAL BODY ELECTRONS
DYNAMIC WEDGE X-RAYS	HIGH DOSE RATE TS e-

5. Select a treatment type and press **ENTER**.

Depending on the type of treatment you select, a series of menus appears that allows you to configure the treatment. These menus are similar to those you use to set up a treatment in the Clinical or Special Procedures mode, detailed in the appropriate accelerator *Instructions for Use*.

- For arc treatments, the system displays a data box for you to type the **START ANGLE** and **STOP ANGLE**.
- For Enhanced Dynamic Wedge treatments, the **FIELD Y** entry should be the same as the entry for the **FIELD SIZE**. When different, expect an IPSN interlock preventing beam-on during the morning checkout of EDW.

6. Make your selections, pressing **ENTER** each time to save the new data and move to the next menu.

When you press **ENTER** in the last menu, the **SELECT OPTION** menu reappears.

7. When you are finished:

- Press **SETUP** to return to the **SELECT OPTION** menu.
- Press **F1** to return to the **SELECT CONFIGURATION** menu.

8. To save changes, press **F5**.

MODIFY A TREATMENT in Morning Checkout

You can use the **MODIFY A TREATMENT** option to change a morning checkout treatment. You can modify any or all of the parameters:

- To leave a parameter unchanged, press **ENTER**.
- To change a parameter, select a different choice or type a new number as appropriate, and then press **ENTER**.

For more information about treatment parameter menus and values, refer to the appropriate accelerator *Instructions for Use*.

To modify a treatment:

1. In the Physics mode **SELECT OPTION** menu, select **CONFIGURATION**, and press **ENTER**.

The **SELECT CONFIGURATION** menu appears.

2. Select **MORNING CHECKOUT** and press **ENTER**.

The **SELECT OPTION** menu appears.

3. Select **MODIFY A TREATMENT**, and press **ENTER**.

The accelerator software prompts you to select the treatment you want to modify. A black box next to a treatment indicates that the treatment is selected.

4. Use the arrow keys to select the treatment, and then press **ENTER**.

The treatment you select appears at the top of the list, with new rows below it.

In addition, the **SELECT TREATMENT** menu appears, so you can confirm or change the treatment type.

5. Select a treatment type and press **ENTER**.

Depending on the type of treatment you select, menus appear that allow you to configure the treatment. These menus are similar to those you use to set up a treatment in the Clinical or Special Procedures mode, detailed in the appropriate accelerator *Instructions for Use*.

- For arc treatments, the system displays a data box for you to type the **START ANGLE** and **STOP ANGLE**.
 - For Enhanced Dynamic Wedge treatments, the **FIELD Y** entry should be the same as the entry for the **FIELD SIZE**. When different, expect an IPSN interlock preventing beam-on during the morning checkout of EDW.
6. Make your selections, pressing **ENTER** each time to save the new data and move to the next menu.

When you press **ENTER** from the last menu, the **SELECT OPTION** menu reappears.

7. When you are finished:
 - Press **SETUP** to return to the **SELECT OPTION** menu.
 - Press **F1** to return to the **SELECT CONFIGURATION** menu.
8. To save changes, press **F5**.

REMOVE A TREATMENT from Morning Checkout

To remove a morning checkout treatment, use the **REMOVE A TREATMENT** option in the Morning Checkout mode **SELECT OPTION** menu:

1. In the Physics mode **SELECT OPTION** menu, select **CONFIGURATION** and press **ENTER**.

The **SELECT CONFIGURATION** menu appears.

2. Select **MORNING CHECKOUT** and press **ENTER**.

The **SELECT OPTION** menu appears.

3. Select **REMOVE A TREATMENT**, and then press **ENTER**.

The accelerator software prompts you to select the treatment you want to remove.

4. Use the arrow keys to select the treatment, and then press **ENTER**.

The accelerator software prompts you to confirm the removal.

5. Confirm removal or cancel:

- To remove the treatment, select **YES**, and then press **ENTER**.
- To cancel the removal, select **NO**, and then press **ENTER**.

The **SELECT OPTION** menu reappears.

6. When you are finished:
 - Press **SETUP** to return to the **SELECT OPTION** menu.
 - Press **F1** to return to the **SELECT CONFIGURATION** menu.
7. To save changes, press **F5**.

Restore a Deleted Morning Checkout Treatment

The last morning checkout treatment you remove is stored in memory and can be restored anytime *before* you exit Physics mode. When you want to change the order of morning checkout treatments, it is necessary to remove the treatment from the set of morning checkout treatments and then restore the treatment at a location that puts the treatment in the proper order.

To restore the last treatment you removed:

1. In the Physics mode **SELECT OPTION** menu, select **CONFIGURATION** and press **ENTER**.
The **SELECT CONFIGURATION** menu appears.
2. Select **MORNING CHECKOUT** and press **ENTER**.
The **SELECT OPTION** menu appears.
3. Select **ADD A TREATMENT** and press **ENTER**.
You are prompted to select a row where want to add (restore) the treatment.
4. Restore the treatment by:
 - Use the arrow keys to select a treatment you want to replace with the restored treatment.
 - Press **F3** to page through the morning treatments, and use the arrow keys to select a blank line.
5. Press **ENTER**.
The **ADD LAST REMOVED?** menu appears.
6. Select **YES** and press **ENTER**.
The treatment reappears in the line you have selected, and the **SELECT OPTION** menu reappears.
7. When you are finished:
 - Press **SETUP** to return to the **SELECT OPTION** menu.
 - Press **F1** to return to the **SELECT CONFIGURATION** menu.
8. To save changes, press **F5**.

CUSTOMER INTERLOCKS

The **CUSTOMER INTERLOCKS** option in the **SELECT CONFIGURATION** menu allows you to define an interlock specifically for your department.

The **CUSTOMER INTERLOCKS** menu offers these options:

- **CUSTOMER MINOR ACTIVE** interlock enables or disables customer-defined interlocks for customized functions, such as a neutron door.
- **CUSTOMER CDOS EVENT LOGS** enable or disable log events related to a customer-defined interlock.

The CDOS interlock is a custom feature that you can configure to support beam control. The default setting for CDOS generates an error log file at every occurrence of the CDOS interlock. Some uses of this function create a high number of CDOS log files. The Varian high energy linear accelerator models allow the CDOS logging function to be turned on and off in Physics mode. Other accelerator models do not have the option to turn off CDOS logging.

SELECT CONFIGURATION
DEFAULT DOSE RATE
TIME FACTOR
PASSWORDS
MAXIMUM DOSES
MORNING CHECKOUT
CUSTOMER INTERLOCKS
DATE FORMAT
TIME FORMAT
ACCESSORIES
DYNAMIC TREATMENTS
REMOTE AUTO MOTIONS
BEAM DELIVERY TIME INTERLOCK
BEAM GATING ICON
RV INTERFACE
BEAMSTOPPER
LOW DOSE RATE
SMALL MOTIONS
SMART CONNECT

Select **CUSTOMER INTERLOCKS**.

CUSTOMER INTERLOCKS
CUSTOMER CDOS EVENT LOGS
CUSTOMER MINOR ACTIVE

Select **CUSTOMER CDOS EVENTS LOGS**.

CUSTOMER CDOS EVENT LOGS
ENABLE
DISABLE

Select **ENABLE** or **DISABLE** and press **ENTER**.

CUSTOMER INTERLOCKS
CUSTOMER CDOS EVENT LOGS
CUSTOMER MINOR ACTIVE

Select **CUSTOMER MINOR ACTIVE**.

CUSTOMER MINOR ACTIVE			
6	MU	X-Rays	YES
6	MU	X-Rays	YES
18	MU	X-Rays	YES
6	MeV	Electrons	YES
9	MeV	Electrons	YES
12	MeV	Electrons	YES
15	MeV	Electrons	YES
18	MeV	Electrons	YES
22	MeV	Electrons	

Toggle **YES** or **NO** for each selection and press **ENTER**.

To configure these options:

1. In the Physics mode **SELECT OPTION** menu, select **CONFIGURATION** and then press **ENTER**.

The **SELECT CONFIGURATION** menu appears.

2. Select **CUSTOMER INTERLOCKS** and press **ENTER**.

The **CUSTOMER INTERLOCKS** menu appears, displaying the current configurations.

3. For high energy accelerator models only: To enable or disable logging of customer-defined interlock events, select **CUSTOMER CDOS EVENT LOGS** and press **ENTER**.

4. Make a selection:

- Select **ENABLE** and press **ENTER**.
- Select **DISABLE** and press **ENTER**.

5. For all accelerator models: To enable or disable a customer-defined interlock, select **CUSTOMER MINOR ACTIVE** and press **ENTER**.

The **CUSTOMER MINOR ACTIVE** data box displays YES next to active interlocks and NO next to inactive ones (for **UNIQUE** accelerator models there is only one interlock available).

6. To navigate the data box, use the arrow keys, or press **ENTER**.
7. Press **F3 TOGGLE** to toggle the status of an interlock between active and inactive.

Each time you press **F3 TOGGLE**, the status changes.

To cancel a change, change the setting back to the original entry by pressing **F3** before you exit the menu.

8. To save changes, press **F5**.

DATE FORMAT

The **DATE FORMAT** option in the **SELECT CONFIGURATION** menu allows you to view and change the format in which the date is displayed.

To view and change the date format:

1. In the Physics mode **SELECT OPTION** menu, select **CONFIGURATION** and press **ENTER**.

The **SELECT CONFIGURATION** menu appears.

2. Select **DATE FORMAT** and press **ENTER**.

The **SELECT DATE FORMAT** menu appears, with the current format selected.



3. Select the date format you want, and press **ENTER**.

The new format is set and the **SELECT CONFIGURATION** menu reappears.

4. To save changes, press **F5**.

TIME FORMAT

The **TIME FORMAT** option allows you to select the format in which the system displays the time. You can choose 12-hour or 24-hour clock format.

To select the time format:

1. In the Physics mode **SELECT OPTION** menu, select **CONFIGURATION** and press **ENTER**.

The **SELECT CONFIGURATION** menu appears.

2. Select **TIME FORMAT** and press **ENTER**.

The **SELECT TIME FORMAT** menu appears, with the current time format selected.



3. Select the time format you want, and press **ENTER**.

The new format is set and the **SELECT CONFIGURATION** menu reappears.

4. To save changes, press **F5**.

ACCESSORIES

The **ACCESSORIES** option in the **SELECT CONFIGURATION** menu allows you to:

- Configure the X and Y field sizes for special accessories
- Select a Target-to-Tray Distance (TTD)

FIELD SIZES

When using special X-ray accessories, you must adjust the collimator to the appropriate field size. If you do not adjust the field size correctly, a COLL interlock prevents you from beaming on. The maximum field size you can specify is 40.0 centimeters. If you type a larger field size, an error message appears. You must type a new value to continue.

For high energy linear accelerator models only, when you install a special electron accessory and select the appropriate electron energy at the console, the accelerator software automatically starts collimator jaw motion to adjust to that accessory's field size.

- If you use custom molds, you may have to change the **ARC E-Tray** field size setting in Physics mode each time you use a different mold.
- If you have the optional 6 x 10 electron applicator, use the **USER 1 E-** setting to establish the field size for it.

To configure collimator jaw openings:

1. In the Physics mode **SELECT OPTION** menu, select **CONFIGURATION** and press **ENTER**.

The **SELECT CONFIGURATION** menu appears.

2. Select **ACCESSORIES** and press **ENTER**.

The **ACCESSORIES** menu appears.



3. Select **FIELD SIZES**.

The **ENTER COLLIMATOR OPENINGS** data box appears, displaying the current values.

**Note:**

- If your machine is factory-configured with custom accessories, CUST appears in the Field Y and Field X data boxes. These custom accessories set the Field Y and Field X values, so you cannot change the collimator values in these spaces.

ENTER COLLIMATOR OPENINGS		
	Field Y	Field X
USER 1 X	CUST	CUST
USER 1 E-	CUST	CUST
TBI X Tray	1.0	1.0
TBI E- Tray	34.0	34.0
HDTSE Tray 6E	36.0	36.0
ARC E- Tray	35.0	10.0

ENTER COLLIMATOR OPENINGS		
	Field Y	Field X
USER X	1.0	1.0
TBI X Tray	1.0	1.0

- Values entered for the USER X and TBI X Tray set the maximum collimator opening allowed. If values are entered during treatment setup that exceed the values set in the ENTER COLLIMATOR OPENINGS dialog box, a COLL interlock is activated.
 - For high energy linear accelerator models only, values entered for the TBI E- Tray, HDTSE Tray, and ARC E- Tray automatically drive the collimators to these set values when the mode is selected and/or tray installed.
4. Use the arrow keys to select the field size you want to change.
 5. Type in the new value.
 - If you make a mistake, type a new value over the error.
 - If you type a value that is out of range, an error message appears. To proceed, you must press **CLEAR** and type a new value.
 6. When you are finished, you can:
 - Return to the **ACCESSORIES** menu by pressing **ENTER** until you exit the data box.
 - Return to the **SELECT CONFIGURATION** menu by pressing **SETUP**.
 7. To save changes, you must press the **F5** function key before you exit Physics mode.

ACCESSORY TTD (Accessory Target-to-Tray Distance)

To set the target-to-tray distance (TTD) on an accelerator *without* an MLC, use the **ACCESSORY TTD** option in the **ACCESSORIES** menu.

Do not change the TTD for an accelerator *with* an MLC; when an MLC is present, the accelerator has a fixed target-to-tray distance of 65.4 cm.

To set the TTD:

1. In the Physics mode **SELECT OPTION** menu, select **CONFIGURATION** and press **ENTER**.

The **SELECT CONFIGURATION** menu appears.

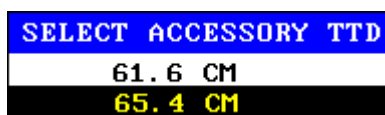
2. Select **ACCESSORIES** and press **ENTER**.

The **ACCESSORIES** menu appears.



3. Select **ACCESSORY TTD** and press **ENTER**.

The **SELECT ACCESSORY TTD** menu appears, with the current TTD selected.



4. Select the TTD you want and press **ENTER**.

The **ACCESSORIES** menu reappears.

5. To return to the **SELECT CONFIGURATION** menu:

- Press **SETUP**.
- Press **F1 EXIT**.

6. To save changes, press **F5**.

DYNAMIC TREATMENTS

The **DYNAMIC TREATMENTS** option in the **SELECT CONFIGURATION** menu provides four menu options:

- **ENHANCED D-WEDGE COMMISSION:** Allows you to commission (enable) and decommission (disable) Enhanced Dynamic Wedge (EDW) treatments for clinical use.
- **EDW ORIENTATION:** Allows you to specify whether users are required to select wedge orientation at both the console and the in-room pendant, or whether users can select wedge orientation at the console only.
- **MLC INTERFACE:** Allows you to disable communication between the accelerator software and the dynamic multileaf collimator (DMLC) software. Communication can be restored only by authorized service personnel.
- **DOSE EXPORT:** Allows you to enable or disable dose export feature.
- **EXTEND ETR LIMIT:** Raises the extended travel range to 90 cm and raises the start of ETR so that it is 10 cm above isocenter.

ENHANCED D-WEDGE COMMISSION

When you commission the Enhanced Dynamic Wedge (EDW) feature, you simultaneously enable EDW treatments for all field sizes and wedge angles.

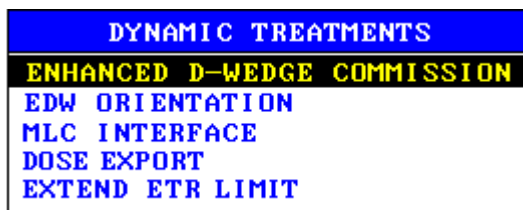
To commission or decommission EDW treatments:

1. In the Physics mode **SELECT OPTION** menu, select **CONFIGURATION** and press **ENTER**.

The **SELECT CONFIGURATION** menu appears.

2. Select **DYNAMIC TREATMENTS** and press **ENTER**.

The **DYNAMIC TREATMENTS** menu appears.



3. Select **ENHANCED D-WEDGE COMMISSION**, and press **ENTER**.

The **ENHANCED D-WEDGE** menu appears, with current status selected.

ENHANCED D-WEDGE
COMMISSION
DE-COMMISSION

4. Make a selection:

- Select **COMMISSION** and press **ENTER**.
- Select **DE-COMMISSION** and press **ENTER**.

A message appears stating that Enhanced Dynamic Wedge is commissioned or decommissioned, respectively.

In addition, the **DYNAMIC TREATMENTS** menu reappears.

5. To return to the **SELECT CONFIGURATION** menu:

- Press **SETUP**.
- Press **F1 EXIT**.

6. To save changes, press **F5**.

EDW ORIENTATION

The Enhanced Dynamic Wedge **EDW ORIENTATION** option in the **DYNAMIC TREATMENTS** menu allows you to specify whether users are required to select orientation at both the console and the in-room pendant, or whether users can select wedge orientation at the console only.



Note: This option applies only to Enhanced Dynamic Wedge.

To specify whether pendant selection of wedge orientation is required:

1. In the Physics mode **SELECT OPTION** menu, select **CONFIGURATION** and press **ENTER**.

The **SELECT CONFIGURATION** menu appears.

2. Select **DYNAMIC TREATMENTS** and press **ENTER**.

The **DYNAMIC TREATMENTS** menu appears.

DYNAMIC TREATMENTS
ENHANCED D-WEDGE COMMISSION
EDW ORIENTATION
MLC INTERFACE
DOSE EXPORT
EXTEND ETR LIMIT

3. Select **EDW ORIENTATION** and press **ENTER**.

The PENDANT SELECTION menu appears, with current status selected:



- **REQUIRED** indicates that users must select orientation both at the console and using the in-room pendant (after setup at the console).
 - **NOT REQUIRED** indicates that users may select orientation at the console, and are not required to also select it using the in-room pendant.
4. Select an option and press **ENTER**.
 - Select **REQUIRED**.

The software displays the message `Orientation Select AND Console`.
 - Select **NOT REQUIRED**.

The software displays the message `Orientation Select Console ONLY`.
 5. When you are finished, return to the **SELECT CONFIGURATION** menu by:
 - Press **SETUP**.
 - Press **F1**.
 6. To save changes, press **F5**.

MLC INTERFACE: Dynamic MLC Communication

An accelerator configured with dynamic MLC must have MLC interface enabled. Communication between the dynamic MLC software and accelerator software is initialized to default to the enabled state.

The **MLC INTERFACE** option in the **CONFIGURATION > DYNAMIC TREATMENTS** menu allows qualified physics and service personnel to disable communication between accelerator software and the dynamic multileaf collimator (DMLC) software for service and testing purposes. Varian recommends that the MLC interface remain in the enabled state.

Disabling dynamic MLC communication triggers an interlock that can be reset only by qualified physics and service personnel. For detailed information, refer to the *MLC File Format Description Reference Guide* and *DMLC Implementation Guide*.

If you try to reconnect communication with dynamic MLC by selecting **ENABLE** in the **MLC INTERFACE** menu, the system again opens the **DYNAMIC TREATMENTS** menu and displays the message `MLC Enabled`

This attempt to reconnect dynamic MLC communications with the accelerator turns on the software only, which is not sufficient to perform treatments. Qualified physics or service personnel must perform additional procedures at the accelerator console to restore the entire system.

DOSE EXPORT

The **DOSE EXPORT** option allows you to export dose information to third-party devices and interfaces.

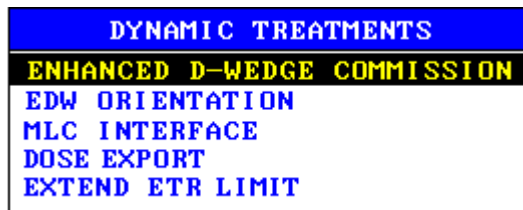
To enable dose export:

1. In the Physics mode **SELECT OPTION** menu, select **CONFIGURATION** and press **ENTER**.

The **SELECT CONFIGURATION** menu appears.

2. Select **DYNAMIC TREATMENTS** and press **ENTER**.

The **DYNAMIC TREATMENTS** menu appears.



3. Select **DOSE EXPORT**, and press **ENTER**.

The **DOSE EXPORT** menu appears.



4. Select an option and press **ENTER**.
5. When you are finished, return to the **SELECT CONFIGURATION** menu by:
 - Press **SETUP**.
 - Press **F1**.
6. To save changes, press **F5**.

EXTEND ETR LIMIT

The **EXTEND ETR LIMIT** option in the **DYNAMIC TREATMENTS** menu raises the extended travel range to 90 cm and raises the start of ETR to be 10 cm above isocenter.

The couch VRT limit specifies the point of maximum vertical couch travel, measured down from isocenter, and does not change when you enable ETR.

The ETR option supports treatment couch accessories such as the Prone Breast Couch Insert with KVUE Couch Top, which may require the couch to rise above the standard ETR for treatment.

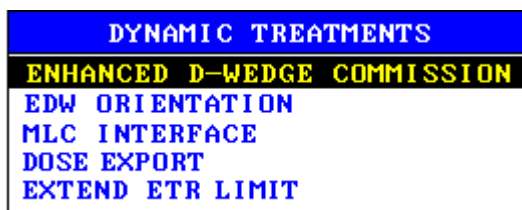
To enable or disable ETR:

1. In the Physics mode **SELECT OPTION** menu, select **CONFIGURATION** and press **ENTER**.

The **SELECT CONFIGURATION** menu appears.

2. Select **DYNAMIC TREATMENTS** and press **ENTER**.

The **DYNAMIC TREATMENTS** menu appears.



3. Select **EXTEND ETR LIMIT** and press **ENTER**.

The **EXTEND ETR LIMIT** menu appears.



4. Select an option and press **ENTER**.
5. When you are finished, return to the **SELECT CONFIGURATION** menu by:
 - Press **SETUP**.
 - Press **F1**.
6. To save changes, press **F5**.

REMOTE AUTO MOTIONS

The **REMOTE AUTO MOTIONS** option in the **SELECT CONFIGURATION** menu opens the **REMOTE AUTO MOTIONS** box. Depending upon your accelerator configuration, the options in this box allow you to:

- **Enable Auto Setup and Auto GoTo.** These capabilities support Auto Field Sequencing by a record-and-verify (RV) system. Auto Field Sequencing automates multiple port treatments (in Clinical or Special Procedures mode) when an accelerator is connected to a compatible RV system. For instructions on enabling the RV interface, see [RV \(Record-and-Verify\) INTERFACE](#) on page 63.

Depending upon your accelerator configuration and options, Varian factory settings determine which parameters are allowed and disallowed for Auto Setup and Auto GoTo.

- **Auto Setup** allows you to drive the gantry, collimator, MLC, and treatment couch to their planned positions from outside the treatment room.

You can enable or disable this capability for each axis.

- **Auto GoTo** allows you to manually enter target positions and then drive the gantry, collimator, MLC, and treatment couch to those target positions.

You can enable or disable this capability for each axis.

- **Configure large and small remote motion ranges** for gantry, upper and lower collimator jaws, and couch axes. The two right columns and the bottom part of this box allow you to:

- Disable or enable large remote motions
- Configure upper motion range limits for small remote motions
- Configure allowable motion range boundaries for large remote auto motions.

You can configure these ranges for individual machine axes. If an operator attempts to move beyond an allowed motion range, the motion is prevented, and an error message appears. For information about these error messages, see [Resolve Remote Motion Error Messages](#) on page 57.

For information about the maximum and minimum range limits for large and small remote motions, see [Range Limits for Large and Small Remote Motions](#) on page 55.

For detailed information about large and small remote auto motions, refer to the appropriate accelerator *Instructions for Use*.



Note: The accelerator software allows you to configure small remote motion range limits, but if you wish to use this capability, you must also enable small remote auto motions as a group. To learn how to enable or disable small motions as a group, see [SMALL MOTIONS](#) on page 72.

- **Specify whether the treatment room door can be opened or closed during remote gantry and couch motions.** This functionality allows you to open the treatment room door while motion is in progress. However, Varian accelerators do not allow you to start motion with the door open; motions must begin before the door opens.

REMOTE AUTO MOTIONS Box

When you select **REMOTE AUTO MOTIONS** from the **SELECT CONFIGURATIONS** menu, the **REMOTE AUTO MOTIONS** box appears. If you have an imaging system installed, the box appears as in [Figure 1](#). If you do not have an imaging system, only the left column under **ALLOWABLE RANGE** appears, as in [Figure 2](#) on page 52.

Values that appear dim or gray are read-only and cannot be changed.

REMOTE AUTO MOTIONS				
	AUTO SETUP	AUTO GOTO	LARGE MOTION	SMALL MOTION
GANTRY RTN	YES	YES	YES	2
COLL RTN	YES	YES		
COLL Y1/Y2	YES	YES		
COLL X1/X2	YES	YES		
COUCH URT	YES	YES	YES	2
COUCH LAT	YES	YES	YES	2
COUCH LNG	YES	YES	YES	2
COUCH RTN	YES	YES	YES	2
IMAGING ARMS RANGE-		RETRACTED		EXTENDED
GANTRY RTN (+/-)		25		10
COUCH URT (-)		5		5
COUCH LAT (+/-)		2		2
COUCH LNG (<)		132		132
EXTEND ZONE MARGIN		COUCH RTN		GANTRY
		0		0
ALLOW DOOR TO OPEN DURING MOTIONS				NO
LARGE GANTRY WITH LIMITED COUCH RTN				NO

1. AUTO SETUP
2. AUTO GOTO
3. LARGE MOTION
4. SMALL MOTION
5. IMAGING ARMS RANGE – EXTENDED
6. EXTEND ZONE MARGIN
7. ALLOW DOOR TO OPEN DURING MOTIONS
8. LARGE GANTRY WITH LIMITED COUCH RTN

Figure 1 REMOTE AUTO MOTIONS Box, Accelerator with an Imaging System

COUCH RTN	NU	NU	NU	Z
		ALLOWABLE	RANGE	
GANTRY RTN (+/-)		180		
COUCH VRT (-)		8		
COUCH LAT (+/-)		2		
COUCH LNG (<)		132		
EXTEND ZONE MARGIN	COUCH RTN		GANTRY	
	0		0	
ALLOW DOOR TO OPEN DURING MOTIONS				NO
LARGE GANTRY WITH LIMITED COUCH RTN				NO

1

1. Sets allowable range boundaries for accelerator with no imaging system.

Figure 2 ALLOWABLE RANGE Area, for Accelerator without an Imaging System

The **REMOTE AUTO MOTIONS** box (the numbers correspond to the numbers in [Figure 1](#)) comprises the following:

1. **AUTO SETUP** allows you to configure whether or not RV target values appear on the F2 Position screen, allowing you to drive to the target positions from outside the treatment room. You can enable or disable this for each axis individually.
 - **NO** indicates that the target value for that axis are still in the system, but the target value does not appear on the F2 Position screen. You are not able to move to the target position from outside the treatment room, but you are still able to move the axis to the target position using the pendant.
 - **YES** indicates that the target value for that axis appears on the F2 Position screen. You are able to move to the target position from outside the treatment room or inside the treatment room using the pendant.
2. **AUTO GOTO** allows you to enter or edit targets manually on the F2 Position screen. You can enable or disable this for each axis individually. Varian configures all these limits to NO by default.
 - **NO** indicates that the RV system can send a target value for that axis to the F2 Position screen, but you are not able to enter or change it.
 - **YES** indicates that the RV system can send a target value for that axis to the F2 Position screen, and you are allowed to enter or change it.
3. **LARGE MOTION** column allows you to enable or disable large remote motion for the gantry and for each couch axis. Default for these axes is NO. Varian sets collimator axes to YES and read-only because these do not pose a risk of collision.
4. **SMALL MOTION** column displays the motion range upper limits for small remote motions. Default values are set at 2 cm (couch) or 2° (gantry) by Varian, but can be set to 0 (no motion allowed), or increased up to 5 cm or 5°. For further information about the difference between large and small remote motions, see [Range Limits for Large and Small Remote Motions](#) on page 55.

To use any small motion capability, you must first enable all small motions as a group. Later, if you wish to disable some individual small motion axes, you can do this by setting the SMALL MOTION range upper limits for those axes to zero. For information about enabling and disabling small motions, see [SMALL MOTIONS](#) on page 72.

For detailed information about large and small remote auto motions, refer to the appropriate accelerator *Instructions for Use*.

5. **IMAGE ARMS RANGE** column allows you to configure motion ranges for the system when all imager arms are extended.

For information about the maximum and minimum range limits for large and small remote motions, see [Range Limits for Large and Small Remote Motions](#) on page 55.

If your accelerator system does not include an imaging system, only the leftmost ALLOWABLE RANGE column appears ([Figure 2](#) on page 52), because an accelerator system without an imager behaves in the same way as an accelerator system with an imager when all imager arms are retracted.

6. **EXTEND ZONE MARGIN** allows you to extend the Gantry Exclusion Zone and/or the Couch Rotational Exclusion Zone. The maximum allowable extension to either the Gantry Exclusion Zone or the Couch Rotational Exclusion Zone is 14°.

Examples:

If you enter 7 under COUCH RTN, then the Couch Rotational Exclusion Zone becomes 7° larger than the default exclusion zone (see [Table 5](#)).

Table 5 Couch Example

COUCH RTN	Scale	Default Couch Ex- tend Zone Margin: (0° extension)	Couch Extend Zone Margin: (7° extension)
Right side	Varian	165–85	172–85
	IEC 60101-2-1	345–265	352–265
	IEC 61217	15–95	8–95
Left side	Varian	195–275	202–275
	IEC 60101-2-1	15–95	8–95
	IEC 61217	345–265	352–265

If you enter 5 under GANTRY, then the Gantry Exclusion Zone becomes 5° larger than the default exclusion zone (see [Table 6](#)).

Table 6 Gantry Example

GANTRY	Scale	Default Gantry Extend Zone Margin: (0° extension)	Gantry Extend Zone Margin: (5° extension)
Right side	Varian	165–5	170–5
	IEC 60101-2-1	15–175	10–175
	IEC 61217	15–175	10–175
Left side	Varian	195–355	190–355
	IEC 60101-2-1	345–185	350–185
	IEC 61217	345–185	350–185

7. **ALLOW DOOR TO OPEN DURING MOTIONS** allows you to enable or disable the capability to continue remote auto motions if the treatment room door is opened while remote auto motions are proceeding. If you enable this capability, the door can only be opened *after* remote auto motion has started. You cannot start remote auto motions with the treatment room door open.

If you disable this capability, remote auto motion is stopped the moment you open the treatment room door.

8. **LARGE GANTRY WITH LIMITED COUCH RTN** allows you to enable or disable gantry motion if the couch is no more than 2° outside its allowable motion range. YES allows this motion; NO disables it.

Configure Remote Auto Motions

To configure remote auto motions and motion range boundaries and limits:

1. In the Physics mode **SELECT OPTION** menu, select **CONFIGURATION** and press **ENTER**.
The **SELECT CONFIGURATION** menu appears.
2. Select **REMOTE AUTO MOTIONS** and press **ENTER**.
The **REMOTE AUTO MOTIONS** box opens, displaying the current settings ([Figure 1](#) on page 51 or [Figure 2](#) on page 52).
3. Select each setting you want to change.

4. Press **F3** to toggle the settings between **YES** or **NO**, or type new values for motion limits.



Note: You can configure small motion range limits, even if small motions are disabled. However, to use any small motion capability, you must first enable all small motions as a group. Later, if you wish to disable individual small motion axes, you can do this by setting the **SMALL MOTION** range upper limits for those axes to zero. For information about enabling and disabling small motions, see [SMALL MOTIONS](#) on page 72.

5. Continue until you have made all changes.

After remote motions are configured, if you enter a remote motion value that is out of the permitted motion range, the motion is prevented and an error message appears. For information about these error messages, see [Resolve Remote Motion Error Messages](#) on page 57.

Range Limits for Large and Small Remote Motions

[Table 7](#) on page 55 and [Table 8](#) on page 56 describe the allowable and default boundary or range limits for large and small remote auto motions. *Maximum limits* are the maximum physical ranges that are allowable for a specific machine axis. *Default limits* are the limits that Varian uses to configure your accelerator before it is shipped from the factory.



WARNING: To avoid risk of collision, you must be aware of any changes to boundary and range limits. Increasing Large Motion axis limits beyond the default values can increase the risk of collision and can result in serious injury or death, as well as equipment damage.

For detailed information about large and small remote auto motions, refer to the appropriate accelerator *Instructions for Use*.

Table 7 Range Limits for Large and Small Remote Auto Motions for Gantry and Collimator

Motion	Limits	Rotation (RTN)		Jaws 1 and 2	
		Gantry	Collimator	Upper	Lower
		Degrees	Degrees	cm	cm
Small	Maximum	5.0	360.0	20.0	20.0
	Default	2.0	360.0	20.0	20.0
Large imager arms retracted (or no imager)	Maximum	185.0	—	—	—
	Default	180.0	—	—	—

Motion	Limits	Rotation (RTN)		Jaws 1 and 2	
		Gantry	Collimator	Upper	Lower
		Degrees	Degrees	cm	cm
Large imager arms extended	Maximum	185.0	—	—	—
	Default	10.0	—	—	—

Table 8 Range Limits for Large and Small Remote Auto Motions for Couch

Motion	Limits	Couch			
		Vertical (Vrt)	Lateral (Lat)	Longitudinal (Lng)	Rotation (RTN)
		cm	cm	cm	Degrees
Small	Maximum	5.0	5.0	5.0	5.0
	Default	2.0	2.0	2.0	2.0
Large imager arms retracted (or no imager)	Maximum	40.0	25.0	158.0	90.0
	Default	8.0	2.0	132.0	2.0
Large imager arms extended	Maximum	40.0	25.0	158.0	90.0
	Default	8.0	2.0	132.0	2.0

Resolve Remote Motion Error Messages

If you attempt a remote auto motion that falls outside the allowable boundary or range, the system prevents the motion and displays an error message on the console screen, as shown in the following figure.

CLINICAL							
MU 1		MU 2		TIME		DOSE RATE	
0		0		0.00		0	
				SYMMETRY			
				+0.0 +0.0			
				TRN RDL			
TREATMENT TYPE	ENERGY	MU	ORIG MU			TIME	ACCESSORY
Press MOTION ENABLE and >> keys							
Simultaneous large GANTRY and COUCH not allowed							
TARGET POSITIONS							
COLL RTN		deg	COLL Y1	cm	COUCH VRT	105.0	cm
FIELD Y		cm	COLL Y2	cm	COUCH LNG		cm
FIELD X		cm	COLL X1	cm	COUCH LAT		cm
GANTRY RTN	182.0	deg	COLL X2	cm	COUCH RTN		deg
COLL RTN	180.1	deg	COLL Y1	cm	COUCH VRT	101.6	cm
FIELD Y	9.9	cm	COLL Y2	cm	COUCH LNG	49.9	cm
FIELD X	10.1	cm	COLL X1	cm	COUCH LAT	99.9	cm
GANTRY RTN	181.1	deg	COLL X2	cm	COUCH RTN	180.0	deg
F1 END		F2 PENDANT		F3		F4	
F5		F6					

The remote motion messages use the following abbreviations to denote machine axes:

- CRTN: Couch rotation
- CVRT: Couch vertical
- CLNG: Couch longitudinal
- CLAT: Couch lateral
- GANT: Gantry

To resolve remote motion error messages, do any of the following:

- Click **CLEAR** to close the message.
- Type a new value in the **TARGET POSITIONS** box, if permitted.

Moving to the target position removes the message.

Table 9 describes remote motion error messages and actions you can take to resolve them.

Table 9 Remote Motion Error Messages

Error Message	Meaning	Action
Simultaneous large GANTRY and COUCH not allowed	Linear accelerator software does not allow you to move the gantry and couch at the same time from outside the treatment room.	Choose an option: ■ In two steps, move the gantry and couch separately, from outside the treatment room. ■ Enter the treatment room and move to the target position either manually (using the pendant or couch controls), or using Auto Setup (using the pendant).
■ CLNG large motion is disabled ■ CLNG small and large motions are disabled	Software received a target value for an axis (in this example, couch longitudinal), but the remote motion you are attempting is disabled in Physics mode.	Enter the treatment room and move to the target position either manually (using the pendant or couch controls), or using Auto Setup (using the pendant).
CVRT CURRENT position outside set range	Current position is outside the zone defined by the large remote motion boundary limits for the axis (in this example, couch vertical).	Choose an option: ■ In the TARGET POSITIONS box, type new values that are within the allowable range, and move to these values. ■ Press Clear. Enter the treatment room and move to the target position manually (using the pendant or couch controls), or using Auto Setup (using the pendant).



Note: To prevent the CLNG and CVRT CURRENT errors, an authorized user can enable large and small remote auto motions for couch motion axes.

Error Message	Meaning	Action
CVRT motion prevented couch in ETR	Couch is in the extended travel range and remote motion is not allowed.	Enter the treatment room and move to the target position manually (using the pendant or couch controls).
CRTN axis is disabled yet axis is Active.	Target value was entered for an axis (in this example, couch rotation), but Auto GoTo, Auto Setup, and Pendant Auto Setup for that axis are all disabled in your linear accelerator's factory configuration.	Enter the treatment room and move to the target position manually (using the pendant or couch controls).

For a case example of large and small remote motions, refer to the appropriate accelerator *Instructions for Use*.

BEAM DELIVERY TIME INTERLOCK

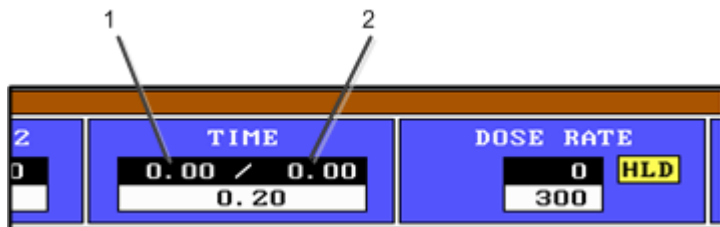
The **BEAM DELIVERY TIME INTERLOCK** option in the **SELECT CONFIGURATION** menu changes the time indicator in Clinical and Special Procedures modes and changes the way the TIME interlock works.

This feature is useful if you are using the Varian RPM™ respiratory gating system to turn the beam on and off in synchronization with the patient's breathing pattern.

BEAM TIME DELIVERY INTERLOCK Enabled

With **BEAM DELIVERY TIME INTERLOCK** enabled:

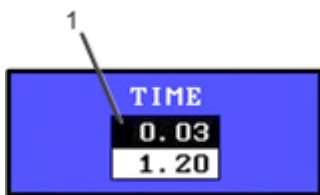
- The system displays two time values in the Time indicator area (see the following image for the item numbers that are referenced in brackets): One value for *beam time* [1] (the amount of time the beam was actually on), and one value for *elapsed time* [2] (total treatment time that includes both beam time and time that no beam was delivered).
- If beam time is exceeded, the system activates the TIME interlock and turns off the beam.
- The system does not monitor elapsed time and does not activate the TIME interlock if elapsed time is exceeded.



BEAM TIME DELIVERY INTERLOCK Disabled

With the **BEAM DELIVERY TIME INTERLOCK** disabled (see the following image for the item numbers that are referenced in brackets):

- The system displays a single time value, elapsed treatment time [1], in the Time indicator area. Beam time is NOT displayed.
- If the elapsed time is exceeded, the system activates the TIME interlock and turns off the beam.
- The system does not monitor beam time and does not activate the TIME interlock if the beam time is exceeded.



ENABLE/DISABLE BEAM TIME DELIVERY INTERLOCK

The default setting for the **BEAM DELIVERY TIME INTERLOCK** is **DISABLE**.

To enable or disable beam time display:

1. In the Physics mode **SELECT OPTION** menu, select **CONFIGURATION** and press **ENTER**.
The **SELECT CONFIGURATION** menu appears.
2. Select **BEAM DELIVERY TIME INTERLOCK** and press **ENTER**.
The **BEAM TIME INTERLOCK** menu appears.

BEAM TIME INTERLOCK
ENABLE
DISABLE

3. Select the appropriate task:

- To activate the interface, select **ENABLE** and press **ENTER**.
- To deactivate the interface, select **DISABLE** and press **ENTER**.

The BEAM TIME INTERLOCK is enabled or disabled, depending on your selection.

4. To return to the **SELECT CONFIGURATION** menu:

- Press **SETUP**.
- Press **F1 EXIT**.

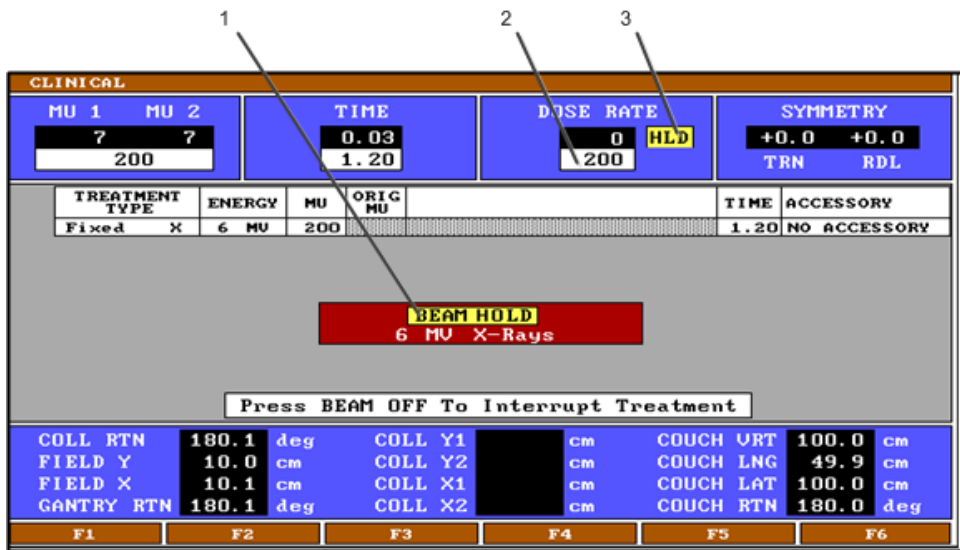
5. To save changes, press **F5**.

BEAM GATING ICON

The **BEAM GATING ICON** indicates when Beam Hold is active, and is enabled by default.

When the beam is held:

- Black-on-yellow BEAM HOLD message [1] appears instead of the BEAM ON message
- Beam hold icon [3] appears next to the dose rate [2]



- When the beam gating icon is disabled, if a dynamic target position is not reached in 15 seconds, the system triggers a DPSN interlock.
- When the beam gating icon is enabled, the system allows 30 seconds to reach a dynamic target position before triggering a DPSN interlock.



Note: Enabling or disabling this beam gating icon does not enable or disable respiratory gating.

To disable or enable beam gating:

1. In the Physics mode **SELECT OPTION** menu, select **CONFIGURATION** and press **ENTER**.

The **SELECT CONFIGURATION** menu appears.

2. Select **BEAM GATING ICON** and press **ENTER**.

The **BEAM GATING ICON** menu appears, with current status selected.



3. Select the appropriate task:

- To activate the interface, select **ENABLE** and press **ENTER**.
- To deactivate the interface, select **DISABLE** and press **ENTER**.

The beam gating icon is enabled or disabled, depending on your selection.

4. To return to the **SELECT CONFIGURATION** menu:

- Press **SETUP**.
- Press **F1 EXIT**.

5. To save changes, press **F5**.

RV (Record-and-Verify) INTERFACE

The **RV INTERFACE** option in the Physics mode **SELECT CONFIGURATION** menu allows you to use the accelerator with a record-and-verify (RV) system by linking to an information management system.

The **RV INTERFACE** option allows you to enable or disable the connection to the RV system.



Note: The RV interface should only be disabled for service or testing purposes. Never treat a patient with the RV interface disabled.

INTERFACE ENABLING

INTERFACE ENABLING allows you to link to an information management system. The system verifies that the accelerator is set up according to the treatment plan before allowing beam-on, and maintains accurate and reliable records of all treatments given on the accelerator.

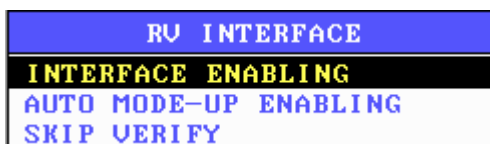
To enable or disable the RV interface:

1. In the Physics mode **SELECT OPTION** menu, select **CONFIGURATION** and press **ENTER**.

The **SELECT CONFIGURATION** menu appears.

2. Select **RV INTERFACE** and press **ENTER**.

The **RV INTERFACE** menu appears.



3. Select **INTERFACE ENABLING** and press **ENTER**.

The **INTERFACE ENABLING** menu appears, with the current status selected.



4. Select the appropriate task:

- To activate the interface, select **ENABLE** and press **ENTER**.
- To deactivate the interface, select **DISABLE** and press **ENTER**.

The **RV interface** is now enabled or disabled, depending on your selection. The **RV INTERFACE** menu reappears.

5. To return to the **SELECT CONFIGURATION** menu:

- Press **SETUP**.
- Press **F1 EXIT**.

6. To save changes, press **F5**.

AUTO MODE-UP ENABLING

AUTO MODE-UP ENABLING allows the RV system to send preplanned beam parameters to the accelerator console. Following operator verification, the system automatically modes up to the plan beam parameters and performs the calibration check cycle. Beam parameters include treatment type, energy, total dose, dose rate, time, accessories, start and stop angles, and field sizes.

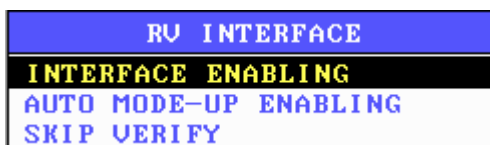
To enable or disable Auto Mode-Up:

1. In the Physics mode **SELECT OPTION** menu, select **CONFIGURATION** and press **ENTER**.

The **SELECT CONFIGURATION** menu appears.

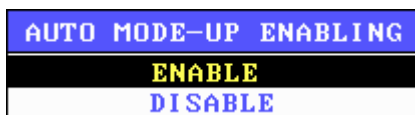
2. Select **RV INTERFACE** and press **ENTER**.

The **RV INTERFACE** menu appears.



3. Select **AUTO MODE-UP ENABLING** and press **ENTER**.

The **AUTO MODE-UP ENABLING** menu appears, with the current status selected.



4. Select the appropriate task:

- To activate the interface, select **ENABLE** and press **ENTER**.
- To deactivate the interface, select **DISABLE** and press **ENTER**.

Auto Mode-Up is now enabled or disabled, depending on your selection. The **RV INTERFACE** menu reappears.

5. To return to the **SELECT CONFIGURATION** menu:

- Press **SETUP**.
- Press **F1 EXIT**.

6. To save changes, press **F5**.

SKIP VERIFY

SKIP VERIFY allows you to skip the **VERIFY OR MODIFY SETUP** step during treatments using a record-and-verify (RV) system, so you can go directly into the calibration and check cycles during mode-up in Clinical and Special Procedures modes. This minimizes the number of button pushes required during treatments.



Note: *SKIP VERIFY works only with fields downloaded by an RV system. If you are doing a manual treatment or using a stand-alone system, you cannot skip the VERIFY OR MODIFY SETUP menu.*

When **SKIP VERIFY** is enabled, after the calibration and check, the software waits for you to remove the KEY interlock by turning the **DISABLE/ENABLE** key switch to **ENABLE**.

If treatment parameters do not agree with the patient's treatment plan, you can still press **F4** to return to the treatment parameters and make changes.

To enable or disable Skip Verify:

1. In the Physics mode **SELECT OPTION** menu, select **CONFIGURATION** and press **ENTER**.

The **SELECT CONFIGURATION** menu appears.

2. Select **RV INTERFACE** and press **ENTER**.

The **RV INTERFACE** menu appears.

RV INTERFACE
INTERFACE ENABLING
AUTO MODE-UP ENABLING
SKIP VERIFY

3. Select **SKIP VERIFY** and press **ENTER**.

The **SKIP VERIFY** menu appears, with the current status selected.

SKIP VERIFY
YES
NO

4. Select the appropriate task:

- To skip the **MODIFY/VERIFY** step during treatments, select **YES** and press **ENTER**.
- To require the **MODIFY/VERIFY** step during treatments, select **NO** and press **ENTER**.

Skip Verify is now enabled or disabled, depending on your selection. The RV INTERFACE menu reappears.

5. To return to the **SELECT CONFIGURATION** menu:

- Press **SETUP**.
- Press **F1 EXIT**.

6. To save changes, press **F5**.

BEAMSTOPPER

The beamstopper is an optional shield that is designed to extend under the treatment couch, preventing accidental radiation exposure in situations where barrier walls or ceiling may be insufficiently shielded..



WARNING: Failure to use the beamstopper when the beam is directed at insufficiently shielded barrier walls or ceiling may expose personnel outside the treatment room to unplanned radiation. The selection of gantry angles and energies that require the use of the beamstopper should be done only by a qualified physicist. The adequacy of the beamstopper positioning must be checked with calibrated survey instruments after setting or resetting any variables.

The **BEAMSTOPPER** option in the Physics mode **SELECT CONFIGURATION** menu opens the **BEAMSTOPPER ACTIVE/RANGES** box. This box allows you to specify energies that require the beamstopper.

If you configure the beamstopper to be extended for a particular energy, you must specify the start and stop angles where the beamstopper must be extended for that energy. Up to ten start and stop sectors are allowed.

Beamstopper modes with a status of **YES** are then constrained by the specified ranges. When the therapist selects an energy and a position that is within one of these specified ranges, if the beamstopper does not extend, the STPR interlock is asserted.

Valid angle ranges lie between 0° and 359°. Ranges may not go through 0° Varian scale or 180° IEC scale. You must specify ranges in the counterclockwise (CCW) direction; that is:

- 0° through 180° rather than 180° through 90°, Varian scale.
- 90° through 0° rather than 0° through 90°, IEC scale.

Setting both the start and stop angles to zero specifies that the beamstopper should be used for 360 degrees of motion. If you type the same values (other than zero) for start and stop angles, your entries are converted to 0° when you exit the **BEAMSTOPPER ACTIVE/RANGES** data box.

To configure the beamstopper:

1. In the Physics mode **SELECT OPTION** menu, select **CONFIGURATION** and press **ENTER**.
The **SELECT CONFIGURATION** menu appears.
2. Select **BEAMSTOPPER** and press **ENTER**.

The **BEAMSTOPPER ACTIVE/RANGES** data box appears, displaying the current values.

This figure shows an example with IEC scale values, that specifies a range starting at 135° and stopping at 45°. A second range is specified starting at 335° and ending at 225°, IEC scale.

BEAMSTOPPER ACTIVE				RANGES		
				Start	Stop	
6 MV	X-Rays	YES		1	135	45
6 MV SRS	X-Rays	YES		2	335	225
18 MV	X-Rays	YES		3		
6 MeV	Electrons	NO		4		
9 MeV	Electrons	NO		5		
12 MeV	Electrons	NO		6		
15 MeV	Electrons	NO		7		
18 MeV	Electrons	NO		8		
22 MeV	Electrons	NO		9		
				10		

You must specify ranges in the counterclockwise (CCW) direction; that is:

- 0° through 180° rather than 180° through 90°, Varian scale.
- 90° through 0° rather than 0° through 90°, IEC scale.

This figure is an example that specifies a range (in Varian scale, while facing the gantry) that starts at 45° and stops at 135°. A second range is specified starting at 225° and ending at 315°, Varian scale. This beamstopper setting *protects* the wall on the right of the accelerator.

BEAMSTOPPER ACTIVE				RANGES		
				Start	Stop	
6 MV	X-Rays	YES		1	45	135
6 MV SRS	X-Rays	YES		2	225	315
18 MV	X-Rays	YES		3		
6 MeV	Electrons	NO		4		
9 MeV	Electrons	NO		5		
12 MeV	Electrons	NO		6		
15 MeV	Electrons	NO		7		
18 MeV	Electrons	NO		8		
22 MeV	Electrons	NO		9		
				10		



Note: To navigate the BEAMSTOPPER ACTIVE/RANGES data box, either press the arrow keys, or press **ENTER**.

3. In the **BEAMSTOPPER ACTIVE** box, you can specify whether the beamstopper must be extended (**YES**) or not extended (**NO**) by pressing **F3 TOGGLE** when the cursor is beside the desired energy.

Each time you press F3, the status changes from YES to NO or NO to YES.

4. To specify a range for the beamstopper:
 - a. Type the start angle and press **ENTER**.
 - b. Type the stop angle and press **ENTER**.
5. To return to the **SELECT CONFIGURATION** menu:
 - Press **SETUP**.
 - Press **F1 EXIT**.
6. To save changes, press **F5**.

LOW DOSE RATE (LDR)

Accelerator software includes a range of extremely low dose rates available on high energy accelerators in all photon energies for fixed X and TBI X. If the Low Dose Rate (LDR) feature is purchased, you can reach the LDR range in Clinical and Special Procedures modes by repeatedly pressing the **F5 DR DOWN** softkey (**F5** on the dedicated keyboard). LDR can be enabled and disabled in Physics.



Note: *PortalVision™ cannot be used while the accelerator is in the LDR range. As a work-around, if a PV image is desired during an LDR treatment, you can break the treatment field into two separate fields and schedule a port film between them.*

The UDRS interlock is enabled for all fixed treatments unless the planned dose rate is below 20 MU/minute. All treatments except gantry arc (ARC) and Dynamic Wedge (EDW) treatments are affected.



Note: *UDR1 and UDR2 are triggered when the dose rate drops below 80% of the set dose rate, as measured at the radial ion chambers and averaged over 600 millisecond periods. UDRS checks on the expected dose rate within a 50 millisecond period.*

Enable and Disable LDR

To enable or disable LDR:

1. In the Physics mode **SELECT OPTION** menu, select **CONFIGURATION** and press **ENTER**.

The **SELECT CONFIGURATION** menu appears.

2. Select **LOW DOSE RATE** and press **ENTER**.

The **LOW DOSE RATE** menu appears.

LOW DOSE RATE	
LOW X	LDR
HIGH X	LDR



Note: The two options in this menu, LOW X LDR and HIGH X LDR, allow you to enable or disable LDR for different energies on your accelerator.

3. Make your selection and press **ENTER**.

- To enable or disable LDR for your lower energy, select **LOW X LDR** and press **ENTER**.

The LOW X LDR menu appears, with current status selected.

LOW X LDR	
ENABLE	
DISABLE	

- To enable or disable LDR for your higher energy, select **HIGH X LDR** and press **ENTER**.

The HIGH X LDR menu appears, with current status selected.

HIGH X LDR	
ENABLE	
DISABLE	

4. Select the appropriate task:

- To enable LDR, select **ENABLE** and press **ENTER**.
- To disable LDR, select **DISABLE** and press **ENTER**.

LDR is now enabled or disabled for high energy or UNIQUE, depending on which options you selected. The LOW DOSE RATE menu reappears.

5. To return to the **SELECT CONFIGURATION** menu:

- Press **SETUP**.
- Press **F1 EXIT**.

6. To save changes, press **F5**.

SMALL MOTIONS

The **SMALL MOTIONS** option in the **SELECT CONFIGURATIONS** menu allows you to use the small remote auto motions feature to make small corrections to the couch position from outside the treatment room.

For information about configuring small motions range limits, see [REMOTE AUTO MOTIONS](#) on page 49. For an explanation of large and small remote auto motions, see [Range Limits for Large and Small Remote Motions](#) on page 55.

To enable or disable remote small motions:

1. In the Physics mode **SELECT OPTION** menu, select **CONFIGURATION** and press **ENTER**.

The **SELECT CONFIGURATION** menu appears.

2. Select **SMALL MOTIONS** and press **ENTER**.

The **SMALL MOTIONS** menu appears, with current status selected.



3. Make your selection and press **ENTER**.

The **SELECT CONFIGURATION** menu reappears.

4. To save changes, press **F5**.

SMART CONNECT

If your system is configured with Smart Connect, the **SMART CONNECT** option appears in your **SELECT CONFIGURATION** menu. This option allows you to enable or disable Smart Connect, the Varian Medical Systems remote service feature.

Smart Connect allows you to enable Varian support representatives to view your system computer interface using a secure internet connection. Varian uses Smart Connect to help troubleshoot your system, minimizing delays and service visits.

The Smart Connect connection is controlled by your authorized department personnel, and all department information is strictly confidential.

For more information about Smart Connect, contact your Varian Service or Sales personnel.

To enable or disable Smart Connect:

1. In the Physics mode **SELECT OPTION** menu, select **CONFIGURATION** and press **ENTER**.

The **SELECT CONFIGURATION** menu appears.

2. Select **SMART CONNECT** and press **ENTER**.

The **SMART CONNECT** menu appears, with current status selected.



3. Make your selection and press **ENTER**.

The **SELECT CONFIGURATION** menu reappears.

4. To save changes, press **F5**.

Save Physics Mode Configuration Data

When the **F5 SAVE** indicator appears at the bottom of the screen, you can save configuration data to a directory in Windows by pressing the **F5** key on the dedicated keyboard.

Using the **F5** function key saves all changes stored in memory since the last time changes were saved. You can either use **F5** once (after you make all changes) or you can use it multiple times (for example, after each change).

When you make configuration changes, the system retains the changes in memory only while you are in Physics mode. To retain the changes for the future, you must save them to the hard disk before you exit Physics mode.

To save the Physics mode configuration data:

1. In the Physics mode **SELECT OPTION** menu, use the dedicated keyboard to select **CONFIGURATION** and press **ENTER**.

The **SELECT CONFIGURATION** menu appears.

2. Press **F5**.
3. Make a selection:
 - To cancel the save, select **NO**.
 - To save configuration changes, select **YES**.
4. Press **ENTER**.

If you have selected **YES**, a message appears stating that the data is being saved.

The **SELECT CONFIGURATION** menu reappears.

For information on printing a report of the Physics mode configuration data, see [.Print Physics Mode Configuration Data](#) on page 199

Chapter 4 PHYSICS Mode: UTILITY Menu

UTILITY Menu

Utilities and functions formerly in the Physics mode **SELECT UTILITY** menu have been moved to the 4DITC workstation or the LogViewer application.

Table 10 Utilities Formerly in the UTILITY Menu

Utility	Location
Set the system date	4DITC workstation (the system date and time are automatically set when 4DITC is opened)
Set the system time	
Back up and restore the machine configuration	LogViewer on 4DITC workstation
View and back up Morning Checkout, Event, and Clinac Dynalogs, and screenshots	LogViewer on 4DITC workstation or customer workstation
Print logs and screenshots	LogViewer on customer workstation connected to a printer



Note: All functions except printing are available the 4DITC workstation using LogViewer. Printing is available on a customer workstation running LogViewer and connected to a printer.

Chapter 5 SERVICE Mode

Service Mode Screen

Service mode is used by authorized personnel to maintain the accelerator, calibrate its readouts, and troubleshoot it. This chapter is intended for service and other authorized personnel only.



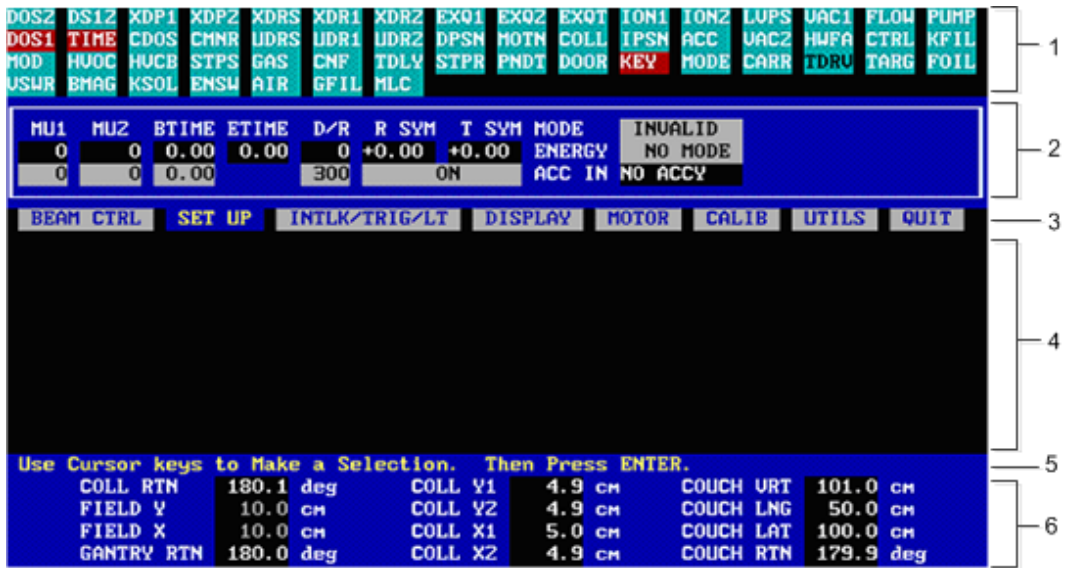
WARNING: Service mode procedures should be used only by qualified maintenance personnel. Unauthorized maintenance or use of the accelerator can result in radiation exposure, machine malfunctions, and other hazards that may result in damage to equipment, injury, or death. When using Service mode, take the following precautions:

- Only authorized Varian personnel should know the Service mode password. Do not divulge the Service mode password to unauthorized individuals.
- When the system is in Service mode and the beam is on, prevent personnel from entering the treatment room.
- Never use Service mode for clinical operation, or to treat patients.



Note: When selecting a new energy to perform dose or QA measurements in Service mode, wait an additional 6 seconds after the Ready condition is reached before beaming on. This allows time for the gun high voltage and bending magnet values to stabilize to provide stable dose measurements. In Clinical mode, this delay is automatically built into the Cal Check cycle during energy selection.

When you enter Service mode, the system displays the Service mode screen. For information about entering Service mode, see [Enter Service Mode](#) on page 83. This section describes the format of the screen.



1. **Interlock status:** Four rows of 3- or 4-character interlock codes
2. **Beam data:** Three rows of beam parameters
3. **Menu bar:** Single row of menu headings
4. **Interactive display:** Menu or data boxes that vary according to function
5. **Message line:** Single line that displays user instructions
6. **Position readouts (PROs):** Four rows and three columns displaying 12 parameters

Figure 3 Service Mode Screen

Interlock Status

The interlock status area at the top of the Service mode screen lists the interlocks that you can access in Service mode. [Table 11](#) lists the colors and backgrounds of the interlocks and their significance.



Note: You cannot use Service mode to access the CKFA, CKTO, CLFA, CLTO, DSFA, EXT, ORNT, and RVBP interlocks. For assistance with these interlocks, contact Varian customer service.

Table 11 Status of Interlocks

Interlock Code Color	Background	Meaning
White (steady)	Cyan	Interlock enabled but not activated
White (flashing)	Red	Interlock enabled and activated

Interlock Code Color	Background	Meaning
Black	Cyan	Interlock overridden but not activated
Black	Red	Interlock overridden and activated

The colors of the interlock code characters and background are interpreted in combination; for example, black letters and a red background indicate that the interlock has been overridden and is active. To reset and override interlocks, use the [Override Intlks Option: Override Interlocks](#) on page 109.

Beam Data

The beam data area displays current setup parameters, real-time exposure parameter values, and the current status of the machine.

Current Setup Parameters

Setup fields in the beam data area display the values that you have specified using the **SET UP** menu (for more information, see [SET UP Menu: Set Beam Test Values](#) on page 85). Setup values appear as black characters against a light gray background.

You must specify the following setup parameters for all treatments:

- **MU1** (primary dose in monitor units)
- **TIME** (exposure time)
- **D/R** (dose rate)
- **MODE** (exposure type)
- **ENERGY** (beam energy)
- **ACCESSORIES**

MU1 in the Current Setup Parameters

When you specify the MU1 value, the software displays a value for MU2 (backup dose) in white against a black background:

- If MU1 is greater than or equal to 1000, MU2 is 110% of MU1
- If MU1 is less than 1000, MU2 is 110% of MU1, or MU1 + 25, whichever is less
- If MU1 is less than 10, MU2 is MU1 + 1

For total body and HDTSe- treatments, the set point for MU2 (displayed in black against a gray background) is always 10% greater than MU1. The minimum value for the set point is 1 monitor unit greater than MU1.

Enhanced Dynamic Wedge and Arc Values in the Current Setup Parameters

When the mode is Enhanced Dynamic Wedge or Arc, the values for additional setup parameters appear at the extreme right of the beam data area. These are listed in [Table 12](#).

Table 12 Parameters Associated with Each Mode

Mode	Parameters
Enhanced Dynamic Wedge	■ ORNT-ANG (wedge orientation and angle) ■ Y1/Y2 (Y1 and Y2 initial positions) ■ ORG MU (original MU1 value for partial treatment)
Dynamic Arc	■ STRT ANG (start angle) ■ STOP ANG (stop angle) ■ ORG MU (original MUI value for partial treatment)

Real-Time Parameters

The real-time fields in the beam data area display current exposure data detected by the machine. Real-time values appear as white characters against a black background. The following real-time parameters are continually updated during beam-on:

- **MU1** (primary dose in monitor units)
- **MU2** (secondary dose in monitor units)
- **B TIME** (beam exposure time, if enabled)
- **E TIME** (elapsed time)
- **D/R** (dose rate)
- **R SYM** (relative beam symmetry in the radial plane)
- **T SYM** (relative beam symmetry in the transverse plane)

The code in the **ACC IN** field represents the accessory currently installed in the machine, not the accessory you select during setup. If no valid accessory is installed, the message **NO ACCY** appears in the field.

System Status in the Beam-On-Parameters Area

Real-time system status appears below the **R SYM** and **T SYM** readouts. They appear as black characters against a light gray background, except for **BEAM ON**, which appears as blinking white letters on a red background.

[Table 13](#) lists codes that indicate the current status of the accelerator.

Table 13 System Status Codes

Indication	Significance	Notes
STANDBY	The system is in Standby mode. Only the console, motor, vacuum, and water cooling systems are powered.	The system enters Standby whenever a major interlock occurs or the user turns the console power keyswitch on the console electronics cabinet to Off (extreme counterclockwise position).
TDLY (time delay)	The system is waiting for the machine to warm up after the power keyswitch is turned to on.	The TDLY status marker appears, with the delay time remaining until the TDLY interlock is deactivated.
ON	The system is on. All machine systems are powered.	The system is on when the time delay (TDLY) ends and no major interlocks are active.
READY	The system is in on, and no interlocks are active.	A full treatment has been entered and the DISABLE/ENABLE key switch on the dedicated keyboard has been turned to ENABLE . The accelerator is ready to beam on.
BEAM ON	The system is in the beam-on state.	Beam-on occurs when: - Valid setup values were set - No interlocks are active DISABLE/ENABLE keyswitch is turned to ENABLE - The system is in a ready state BEAM ON has been pressed

Menu Bar

The next area of the Service mode screen is the menu bar. When you first enter Service mode, the **SET UP** option in the menu bar is selected.



When you select a menu heading, a menu appears with available functions that you can select using the cursor (arrow) and **Enter** keys. [Table 14](#) briefly describes each of the Service mode menus.

Table 14 Service Mode Menu Bar Options

Option	Function
BEAM CTRL	■ Initiates the beam when the machine is in the ready state. ■ Terminates the beam when the beam is on.
SET UP	■ Sets the parameters required for a beam test exposure, including mode, energy, rep rate (dose rate), dose, time, and accessories. ■ Resets parameters to repeat a test.
INTLK/ TRIG/LT	■ Overrides and resets interlocks. ■ Controls trigger delays. ■ Disables system triggers. ■ Switches various lights on and off.
DISPLAY	Displays process data, parameters, readings, and status.
MOTOR	Moves gantry, collimator, and treatment couch axes in manual or automatic mode.
CALIB	■ Verifies operation of the dosimetry system. ■ Displays the position readouts (PRO) system and identifies the PRO scale. ■ Displays the motor parameters and the dynamic beam delivery (DBD) parameters. ■ Displays the current pulse repetition frequency (PRF) [for UNIQUE accelerators only]. ■ Allows adjustment of calibration values. ■ Saves calibration values to hard disk. ■ Allows you to select and display site-specific information: PRO scale setting selected for clinical modes, and software registration information.
UTILS	■ Displays the status of input/output (I/O) data or writes password-protected data to disk. ■ Changes the value of the communication time-out. ■ Runs controller diagnostics. ■ Downloads the current configuration. ■ Restores the Service mode configuration from the hard disk. ■ Activates keys on the hand pendant. ■ Tests mechanisms that read Type III accessory codes.
QUIT	Exits Service mode.

Availability of Options

There are two levels of password protection for the options described in this manual: Customer and Varian service. Default customer passwords can be changed in Physics mode (see [PASSWORDS](#) on page 27). Service passwords cannot be changed. [Table 15](#) lists the password protection level of the various options.

Items listed as NA are accessible only to Varian factory personnel.

Table 15 Password Protection Level of Service Mode Options

Option	Customer Password Protected	Varian Service Password Protected
Write I/O	No	Yes
Calibrate meters	No	Yes
Adjust motor parameters	NA	NA
Adjust dynamic motor parameters	NA	NA
Change axis motion limits and calibrate positions	NA	NA
Set PRF frequency	No	No
Other options	Yes	Yes

Interaction and Display

The interaction and display area displays menus, data entry boxes, and various kinds of process data in the center of the screen.

When you select an option from a menu, a data box may appear with one or more input fields, and sometimes an additional menu with more choices.

Message Line

Below the interaction and display area is a single text line where system messages appear. These messages provide error explanations and additional instructions for data entry.

Position Readouts (PRO)

Below the message line at the bottom of the Service mode screen is the position readout (PRO) area. This area provides a real-time digital display of the positions of each motor axis.



Note: For information about changing the scale in which the readouts are displayed, see [Select PRO Scale Option](#) on page 146.

Use Cursor keys to Make a Selection. Then Press ENTER.					
COLL RTN	180.1 deg	COLL Y1	4.9 cm	COUCH URT	101.0 cm
FIELD Y	10.0 cm	COLL Y2	4.9 cm	COUCH LNG	50.0 cm
FIELD X	10.0 cm	COLL X1	5.0 cm	COUCH LAT	100.0 cm
GANTRY RTN	180.0 deg	COLL X2	4.9 cm	COUCH RTN	179.9 deg

Enter Service Mode

After you finish daily start-up of the accelerator, or when you exit from all major modes, the system displays the **SELECT MAJOR MODE** menu.

- Select **SERVICE** and press **ENTER**.

You must use the standard alphanumeric (service) keyboard to continue in Service mode. However, device/motion and beam controls (such as gantry rotation, the MOTION ENABLE bar, and the BEAM OFF button) are still controlled from the dedicated keyboard.

To return to the **SELECT MAJOR MODE** menu without entering Service mode, press **Esc**.

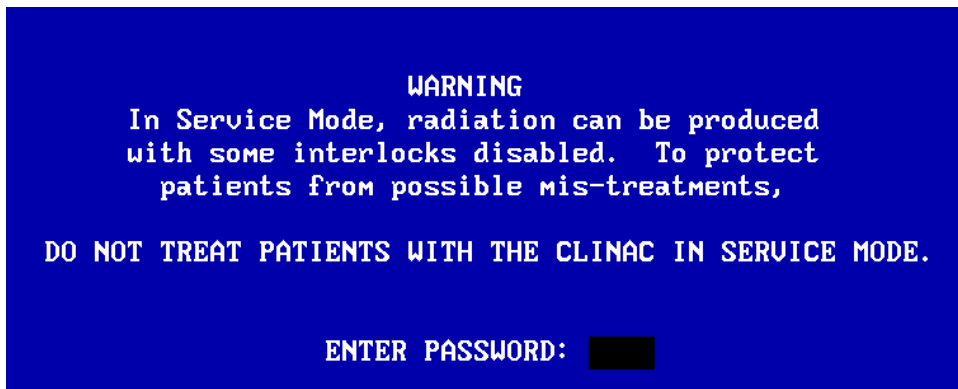
Password Protection in Service Mode

There are two levels of password protection available on-site for Service mode options: Customer and Varian service. (A third level is used at the factory.) The password used determines the Service mode options available to you. [Table 15](#) on page 82 lists the password protection level of the various options.

To enter Service mode:

1. In the **SELECT MAJOR MODE** menu, use the dedicated keyboard to select **SERVICE** and press **ENTER**.

The Service mode password data box appears.



You must use the standard alphanumeric (service) keyboard to continue in Service mode. However, device/motion and beam controls (such as gantry rotation, the MOTION ENABLE bar, and the BEAM OFF button) are still controlled from the dedicated keyboard.

Keys on the service keyboard are labelled in upper and lower case, and are depicted in upper and lower case in the following text; for example, **Enter** indicates the Enter key on the service keyboard.

To return to the SELECT MAJOR MODE menu without entering Service mode, press Esc.

2. On the service keyboard, type the password for Service mode and press **Enter**.

If you type an incorrect password, the system displays an error message. If necessary, retype the password to continue. If you have not typed the password correctly after four attempts, the SELECT MAJOR MODE menu reappears.

After you type the correct password, the Service mode screen appears.

Exit Service Mode

The **QUIT** option allows you to exit Service mode and return to the **SELECT MAJOR MODE** menu. To save configuration changes, you must save them before you exit Service mode (See [Write Data to Media Option: Save Service Mode Data](#) on page 163). After you exit Service mode, interlocks do not remain overridden, and the console gun delay reverts to its original value.

To exit Service mode:

1. Select **QUIT** and press **Enter**.
2. Select **YES** and press **Enter**.

Chapter 6 SERVICE Mode: BEAM TEST Menus

Beam Test Menu

Service mode is used by authorized personnel to maintain the accelerator, calibrate its readouts, and troubleshoot it. This chapter is intended for service and other authorized personnel only. It describes the following Service mode functions:

- **SET UP Menu: Set Beam Test Values** on page 85
- **BEAM CTRL Menu: Turn the Beam On and Off** on page 105
- **INTLK/TRIG/LT Menu: Interlocks, Triggers, and Lights** on page 107

Service mode also provides the means to specify therapy values for a beam test or exposure. These additional Service mode functions are described in [Meter Values Menus](#) on page 115.



WARNING: Service mode procedures should be used only by qualified maintenance personnel. Unauthorized maintenance or use of the accelerator can result in radiation exposure, machine malfunctions, and other hazards that may result in damage to equipment, injury, or death. When using Service mode, take the following precautions:

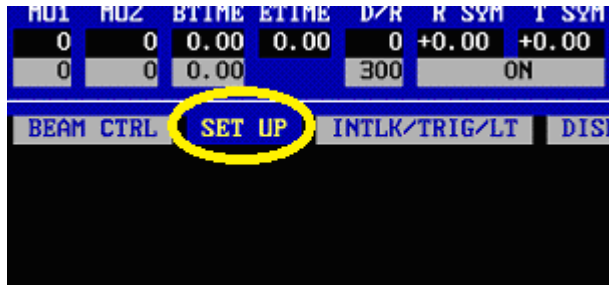
- Only authorized Varian personnel should know the Service mode password. Do not divulge the Service mode password to unauthorized individuals.
- When the system is in Service mode and the beam is on, prevent personnel from entering the treatment room.
- Never use Service mode for clinical operation, or to treat patients.



Note: When selecting a new energy to perform dose or QA measurements in Service mode, wait an additional 6 seconds after the Ready condition is reached before beaming on. This allows time for the gun high voltage and bending magnet values to stabilize to provide stable dose measurements. In Clinical mode, this delay is automatically built into the Cal Check cycle during energy selection.

SET UP Menu: Set Beam Test Values

The **SET UP** menu heading opens the **SET UP** menu, which allows you to set the parameters required for a beam test or exposure.



After you select the **SET UP** menu heading and press **Enter**, the **SET UP** menu appears.



Using the **SET UP** menu, you can select options individually in any order, or use the **Set all** option to automatically step through them:

- **Set all** leads you through all the parameters one by one ([Set All](#) on page 86)
- **Mode** ([Mode](#) on page 87) allows you to program the treatment type
- **Energy** ([Energy](#) on page 94) allows you to select the energy
- **Rep rate** ([Rep Rate](#) on page 96) allows you to set the dose rate
- **Dose** ([Dose \(MU\)](#) on page 97) allows you to set the MU
- **Time** ([Time](#) on page 98) allows you to set the exposure time
- **Accessories** ([Accessories](#) on page 98) allows you to select accessories
- **Perform repeat** ([Perform Repeat \(Repeat an Exposure\)](#) on page 101) allows you to repeat the previous exposure.

Set All

The **Set all** option sequentially presents each of the other setup parameters. In this way, you do not have to choose each option separately from the **SET UP** menu. If you intend to beam on, using **Set all** ensures that you set all required parameters.

Mode

The **Mode** option allows you to select the treatment type (in these menus, *Mode* refers to treatment type, rather than the functional state of the accelerator).



WARNING: Service mode removes some safety controls from the accelerator to allow service, repair, and calibration. Never give patient treatments in Service mode. Patient treatments delivered in Service mode may expose patients and others to unsafe radiation doses or other risks resulting in injury or death.

When you select **Mode** in the **SET UP** menu, the system displays the **SELECT MODE** menu; *Fixed* is selected by default.



The **SELECT MODE** menu offers these treatment types:

- **Fixed**
- **Arc**
- **Enh Dyn Wdg** (Enhanced dynamic wedge)

Plus these treatment types for high energy accelerators, if configured:

- **Total body**
- **HDTSe-**
- **SRS Fixed**
- **SRS ARC**
- **FFF Fixed**
- **FFF Arc**



Note: If the system does not accept entries for the **SET UP** parameters, check the status indicator in the beam data area. When the machine is in Standby mode, the system ignores setup commands and does not display error messages.

Before a mode is chosen, **INVALID** appears in the mode setup field in the beam data area. After you choose a mode, the treatment mode you have specified appears.

Select a Treatment Mode

To set a treatment mode:

- Select it from the **SELECT MODE** menu and press **Enter**.
 - For the **Fixed**, **Total Body**, or **HDTSe-** options, the specification is complete; there are no further parameters to select.
 - For the **Arc** and **Enh Dyn Wdg** [Enhanced Dynamic Wedge (EDW)] options, you must supply additional information:
 - To complete the Arc mode setup, see [Enter Arc Angles](#) on page 89.
 - To complete the EDW setup, see [Set Up Enhanced Dynamic Wedge Mode](#) on page 92.
 - When you select the **HDTSe-** mode, the system automatically sets the HDTSe- energy (6 or 9 MeV) and the dose rate (888 MU/min). The system displays these values in the beam data area.

Dynamic Treatment Modes

Arc and **Enh Dyn Wdg** (EDW) are dynamic beam delivery (DBD) modes in which a motor axis (gantry or upper collimator jaw) moves during beam-on. During a dynamic treatment, the control software of the accelerator continuously adjusts both the dose rate and the speed of the dynamic motor axis. When you select **Arc** or **Enh Dyn Wdg** (EDW) mode, you must specify additional setup parameters as part of the **Mode** option to define the unique elements for dynamic exposures.



Note: *Arc mode or EDW mode cannot be selected whenever the position readout (PRO) is displayed in raw scale.*

Dynamic treatments that are interrupted before completion can resume in partial treatment mode, even if other treatments have been given in the meantime. For instructions on delivering a partial treatment, refer to the appropriate linear accelerator *Instructions for Use*.

Specify Complete or Partial Treatment

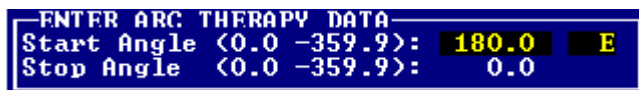
After you select a dynamic treatment mode, the system displays the **SELECT TRMT MODE** menu. Use this menu to specify a new or partial treatment.



- Select **New Treatment** when you are setting up a complete treatment.
- Select **Partial Treatment** when you are resuming an interrupted treatment (the software later prompts you to specify the MU remaining and the original MU, as part of the Dose option).

Enter Arc Angles

When you choose **Arc** mode, the **ENTER ARC THERAPY DATA** box appears, prompting you to specify the gantry start and stop angles.



The numbers in parentheses show the range of valid entries for the normal gantry range. Valid entries in the extended range for start angle are from 355.0°E to 359.9°E and from 0°E to 5.0°E Varian scale, and between 175.0°E and 185.0°E IEC scale. Valid entries in the extended range for stop angle are 0°E Varian scale and 180°E IEC scale. The E must be typed in for extended range.

Specify Extended Ranges

To specify an angle in the extended range, type in the angle and press **ENTER**. The data-entry space on the right is selected:

- For extended range, type the letter E.
- For normal range, leave it blank or type a space to erase a previous value.

The accelerator software determines the arc direction from the values you type. For example, [Table 16](#) shows sample angle entries used for a 360° rotation. Type the start angle and type the E in the extended range box. Specify the stop angle by typing the same angle value in the normal range and leaving the extended range box empty.

Table 16 Sample Values for a 360° Rotation

Angle	Varian	IEC	Rotation Direction
Start Angle	355E	185E	—
Stop Angle	355	185	counterclockwise
Start Angle	5E	175E	—
Stop Angle	5	175	clockwise

If you use 0° Varian scale or 180° IEC scale as the start and stop angles for a 360° rotation, the direction of rotation is determined by the angle in the extended range. An example is shown in [Table 17](#).

Table 17 Additional Sample Values for a 360 Rotation

Angle	Varian	IEC	Rotation Direction
Start Angle	0	180	—
Stop Angle	0E	180E	counterclockwise
Start Angle	0E	180	—
Stop Angle	0	180E	clockwise

To set up for Arc mode:

1. Select the **SET UP** menu heading and press **Enter**.

The SET UP menu opens.



2. Select **Mode** and press **Enter**.

The SELECT MODE menu opens.



3. Select **Arc** and press **Enter**.

The SELECT TRMT MODE menu opens.



4. Select a treatment type:
 - Select **New Treatment** and press **Enter**.
 - Select **Partial Treatment** and press **Enter**.

The ENTER ARC THERAPY DATA box appears.



5. Type the start angle (for a partial treatment, this is the start angle of the original complete treatment you want to resume) and press **Enter**.
6. For extended range, type E. For normal range, leave the E data-entry field blank. Press **Enter**.

The value appears in STRT ANG in the beam data area.

7. Type the stop angle (for a partial treatment, this is the stop angle of the original complete treatment you want to resume) and press **Enter**.
8. For extended range, type E. For normal range, leave the E data-entry field blank. Press **Enter**.

The value appears in STOP ANG in the beam data area.

9. To complete the set up:

- Press **Enter**.
- Press **ESC**.

The menu closes.

Set Up Enhanced Dynamic Wedge Mode

When you choose **Enhanced Dynamic Wedge** mode, the **ENTER ENHANCED DYNAMIC WEDGE DATA** box appears, allowing you to specify wedge orientation, Y1 and Y2 jaw positions, and wedge angle.

The wedge orientation is used to identify the upper jaw that moves during the exposure, **Y1 (IN)** or **Y2 (OUT)**. Select wedge orientation by typing either 1 or 2 when the Orientation field is selected. When you select **Y1 (IN)** or **Y2 (OUT)**, a corresponding image, on the collimator assembly gantry display, indicates the orientation of the wedge.

Coll Y1 and **Coll Y2** are the initial position of each jaw. Valid entries lie between 0.0 and 20.0 centimeters.

Valid entries for wedge angle are 10, 15, 20, 25, 30, 45, and 60.

To set up for Enhanced Dynamic Wedge treatment:

1. Select the **SET UP** menu heading and press **Enter**.

The SET UP menu opens.



2. Select **Mode** and press **Enter**.

The SELECT MODE menu opens.



3. Select **Enh Dyn Wdg** and press **Enter**.

The SELECT TRMT MODE menu appears.



4. Select a treatment type:
 - Select **New Treatment** and press **Enter**.
 - Select **Partial Treatment** and press **Enter**.

The ENTER DYNAMIC WEDGE DATA box appears.



5. Select the wedge orientation:
 - Type 1 (for in) and press **Enter**.
 - Type 2 (for out) and press **Enter**.
6. Type the **Coll Y1** position and press **Enter**.
7. Type the **Coll Y2** position and press **Enter**.
8. Type the wedge angle and press **Enter**.

The values you typed appear in the ORNT-ANG and Y1/Y2 setup fields.

Energy

The **Energy** option in the **SETUP** menu opens the **BCD#/ENERGY** menu, which allows you to select an X-ray or electron energy.

There is a separate menu for Flattening Filter-Free (FFF) treatments.



WARNING: Use of the Flattening Filter Free (FFF) feature without adequate site shielding can result in accidental exposure to radiation causing serious injury or death. Before installing FFF, ensure that your facility is properly shielded for FFF radiation.

In Service mode, you do not have to go through no mode or set mode before changing energies.

If you select an energy that does not support the existing dose rate, the dose rate is reset to the lowest available normal dose rate.

1. To open the **BCD#/ENERGY** menu, select **Energy** from the **SETUP** menu.

The BCD#/ENERGY menu appears, showing the available energies on your accelerator. If you have a non-FFF accelerator, then your BDC#/ENERGY menu should look like the menu on the left. If you have an FFF accelerator, then your BDC#/ENERGY menu should look like the menu on the right.

BCD#/ENERGY	
0	NO MODE
2	6 MV
1	18 MV
3	6 MeV
4	9 MeV
5	12 MeV
6	15 MeV
7	18 MeV

BCD#/ENERGY	
9	6 M
1	10 M

If you have not set energy in the current session, NO MODE is selected by default. Otherwise, the energy level is set to the last energy level that you selected.

2. To select an energy from the **BCD#/ENERGY** menu, do one of the following:
 - Select the energy and press **Enter**.
 - Type the BCD number that precedes the energy you want (for example, type 1 for 18 MV or 10M) and press **Enter**.

The energy you selected appears in the ENERGY setup field in the beam data area.



WARNING: It is possible to beam on with an invalid dose rate or assert an ACC interlock if the user does not perform `no mode` or `set mode` before changing energies. This can happen if the user types keystroke commands too fast. Arrays, such as the Trigger structure and the Accessory structure, may not download into common RAM properly.

In contrast, when you set up a treatment in Clinical or Morning Checkout modes, each selection prompts you for a specific response. Only proper responses are accepted and they must occur in a fixed order.

In Clinical mode, after treatment setup, the software calibrates, checks, and then verifies that all parameters are properly specified. The software always goes through the no mode, set mode, calibrate, check, and verify functions. To review the correct operating procedures, refer to the appropriate linear accelerator *Instructions for Use*.

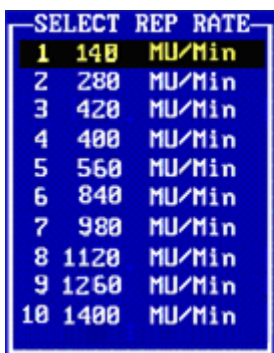
Rep Rate

Service mode uses the term *rep rate* (repetition rate) for dose rate. The **Rep rate** option in the **SET UP** menu opens a **SELECT REP RATE** menu that allows you to control the dose rate output.

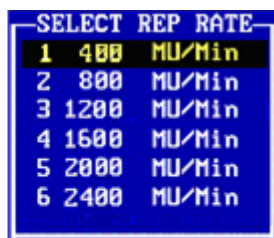
To set a rep rate:

1. From the **SETUP** menu, select **Rep rate**.

The **SELECT REP RATE** menu appears. (Depending upon your features and configuration, your menu may include options that differ from those shown in these illustrations.)



SELECT REP RATE		
1	140	MU/Min
2	280	MU/Min
3	420	MU/Min
4	400	MU/Min
5	560	MU/Min
6	840	MU/Min
7	980	MU/Min
8	1120	MU/Min
9	1260	MU/Min
10	1400	MU/Min



SELECT REP RATE		
1	400	MU/Min
2	800	MU/Min
3	1200	MU/Min
4	1600	MU/Min
5	2000	MU/Min
6	2400	MU/Min

The dose rate corresponding to the selected rep rate appears in the D/R (dose rate) setup field in the beam data area.

If you select either an X-ray energy or a dynamic treatment when the rate is already set at 1000 (for electron treatments), the entry in the D/R field automatically changes to the lowest available normal repetition rate.

2. To select an energy from the **SELECT REP RATE** menu, do one of the following:
 - Select the energy and press **Enter**.
 - Type the index number that precedes the rate you want (for example, type 2 for 280 MU/Min or 800MU/Min) and press **Enter**.

Dose (MU)

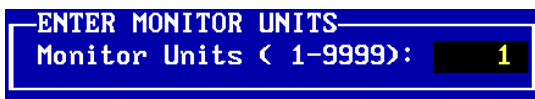
The **Dose** option on the **SET UP** menu opens two data boxes:

- For all treatments, the **ENTER MONITOR UNITS** box opens and allows you to enter the dose in monitor units (MU) for the current exposure.
- For partial treatments, the **ENTER ORIGINAL MONITOR UNITS** box opens and allows you to enter the original dose in monitor units (MU) for a partial dynamic exposure (if you have selected a partial exposure).

To set the dose:

1. From the **SETUP** menu, select **Dose** and press **Enter**.

The ENTER MONITOR UNITS box appears.



Numbers in parentheses show the valid range of entries. The default value is the last dose you typed. If you have not yet set the dose during the current Service mode session, the default is 1.

2. Type the dose for the current exposure, and press **Enter**.

Enter the Original Dose in a Partial Dynamic Treatment

For a partial dynamic treatment, after you specify the MU for the current exposure, the **ENTER ORIGINAL MONITOR UNITS** data box appears.



1. Type the MU *remaining* from the interrupted exposure in the **ENTER MONITOR UNITS** data box and press **Enter**.
2. Type the dose intended for the *original*, interrupted exposure in the **ENTER ORIGINAL MONITOR UNITS** data box and press **Enter**.

The monitor unit value specified for the dose in the MU 1 setup field and the monitor unit value specified for original dose in the ORIG MU setup field appear in the beam data area.



Note: When the mode is Enhanced Dynamic Wedge and you type a value for MU that is less than the minimum 20 MU required for one of these treatment types, a default value of 20 appears in the ORIG MU setup field, reminding you that this default value is already set.

Time

The **Time** option in the **SET UP** menu opens an **ENTER TIME** data box that allows you to specify an exposure time, in minutes.

To specify an exposure time:

1. From the **SETUP** menu, select **Time**.

The ENTER TIME box appears.

ENTER TIME
Minutes (0.01-99.99): 0.01

The numbers in parentheses show the valid range of entries.

2. Type a time value in the box, including a decimal point.
 - For fixed treatments, the default value is the last exposure time you specified. If you have not yet set time during the current Service mode session, the default is 0.01.
 - For dynamic treatments, the default value is a calculated value based on the dynamic parameters, MU and original MU.

After you type a value, the system displays the specified time in the TIME setup field in the beam data area.

Accessories

Accessories are beam collimation devices for treatment (such as wedges and applicators) or for other functions, such as port film exposure.

- To select an accessory, from the **SETUP** menu, select **Accessories**.

One of the following two menus appears, depending upon your energy selection:

- SELECT WEDGE menu, for X-ray energies
- SELECT APPLICATOR menu, for electron energies

SELECT WEDGE Menu

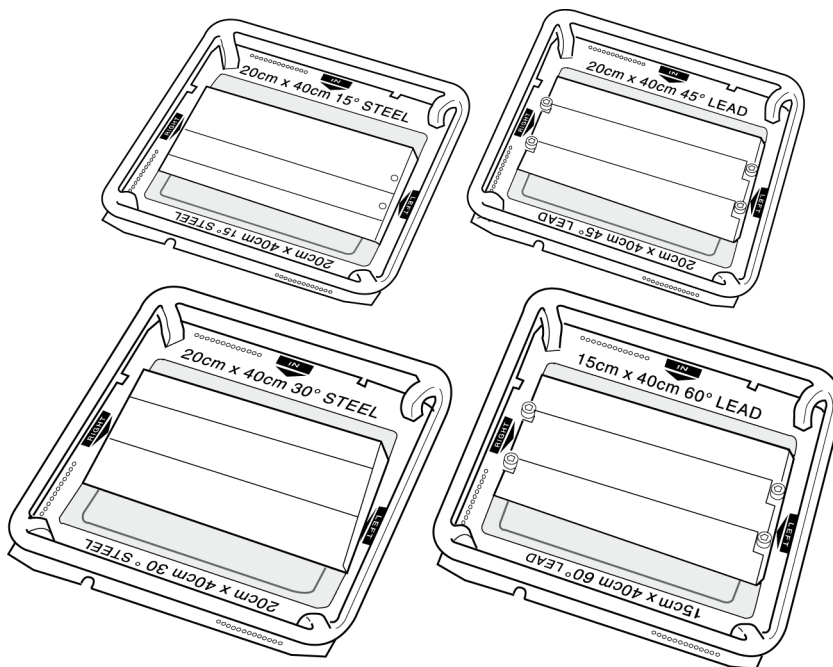
To select a wedge:

1. Select an X-ray energy in the **BCD#/ENERGY** menu.
2. Select **Accessories**.

The SELECT WEDGE menu appears.

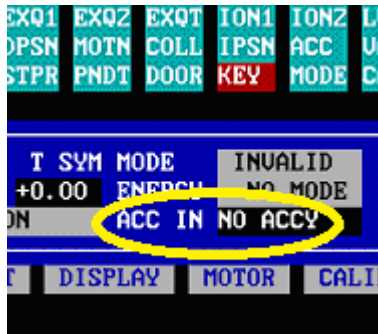
SELECT WEDGE				
15° IN/0°	30° IN/0°	45° IN/0°	60° IN/0°	NO ACCY
15° OUT/180°	30° OUT/180°	45° OUT/180°	60° OUT/180°	USER X
15° LEFT/270°	30° LEFT/270°	45° LEFT/270°	60° LEFT/270°	HDMLC Tray
15° RIGHT/90°	30° RIGHT/90°	45° RIGHT/90°	60° RIGHT/90°	MLC Tray

Each item in the SELECT WEDGE menu is either a standard wedge tray or other X-ray accessory. The designations IN, OUT, LEFT and RIGHT refer to wedge heel orientation in relation to the collimator head when you stand facing the yoke of the gantry. The wedge heel is the thickest part of the wedge. The arrow located on the wedge tray indicates the direction (IN, OUT, LEFT, or RIGHT) of the wedge heel for the wedge to be inserted into the accessory mount.



3. Select the wedge (or **NO ACCY** for no accessory), and press **Enter**.

The entry in the ACC IN field in the beam data area reflects the accessory installed, not the accessory selected at the console.



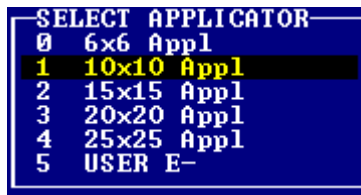
When your selection does not match the accessory that is installed on the machine, the system activates the ACC interlock to prevent beam-on. To proceed, you must correct the mismatch.

SELECT APPLICATOR Menu

To select an applicator:

1. Select an electron energy in the **BCD#/ENERGY** menu.
2. Select **Accessories**.

The SELECT APPLICATOR menu appears.



Each item in the SELECT APPLICATOR menu is either an applicator or other electron accessory. If you are using the optional 6 x 10 applicator, select USER E-.

3. Select an applicator from the **SELECT APPLICATOR** menu:

- Select the applicator and press **Enter**.
- Type the index number that precedes the applicator name in the menu list (for example, 0 for 6x6 Appl) and press **Enter**.

After you select the installed applicator, the collimator jaws move to the appropriate position.

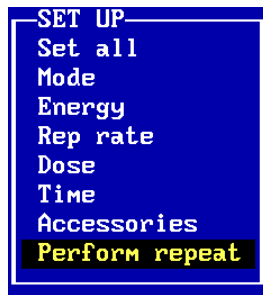
If your selection does not match the applicator that is installed on the machine, the ACC interlock appears until the mismatch is corrected.

Perform Repeat (Repeat an Exposure)

The **Perform repeat** option in the **SET UP** menu allows you to reuse the parameters of the immediately preceding exposure for another exposure.

To set up for a repeat exposure:

- From the **SET UP** menu, select the **Perform repeat** option and press **Enter**.



Set Up Test Treatments: Sample Scenarios

The following sample scenarios provide examples of how to set up two types of test treatments:

- Stereotactic fixed treatments
- Stereotactic arc treatments

Set Up a Stereotactic Fixed Treatment

To set up stereotactic fixed treatments:

1. From the **SET UP** menu:

- To view and program the menus for all parameters available for this treatment type, select **Set All**.
- To program individual parameters, select **Mode** and press **Enter**.

The SELECT MODE menu appears.

2. Select **SRS Fixed** and press **Enter**.

- If you selected Mode from the SET UP menu, the menus disappear, and SRS fixed treatment values appear in the treatment summary area. To program MU and time, you must select **Dose** and **Time** from the **SET UP** menu.
- If you selected Set All from the SET UP menu, the ENTER MONITOR UNITS box appears.

3. Type the MU value for the treatment and press **Enter**.

The ENTER TIME box appears.

4. Type the time value for the treatment and press **Enter**.

The menus disappear, and SRS fixed treatment values appear in the treatment summary area.



Set Up a Stereotactic Arc Treatment

To set up a stereotactic arc treatment:

1. From the **SET UP** menu, choose one of the following:
 - To view and program the menus for all parameters available for this treatment type, select **Set All**.
 - To program individual parameters, select **Mode** and press **Enter**.

The SELECT MODE menu appears.

2. Select **SRS Arc** and press **Enter**.

The SELECT TRMT MODE menu appears.

3. Select a treatment type.
 - Select **New Treatment** and press **Enter**.
 - Select **Partial Treatment** and press **Enter**.

The ENTER ARC THERAPY DATA box appears.

4. Type the start angle (For a partial treatment, this is the start angle of the original complete treatment you want to resume.) and press **Enter**.
5. Define extended range:
 - For extended range, type E.
 - For normal range, leave the E data-entry field blank and press **Enter**.

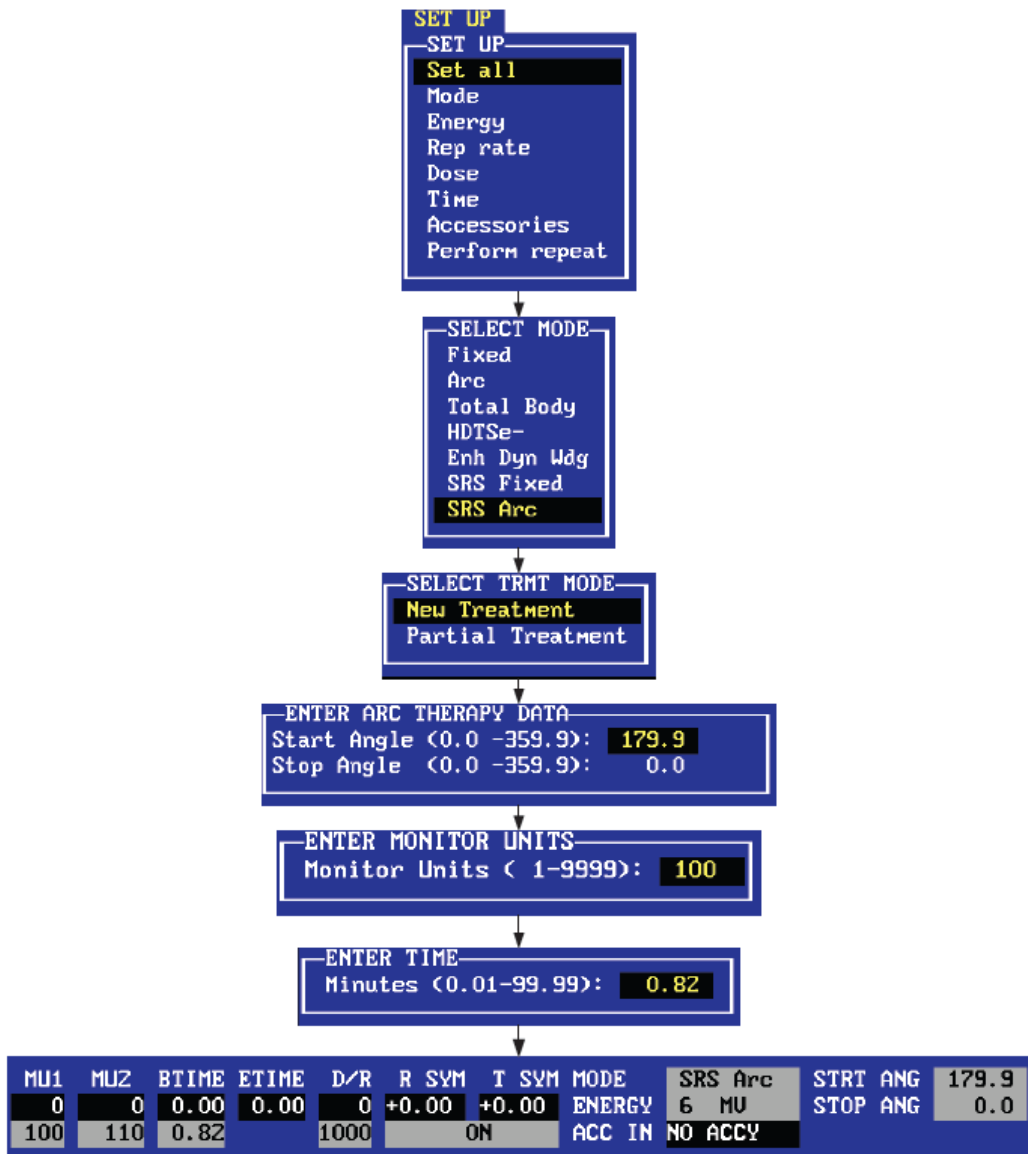
The value appears in STRT ANG in the beam data area.

6. Type the stop angle (For a partial treatment, this is the stop angle of the original complete treatment you want to resume.) and press **Enter**.
7. Define extended range:
 - For extended range, type E.
 - For normal range, leave the E data-entry field blank and press **Enter**.

The value appears in STOP ANG in the beam data area.

- If you selected Mode from the SET UP menu, the menus disappear, and SRS arc treatment values appear in the treatment summary area. To program MU and time, you must select Dose and Time from the SET UP menu.
 - If you selected Set All from the SET UP menu, the ENTER MONITOR UNITS box appears.
8. Type the MU value for the treatment and press **Enter**.
- The ENTER TIME box appears.
9. Type the time value for the treatment and press **Enter**.

The menus disappear, and SRS arc treatment values appear in the treatment summary area.



BEAM CTRL Menu: Turn the Beam On and Off

The **BEAM CTRL** menu heading opens the **CONFIRM BEAM ON** menu, which allows you to beam on when the machine is in the ready state.



Figure 4 CONFIRM BEAM ON Menu

If the beam is off when you select this option, **NO** is selected. If the beam is already on, selecting the **BEAM CTRL** menu heading terminates the beam.

Beam On

To beam on:

- In the **CONFIRM BEAM ON** menu (Figure 4):
 - Select **YES** and press **Enter**.
 - To exit without starting the beam:
 - Select **NO**.
 - Press **Esc**.



Note: During beam-on, you can select other menus, such as the **INTLK/TRIG/LT** and **DISPLAY** menus, and you can change most parameters. Dose rate can also be changed during beam on. However, you cannot change any of the **SET UP** parameters while beam is on.

When you select **YES** to beam on, the software checks the setup values.

If you typed the setup values correctly, the accelerator beams on. To indicate that the beam is on, the system flashes the words **BEAM ON** in white letters on a red background in the beam data area.

In addition, the accumulated dose, treatment time, current dose rate, and beam symmetry data change in real time to reflect the progress of the exposure while the beam is on.



Note: If any exposure value is not set, a message appears with the names of the missing parameters. (For example: **BEAM ON did NOT Occur! CAUSE: Dose Not Set.**) You must enter the missing values before you can beam on.

Beam Off

To beam off:

- Perform one of the following:
 - Select the **BEAM CTRL** menu heading.
 - Type the letter B.
 - Press **BEAM OFF** on the dedicated keyboard.

No menu appears, and the software turns off the beam.

INTLK/TRIG/LT Menu: Interlocks, Triggers, and Lights

The INTLK/TRIG/LT menu in Service mode allows you to:

- **Override or reset interlocks.** *Interlocks* are electrical, software, or mechanical precautionary functions that prevent accelerator operation or the application of power to accelerator primary systems until one or more conditions have been met. For further details, and a guide to the meaning of individual interlocks, see [Interlocks](#) on page 207.



WARNING: Overriding an interlock on the accelerator could cause a machine malfunction that results in damage to equipment, bodily injury, or death. Only authorized knowledgeable persons should clear or override interlocks.

- **Disable system triggers and change gun delays.** *Triggers* are timing signals that coordinate beam production and dosimetry.
- **Switch lights on and off.** Lights controlled by this menu include: Field lights, range finder projectors, back pointer, various laser beams, and the treatment room lights.

View the Interlock Status Area

The interlock status area at the top of the Service mode screen lists interlocks you can access in Service mode. The software selects the last interlock selected (yellow characters on a black background). If no interlock has been selected in the current Service mode session, the interlock at the top left is selected. Interlocks can be enabled or disabled, active or not active, or overridden. The display indicates the interlock status. For example, white letters flashing on a red background indicate that the interlock is enabled and active. [Table 18](#) lists the meaning of each foreground and background color combination.

Table 18 Status of Interlocks

Interlock Code Color	Background	Meaning
White (steady)	Cyan	Interlock enabled but not activated
White (flashing)	Red	Interlock enabled and activated
Black	Cyan	Interlock overridden but not activated
Black	Red	Interlock overridden and activated

If you have overridden an interlock and it is still active, the system displays it as black letters on a red background. When you have overridden it and it is not active, the system displays it as black letters on a cyan background. For additional information, refer to the appropriate *Instructions for Use*.

There are three categories of interlocks:

- Major
- Minor
- Dosimetry

Table 19 Types of Interlocks

Interlock Type	Description
Major	Identify conditions that could damage the accelerator if not corrected (for example, a PUMP interlock indicates the water level is low or the water temperature is too high).
Minor	Identify system conditions that you can usually correct (for example, a GAS interlock indicates that the waveguide dielectric gas pressure is too low).
Dosimetry	Identify conditions that may impair the ability of the accelerator to deliver or measure the radiation dose (for example, an XDRS interlock indicates that the actual servoed dose rate, over a given period of time, was above that which was intended).

Each type of interlock becomes active during different operating cycles.

Override Intlks Option: Override Interlocks

When you select the INTLK/TRIG/LT menu heading, the INTLK/TRIG/LT menu appears.

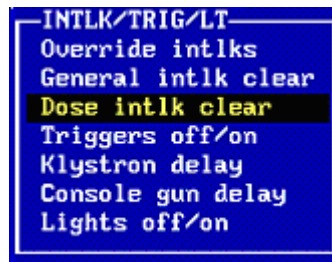
Override intlks allows you to override interlocks that are active or that might become active during an exposure. Overriding an interlock means that when the system activates the interlock, it does not terminate the beam.

This menu also allows you to reactivate an interlock that was overridden. Selecting **Override intlks** opens a **KEYBOARD OVERRIDE** message that displays instructions explaining how to override interlocks.

To override or reactivate an interlock:

1. Select the INTLK/TRIG/LT menu heading and press **Enter**.

The INTLK/TRIG/LT menu opens.



2. Select **Override intlks** and press **Enter**.

The KEYBOARD OVERRIDE menu appears.



3. In the interlock status area, use the cursor (arrow) keys to select the interlock to override and press **Enter**.

The interlock appears in black letters.

4. To reactivate an overridden interlock, select it and press **Enter**.

The interlock appears in white letters.

5. When you are finished, press **Esc**.

General Intlk Clear and Dose Intlk Clear Options

A latched interlock does not clear itself even when the condition that created the interlock is met.

To resolve a latched interlock, you must reset or deactivate it:

- Reset or deactivate an interlock:
 - For *nondosimetry* interlocks, from the **INTLK/TRIG/LT** menu, select **General intlk clear**.
 - For *dosimetry* interlocks, from the **INTLK/TRIG/LT** menu, select **Dose intlk clear**.

If the conditions that caused the interlock persist, the interlock can become active again even though it was reset or deactivated. For more information, see the descriptions in [Table 31 Interlocks, Conditions to Correct, and Actions](#) on page 209.

Triggers Off/On Option

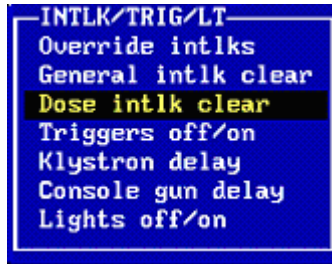
A *trigger* is a timing signal that coordinates beam production and dosimetry. The **Triggers off/on** option in the **INTLK/TRIG/LT** menu allows you to deactivate selected triggers while you perform maintenance and troubleshooting procedures. Selecting the **Triggers off/on** option opens the version of the **TRIGGER ENABLE** menu that is appropriate for your accelerator model.

The system displays the real-time status of each trigger as **ON** or **OFF**. Any trigger status changes you make from the **TRIGGER ENABLE** menu take effect when you exit the menu, and remain in effect only during the current Service mode session. (You cannot disable triggers in Clinical mode.) The system resets all triggers when you exit Service mode.

To set the trigger status:

1. Select the **INTLK/TRIG/LT** menu heading and press **Enter**.

The **INTLK/TRIG/LT** menu appears.



2. Select **Triggers off/on** and press **Enter**.

The TRIGGER ENABLE menu opens.



3. Select the trigger and press **Enter**.

Each time you press Enter, the status toggles from OFF to ON or from ON to OFF.

4. When the status of all triggers is set as desired, press **Esc**.

The changes are implemented, and the menu closes.



Note: On the Clinac 600C, 2100C/D, and 2300C/D, depending on factory configuration, if you have selected the highest rep rate, you may not be able to turn off the dose servo. Even if you can turn the dose servo off, you may be unable to use the highest rep rate.

Klystron Delay Option

Klystron trigger timing must be precise to maximize the beam output. The **Klystron delay** option in the **INTLK/TRIG/LT** menu allows you to fine-tune the timing of the klystron trigger pulse train. This option is only available on high energy accelerators.

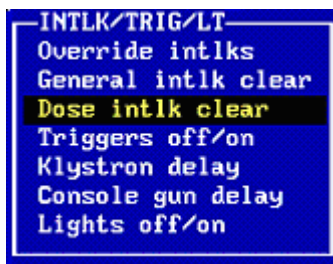
Selecting the **Klystron delay** option opens the **ENTER KLYSTRON TRIGGER DELAY** data box.

The default value is the klystron trigger delay currently used in Clinical mode. The delay is adjustable in quarter-microsecond (μsec) increments. Changing the delay is temporary. Unless you save the new value, a change affects only the current Service mode session. (To save the adjusted delay setting, see [Write Data to Media Option: Save Service Mode Data](#) on page 163.)

To adjust the klystron trigger delay:

1. Select the **INTLK/TRIG/LT** menu heading and press **Enter**.

The INTLK/TRIG/LT menu appears.



2. Select **Klystron delay** and press **Enter**.

The ENTER KLYSTRON TRIGGER DELAY data box appears.



3. Enter the delay you want (in 0.25 μsec) and press **Enter**.

The data box closes.

Console Gun Delay Option

The **Console gun delay** option in the INTLK/TRIG/LT menu allows you to control the electron gun using a console-generated trigger instead of the Klystron-generated trigger. Selecting this option opens the **ENTER CONSOLE GUN TRIGGER DELAY** data box, where you can enter a delay value for the console gun trigger.

The console gun delay option is available only on high energy accelerators.

The default value is the console gun trigger delay currently used in Clinical mode. The delay is adjustable in quarter-microsecond (μsec) increments.

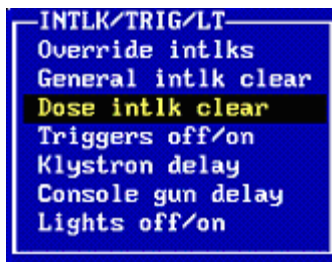


Note: Changing the console gun delay is temporary and affects the current Service mode session only. The software resets the triggers when you exit Service mode; you cannot change the console gun delay value permanently.

To adjust the console gun trigger delay:

1. Select the **INTLK/TRIG/LT** menu heading and press **Enter**.

The INTLK/TRIG/LT menu appears.



2. Select **Console gun delay** and press **Enter**.

The ENTER CONSOLE GUN TRIGGER DELAY data box opens.



3. Enter the delay you want (in 0.25 μsec) and press **Enter**.

The data box closes.

Lights Off/On Option

The **Lights off/on** option in the **INTLK/TRIG/LT** menu allows you to control the field light, range finder projectors, back pointer, laser beams, and the treatment room lights. Selecting the **Lights off/on** option opens the **LIGHTS** menu showing the current status (**OFF** or **ON**) of each light.

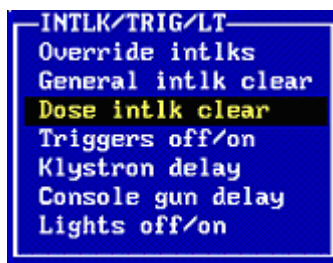


Note: If you turn the lights on or off using the light controls on the hand pendant, the system does not update the status of the Field light setting on the console screen.

To turn the lights and lasers on or off:

1. Select the **INTLK/TRIG/LT** menu heading and press **Enter**.

The **INTLK/TRIG/LT** menu appears.



2. Select **Lights off/on** and press **Enter**.

The **LIGHTS** menu appears.



3. To turn off or on:
 - Select the light or laser and press **Enter**.
 - Type the first letter of the item name (for example, type **F** for field light), and press **Enter**.

The **INTLK/TRIG/LT** menu reappears.

Chapter 7 SERVICE Mode: METER VALUES Menus

Meter Values Menus

Service mode is used by authorized personnel to maintain the accelerator, calibrate its readouts, and troubleshoot it. This chapter is intended for service and other authorized personnel only. It describes the remaining Service mode functions:

- [DISPLAY Menu: View Process Data](#) on page 115
- [CALIB Menu: Calibrate Process Values](#) on page 143
- [UTILS Menu: Service Mode Utilities](#) on page 173



WARNING: Service mode procedures should be used only by qualified maintenance personnel. Unauthorized maintenance or use of the accelerator can result in radiation exposure, machine malfunctions, and other hazards that may result in damage to equipment, injury, or death. When using Service mode, take the following precautions:

- Only authorized Varian personnel should know the Service mode password. Do not divulge the Service mode password to unauthorized individuals.
- When the system is in Service mode and the beam is on, prevent personnel from entering the treatment room.
- Never use Service mode for clinical operation, or to treat patients.



Note: Service mode functions are controlled primarily from the alphanumeric (service) keyboard. However, you must use the dedicated keyboard to enter Service mode. Once in Service mode, you must use the dedicated keyboard to operate the beam and to direct device motion.

DISPLAY Menu: View Process Data

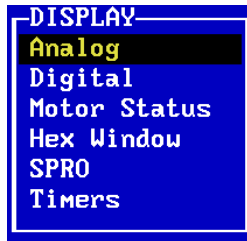
You can change the settings for most Service mode menu options while a data display appears on the screen. This allows you to observe real-time changes. Displays are controlled by toggle functions: A display stays on the screen until you remove it by selecting the same option again, which toggles the status between displayed and not displayed.

Selecting the **DISPLAY** menu heading in Service mode allows you to view various kinds of process data, including:

- Current values of analog meters
- Status of all digital I/O lines and control timers
- Current drive and control status of each positioning motor

- Contents of an area of the common RAM memory
- Secondary position readout readings (SPROs)
- Status of control system timers

Each function is described in the order in which it occurs in the **DISPLAY** main menu.



Analog Option: Analog Meters and Position Readouts (PROs)

The **Analog** option in the **DISPLAY** menu allows you to view the current values of the analog meters. Selecting the **Analog** option opens the **ANALOG DISPLAY** menu, which offers two options:

- **Machine**
- **PRO**



View Analog Meters

When you select the **Machine** option from the **ANALOG DISPLAY** menu, the system opens the **ANALOG METERS** data display, which reflects the current analog meter values.

DOSZ	DS1Z	XDP1	XDPZ	XDRS	XDR1	XDRZ	EXQ1	EXQZ	EXQT	ION1	IONZ	LUPS	VAC1	FLOW	PUMP
DOS1	TIME	CDOS	CMNR	UDRS	UDR1	UDRZ	DPSN	MOTN	COLL	IPSN	ACC	VACZ	HWFA	CTRL	KFIL
MOD	HVOC	HUCB	STPS	GAS	CMF	TDLY	STPR	PNDT	DOOR	KEY	MODE	CARR	TDRU	TARG	FOIL
USWR	BMAG	KSOL	ENSU	AIR	GFIL	MLC									

MU1	MU2	BTIME	ETIME	D/R	R SYM	T SYM	MODE	INVALID
0	0	0.00	0.00	0	+0.00	+0.00	ENERGY	NO MODE
1	2	0.01		300		ON	ACC IN	NO ACCY

BEAM CTRL	SET UP	INTLK/TRIG/LT	DISPLAY	MOTOR	CALIB	UTILS	QUIT
-----------	--------	---------------	---------	-------	-------	-------	------

ANALOG METERS							
P5UPS	0.00	BUNCH RD	-0.24	GUN I	-3.15	AFC DIFF	6.77
P12UPS	0.09	BUNCH TR	-0.04	SOL I	9.33	MNL FREQ	6.85
M12UPS	0.09	ANGLE RD	-0.47	TARGET I	-22.35	ACL VAC	-11.33
P15UPS	0.31	ANGLE TR	-0.80	GUN U	19.58	GUN VAC	39.35
M15UPS	0.40	POS RDL	-0.31	BMAG I	-10.25	KLY VAC	51.71
P24UPS	0.32	POS TRN	-0.56	RF DRIVE	-3.05	NPSA PS	0.68
SYM RDL	0.06	TRIM	0.00	HUPS I	13.56	NCOLL PS	1.01
SYM TRN	0.06	PFN VOLT	9.81	H2O TEMP	-17.94	NGAN PS	-0.89

COLL RTN	180.1 deg	COLL Y1	5.1 cm	COUCH URT	101.1 cm
FIELD Y	9.9 cm	COLL Y2	4.9 cm	COUCH LNG	50.1 cm
FIELD X	9.9 cm	COLL X1	4.9 cm	COUCH LAT	100.0 cm
GANTRY RTN	180.1 deg	COLL X2	5.0 cm	COUCH RTN	180.0 deg

The two columns on the left display scaled meter values for low-voltage power supply voltage, steering coil current, and beam symmetry. The two columns on the right report meter values for process parameters from a variety of sources in the accelerator.

The machine meter values you see depend on the gain and offset factors set at the factory for each meter. To view the gain and offset factors, use the **Alg Meter Calibration** option (see [PRO Meter Calibration Option: PRO Meter Gain and Offset](#) on page 156).

To display analog machine meter values:

1. Select the **DISPLAY** menu heading and press **Enter**.

The **DISPLAY** menu appears.



2. Select **Analog** and press **Enter**.

The ANALOG DISPLAY menu appears.



3. Select **Machine** and press **Enter**.

The ANALOG METERS data display appears.

ANALOG METERS							
P5UPS	0.00	BUNCH RD	-0.24	GUN I	-3.15	AFC DIFF	6.77
P12UPS	0.09	BUNCH TR	-0.04	SOL I	9.33	MNL FREQ	6.85
M12UPS	0.09	ANGLE RD	-0.47	TARGET I	-22.35	ACL VAC	-11.33
P15UPS	0.31	ANGLE TR	-0.80	GUN U	19.58	GUN VAC	39.35
M15UPS	0.40	POS RDL	-0.31	BMAG I	-10.25	KLY VAC	51.71
P24UPS	0.32	POS TRN	-0.56	RF DRIVE	-3.05	NPSA PS	0.68
SYM RDL	0.06	TRIM	0.00	HUPS I	13.56	NCOLL PS	1.01
SYM TRN	0.06	PFN VOLT	9.81	H2O TEMP	-17.94	NGAN PS	-0.89

To close the display, repeat these steps.

View PRO Meters

When you select the **PRO** option from the **ANALOG DISPLAY** menu, the system opens the **PRO METERS** data display, which reflects the current analog position readouts.

DOSZ	DS1Z	XDP1	XDPZ	XDRS	XDR1	XDRZ	EXQ1	EXQZ	EXQT	ION1	IONZ	LUPS	VAC1	FLOW	PUMP
DOS1	TIME	CDOS	CMNR	UDRS	UDR1	UDRZ	DPSN	MOTN	COLL	IPSN	ACC	VACZ	HWFA	CTRL	KFIL
MOD	HVOC	HUCB	STPS	GAS	CMF	TDLY	STPR	PNDT	DOOR	KEY	MODE	CARR	TDRU	TARG	FOIL
USWR	BMAG	KSOL	ENSU	AIR	GFIL	MLC									

MU1	MU2	BTIME	ETIME	D/R	R SYM	T SYM	MODE	INVALID
0	0	0.00	0.00	0	+0.00	+0.00	ENERGY	NO MODE
1	2	0.01		300		ON	ACC IN	NO ACCY

BEAM CTRL	SET UP	INTLK/TRIG/LT	DISPLAY	MOTOR	CALIB	UTILS	QUIT
-----------	--------	---------------	---------	-------	-------	-------	------

PRO METERS							
COLL RTN	0.18	COLL X1	-3.11	CCH URT	-3.24	CCH LNG	3.15
SPR COLL	1.12	SPRO X1	4.87	SPR CURT	0.62	SPR CLNG	0.31
COLL Y1	-4.51	COLL X2	-7.02	CCH RTN	1.90	CCH LAT	3.93
SPRO Y1	-1.22	SPRO X2	0.96	SPR CRTN	1.10	SPR CLAT	-3.11
COLL Y2	-1.69	SPARE 1	0.00	GANTRY R	3.47	ANA TEST	-1.38
SPRO Y2	3.78	COL MOTI	0.81	SPRO GAN	1.27	GAN MOTI	-5.51
UJ1 MOTI	4.58	LJ1 MOTI	-4.04	LAT MOTI	0.34	URT MOTI	-7.90
UJ2 MOTI	2.99	LJ2 MOTI	-2.78	LNG MOTI	1.77	RTN MOTI	1.61

COLL RTN	180.1 deg	COLL Y1	5.1 cm	COUCH URT	101.0 cm
FIELD Y	9.9 cm	COLL Y2	5.0 cm	COUCH LNG	50.1 cm
FIELD X	10.0 cm	COLL X1	5.0 cm	COUCH LAT	99.9 cm
GANTRY RTN	179.9 deg	COLL X2	5.1 cm	COUCH RTN	179.9 deg

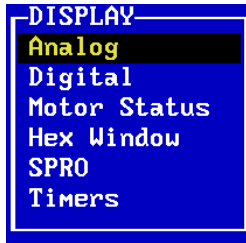
The data display shows PRO potentiometer wiper voltage and motor current readings from all ten motor axes, and secondary PRO values for the gantry and the upper collimator jaws.

The PRO meter values you see depend on the gain and offset factors set at the factory for each meter. For instructions on how to view the gain and offset factors, see [PRO Meter Calibration Option: PRO Meter Gain and Offset](#) on page 156.

To display analog position readouts:

1. Select the **DISPLAY** menu heading and press **Enter**.

The **DISPLAY** menu appears.



2. Select **Analog** and press **Enter**.

The ANALOG DISPLAY menu appears.



3. Select **PRO** and press **Enter**.

The PRO METERS data display appears.

PRO METERS							
COLL RTN	0.18	COLL X1	-3.11	CCH URT	-3.24	CCH LNG	3.15
SPR COLL	1.12	SPRO X1	4.87	SPR CURT	0.62	SPR CLNG	0.31
COLL V1	-4.51	COLL X2	-7.02	CCH RTN	1.90	CCH LAT	3.93
SPRO V1	-1.22	SPRO X2	0.96	SPR CRTN	1.10	SPR CLAT	-3.11
COLL V2	-1.69	SPARE 1	0.00	GANTRY R	3.47	ANA TEST	-1.38
SPRO V2	3.78	COL MOTI	0.81	SPRO GAN	1.27	GAN MOTI	-5.51
UJ1 MOTI	4.58	LJ1 MOTI	-4.04	LAT MOTI	0.34	URT MOTI	-7.90
UJ2 MOTI	2.99	LJ2 MOTI	-2.78	LNG MOTI	1.77	RTN MOTI	1.61

To close the display, repeat these steps.

Digital Option: Digital I/O Lines

The **Digital** option in the **DISPLAY** menu allows you to display the current status of all digital input/output (I/O) lines. Selecting the **Digital** option opens the **DIGITAL DISPLAY** menu.



The **DIGITAL DISPLAY** menu offers these options:

- **Inputs** displays the current state of digital input signals to the controller (see [Digital Inputs](#)).
- **Outputs** displays the current status of all digital output signal lines (see [Digital Outputs](#) on page 121).
- **Accessory** displays the status of the type III serial accessory slots (see [Accessory Slots](#) on page 124).

Digital Inputs

Selecting the **Inputs** option opens the **DIGITAL INPUTS** data display, which displays the current state of digital input signals to the controller. The bit identified lights up if the digital state is on (1). Each column in the data display presents a port address containing 8 bits of digital I/O data. The 8 bits appear in separate rows labeled b0 through b7. Bit 0 appears in the lower right corner of the display.

Port Addresses:

		10	11	12	13	14	15	16	17	
DIGITAL INPUTS										
10	7	KEY	USWR	DOOR	TGT	CarrChk8	ObiRdyCt	Acc80	b7	
	6	CHNR	HUOC	PNDT	EXQT	CarrChk4	OB_RET	Acc40	b6	
	5	KSOL	MOD	MLC	EXQ2	CarrChk2	ETR	Acc20	b5	
	BO_Enble	HUCB	GFIL	ENSW	EXQ1	CarrChk1	ExHoldL	Acc10	b4	
	SetEnble	STPS	KFIL	FOIL	CDOS	CR_Chk8	StprRet	Acc08	b3	
	AuxEnble	AIR	PUMP	PU_EXT	LUPS	CR_Chk4	StprExt	Acc04	b2	
	MjrEnble	GAS	FLOW	TARG	ION2	CR_Chk2	Appl_In	Acc02	b1	
	DosEnble	VACZ	VAC1	BMAG	ION1	CR_Chk1	WedgeIn	Acc01	b0	

When a bit is not used, only the bit number is displayed. For example, bits 7, 6, and 5 in rows b7, b6, and b5 at port address 10 signify that those bits are not used.

Digital Outputs

Selecting the **Outputs** option opens the **DIGITAL OUTPUTS** data display, showing the state of all digital output signals from the controller. Each column in the data display represents a port of digital I/O data (one byte). The port on the far left of the inputs data display shows output bits, and the port on the far right of the outputs data display shows input bits.

Port Addresses:

18	19	1A	1B	1C	1D	1E	1F
DIGITAL OUTPUTS							
18BP_Laser	CR_Reset	WDEnable	DBDSrvEnc	?	RtnBrakeB	?	FLG_Stat2 b7
RangeLt	Cal_Cmd	HW_Ovrd	GunDlyEn	RoomLt	LatClitch	6	LG_Stat1 b6
FieldLt	ModeCmd	BmOffRst	ConTrgEn	Lasers	LatBrakL	StprIn	DosBmOff b5
GPotPur	HDTSeEn	DynMotEn	DR_SrvEn	P24uOn	LngClitch	StprOut	PurKey b4
GunHUON	Mode8	WDStrobe	LouPrfEn	M12uOn	LngBrakL	StyRotCt	MotionEn b3
GunMode	Mode4	BC_Reset	Rf_TrgEn	Ready	Couch_En	CT_State	DkbBmOff b2
ModMode8	Mode2	BeamOn2	GunPrfEn	Fill_Pur	Collm_En	LG_Ctr12	WDStatus b1
ModModeA	Mode1	BeamOn1	KlyTrgEn	WaterVac	GantryEn	LG_Ctr11	HUON b0

Each code or number in a column represents a bit of the port data byte. For example, BO_Enable on the input board is bit 4 of I/O port 10 and BP_Laser is bit 7 of I/O port 18 on the output board. The system displays the bit number itself in the column if the bit has not been assigned to a particular function. For example, unused bit 7 of I/O port C (actually port 1C) on the output board is labeled 7. The bit (input or output) and its real-time status (active or inactive) are indicated by the color coding listed in Table 20.

Table 20 Input/Output Display Color Coding

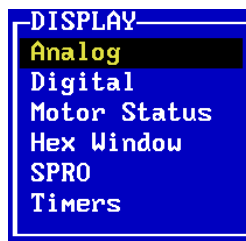
Type of Bit	Color Coding
Active output	Light blue characters on a black background
Active input	Yellow characters on a black background
All inactive	Black characters on a light grey background

Display Digital Inputs or Outputs Lines

To display the status of the digital I/O lines:

1. Select the **DISPLAY** menu heading and press **Enter**.

The DISPLAY menu appears.



2. Select **Digital** and press **Enter**.

The DIGITAL DISPLAY menu appears.



3. Select a DIGITAL DISPLAY:

- Select **Inputs** and press **Enter**.
- Select **Outputs** and press **Enter**.

The DIGITAL INPUTS or the DIGITAL OUTPUTS data display appears.

DIGITAL INPUTS with Port Addresses shown:

10	11	12	13	14	15	16	17
DIGITAL INPUTS							
10	7	1	KEY	2	USWR	3	DOOR
	6		CMNR		HUOC		PNDT
	5		KSOL		MOD		MLC
	BO_Enble		HUCB		GFIL		ENSW
	SetEnble		STPS		KFIL		FOIL
	AuxEnble		AIR		PUMP		PU_EXT
	MjrEnble		GAS		FLOW		TARG
	DosEnble		VAC2		VAC1		BMAG
					ION1		CR_Chk1
					CR_Chk2		CR_Chk4
					CR_Chk8		StprRet
					CR_Chk1		ExHoldL
					CR_Chk2		OB_RET
					CR_Chk4		Acc80
					CR_Chk8		Acc40
					CR_Chk1		Acc20
					CR_Chk2		Acc10
					CR_Chk4		Acc08
					CR_Chk8		Acc04
					CR_Chk1		Acc02
					CR_Chk2		Acc01
					CR_Chk4		b7
					CR_Chk8		b6
					CR_Chk1		b5
					CR_Chk2		b4
					CR_Chk4		b3
					CR_Chk8		b2
					CR_Chk1		b1
					CR_Chk2		b0

DIGITAL OUTPUTS with Port Addresses shown:

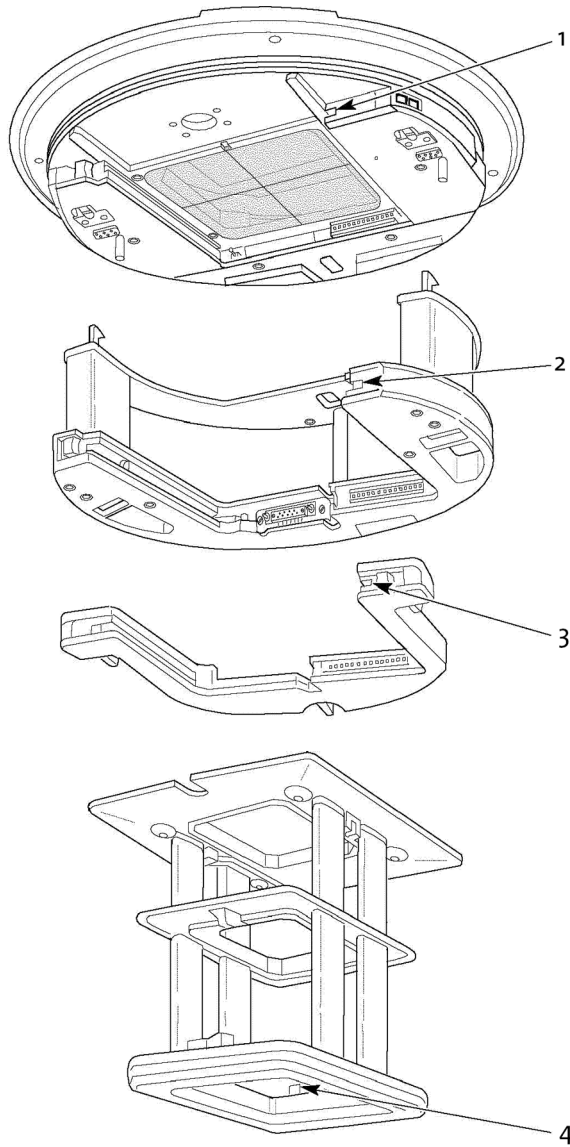
18	19	1A	1B	1C	1D	1E	1F
DIGITAL OUTPUTS							
18	BP_Laser	CR_Reset	UDEnable	DBDSrvEnc	7	DRtnBrake	7
	RangeLt	Cal_Cmd	H4_Ovrd	GunDlyEn	RoomLt	LatClitch	5
	FieldLt	ModeCmd	BmOffRst	ConTrgEn	Lasers	LatBrakL	StprIn
	GPotPur	HDTSeEn	DynMotEn	DR_SrvEn	P24vOn	LngClitch	StprOut
	GunHUON	Mode8	WDStrobe	LouPrfEn	MIZvOn	LngBrakL	GtyRotCt
	GunMode	Mode4	BC_Reset	RF_TrgEn	Ready	Couch_En	CT_State
	ModModeB	Mode2	BeamOnZ	GunPrfEn	Fill_Pur	Collm_En	LG_Ctr12
	ModModeA	Mode1	BeamOn1	KlyTrgEn	WaterVac	GantryEn	LG_Ctr11
							HUON
							b7
							b6
							b5
							b4
							b3
							b2
							b1
							b0

To close the display, press **Enter** four times, or repeat these steps.

Accessory Slots

The Type III Serial Accessories system and the custom coding option allow you to have beam accessories mounted in up to four slots on the accessory mount. The slots are shown in [Figure 5](#). Each slot has two switches:

- Accessory-installed switch
- Accessory-latched switch



- | | |
|-----------------------------|---------------------------------|
| 1. Interface Mount — Slot 1 | 3. Compensator Mount — Slot 4 |
| 2. Accessory Mount — Slot 2 | 4. Electron Applicator — Slot 3 |

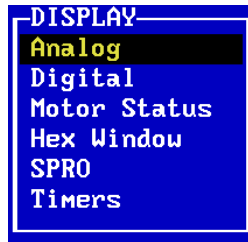
Figure 5 Accessory Mounts for the Type III Accessory System

The **Digital** option in the **DISPLAY** menu allows you to view the code of each installed accessory (accessory data) and the current status of the slot switches (error data).

To view accessory codes and status:

1. Select the **DISPLAY** menu heading and press **Enter**.

The DISPLAY menu appears.



2. Select **Digital** and press **Enter**.

The DIGITAL DISPLAY menu appears.



3. Select **Accessory**.

The ACCESSORY menu appears.



4. Make a selection:

- Select **Accessory Data** and press **Enter**.

If you select Accessory Data, the ACCESSORY DATA display appears. For details about the ACCESSORY DATA display, see [Interpret the ACCESSORY DATA Display](#) on page 127.

ACCESSORY DATA															
b7	I	7	Code	AEA	Av	L	Code	E	7	Code	C	7	Code	CEA	Av
b6	M	6	OH	MCM	Av	L	OH	AS3	Ins	OH	M	6	OH	TCM	Av
b5		5	0	Tch	Grd		0	Tch	Grd	0		5	0	LAM	Av
b4	s	Mot Rst		s	4			s	4		s	4		IM	Av
b3	w	S1 LatL	Config	w	S2 LatL	Config	w	3		Config	w	S4 LatL	Config	s	ERN Lit
b2	t	S1 InsL	1H	t	S2 InsL	1H	t	FSRfail		1H	t	S4 InsL	1H	t	RED Lit
b1	c	AM LatB		c	S3 Ins		c	S3 LatL			c	1		aEA	Inst
b0	h	AM LatA		h	S3 LatL		h	0			h	0		tacc	RED
															80H

- Select **Error Data** and press **Enter**.

If you select Error Data, the ACCESSORY ERROR DATA display appears. For details about the ACCESSORY DATA display, see [Interpret the ACCESSORY ERROR DATA Display](#) on page 130.

ACCESSORY ERROR DATA													
b7	I	?	A	?	E	?	C	?	C	?	A	Lost Lnk	
b6	MAM	LatchB	M	6	A	6	M	6	T	6	C	6	
b5	AM	LatchA		5	Tch	Guard		5	R	5	C	5	
b4	Inv	Conf	Inv	Conf	Inv	Conf	Inv	Conf	L	4	EA	Inst	
b3	Inv	Code	Inv	Code	Inv	Code	Inv	Code		Lost Link	LHost	RED	
b2	ES1	Inst	ES2	Inst	ES3	Inst	ES4	Inst	E	Inv	Conf	I	Mot Dsbl
b1	RS1	Latch	RS2	Latch	RS3	Latch	RS4	Latch	R	CRC	Fault	N	Beam St
b0	R	Not Activ	R	Not Activ	R	Not Activ	R	Not Activ	R	Acc	Fault	K	Mrgn Mod



Note: To close either display, press **Enter** four times.

Interpret the ACCESSORY DATA Display

Selecting the **Accessory Data** option opens the **ACCESSORY DATA** display. The **ACCESSORY DATA** display shows the bit status associated with the four accessory slots and the accessory controller status.

ACCESSORY DATA									
b7	I	7	Code	AE	Av	L	Code	E	7
b6	M	6	OH	MCM	Av	L	OH	AS	3
b5		5	0	Tch	Grd		0	Tch	Grd
b4	s	Mot	Rst	s	4		s	4	
b3	w	S1	LatL	Conf	w	S2	LatL	Conf	w
b2	t	S1	InsL	1H	t	S2	InsL	1H	t
b1	c	AM	LatB	c	S3	Ins		c	S3
b0	h	AM	LatA	h	S3	LatL		h	0

Figure 6 ACCESSORY DATA Display

The **ACCESSORY DATA** display shows information grouped into five columns, shaded in grey. The label of each column appears vertically in white font at the left of each grey box. The five areas are: the interface mount signals (IM switch), the accessory mount signals (AM switch), the electron aperture signals (EA switch), the compensator mount signals (CM switch), and the controller status (CTL stat). The leftmost column of the display identifies each row with a bit number (b0 through b7). Where a bit number is unused, the system displays the bit number itself as an integer value. Four areas labeled Code display the coded number of each installed accessory using a hexadecimal number in row b6 and a decimal number in row b5. The five configuration areas (Conf) display the information in hexadecimal number format in row b2.

Each column contains a colored number or code sent from the controller.

- The system displays a bit number when the bit is unused. For example, in [Figure 6](#), the unused bit 7 (b7) of the first column, the interface mount (code IM switch) column, is labeled 7.
- The meaning of the colors in the display is described in [Table 21](#).
- The system displays a code that shows specific information about the status of the accessory switches. These codes are listed in [Table 22](#) on page 128.

Table 21 Accessory Slot Switch Color Coding

Meaning	Color Coding
Signal is TRUE	Yellow characters on a black background
Signal is FALSE	Black characters on a light grey background

Table 22 Slot Switch Codes and Abbreviations

Code	Meaning
Acc Fault	Accessory controller fault
ACC LINK	Status of the link with the accessory controller
Acc RED	Notification that there is an accessory fault.

Code	Meaning
AM Av	Accessory mount available
AM LatA	Switch A on accessory mount is not completely closed.
AM LatB	Switch B on accessory mount is not completely closed.
Beam St	Unused
CM Av	Compensator mount available
CM Av L	Compensator mount available; signal low
CRC Fault	Cyclic redundancy check fault
CTL stat	Accessory controller status
CTRL ERR	Accessory controller error
EA AV L	Electron aperture available; signal low
EA Inst	Electron aperture installed
FSRfail	One of five touch force sensitive sensors on touch guard failed.
GRN Lit	Reports the status of the controller light.
Host Red	Host computer is setting the light to red.
IM Av	Interface mount available
Inv Code	Invalid code (for example, a parity error)
Inv Confg	Invalid configuration, controller fault
Lost Link	Lost data link. The communication with the accessory controller was lost.
Mot Dsbl	Motor disabled
Mot Rst	Motor reset (active when the reset button on collimator is red)
Mrgn Mod	Working mode is set to marginal conditions (for test purposes).
Not Activ	Not active
RED Lit	Reports the status of the controller light.
S1 InsL	Slot 1 accessory installed; signal low
S1 LatL	Slot 1 accessory latched; signal low
S2 InsL	Slot 2 accessory installed; signal low
S2 LatL	Slot 2 accessory latched; signal low

Code	Meaning
S3 Ins	Slot 3 accessory installed
S3 LatL	Slot 3 accessory latched; signal low
S4 InsL	Slot 4 accessory installed; signal low
S4 LatL	Slot 4 accessory latched; signal low
Tch Grd	Touch guard is being pressed.

Interpret the ACCESSORY ERROR DATA Display

Selecting the **Error Data** option opens the **ACCESSORY ERROR DATA** display, which shows certain errors associated with the slots and the accessory controller.

ACCESSORY ERROR DATA												
b7	I	?	A	?	E	?	C	?	C	?	A	Lost Lnk
b6	MAM	LatchB	M	6	A	6	M	6	I	6	C	6
b5	AM	LatchA		5				5	R	5	C	5
b4	Inv	Confg	Inv	Confg	Inv	Confg	Inv	Confg	L	4	EA	Inst
b3	Inv	Code	Inv	Code	Inv	Code	Inv	Code		Lost Link	LHost	RED
b2	ES1	Inst	ES2	Inst	ES3	Inst	ES4	Inst	E	Inv Confg	I	Mot Dsbl
b1	RS1	Latch	RS2	Latch	RS3	Latch	RS4	Latch	RCRC	Fault	N	Beam St
b0	R	Not Activ	R	Not Activ	R	Not Activ	R	Not Activ	R	Acc Fault	R	Mrgn Mod

Figure 7 ACCESSORY ERROR DATA Display

The **ACCESSORY ERROR DATA** display uses the same colors and codes as the **ACCESSORY DATA** display (Table 21 and Table 22 on page 128).

To display the status of type III accessory slots, see [Accessory Slots](#) on page 124.

Motor Option: Motor Status

The **Motor Status** option in the **DISPLAY** menu allows you to view the current drive and control parameters for each positioning motor. This display is useful for troubleshooting motion problems. For example, if the controller is not putting out a **Drv** (DAC) signal, the controller is preventing motion through software. If a **Drv** signal is present yet motion does not occur, the problem is beyond the controller.

Selecting the **Motor Status** option opens the **MOTOR STATUS** display.

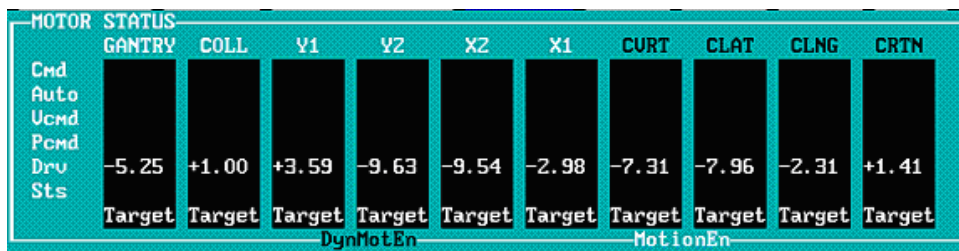


Figure 8 MOTOR STATUS Display

At the bottom of the display, above the label DynMotEn, the system displays the status of *controller-initiated* dynamic motion enable *output*. Above the label MotionEn, the system displays the status of *operator-initiated* motion enable *input*. These labels are black when motion enable is active and white when motion enable is inactive. When there is no motion, the system displays only the Drv status indicators.

When MotionEn is activated, one of three motion enable outputs is asserted. These are GantryEn for gantry rotation, Collm_En for collimator rotation and jaw movement, or Couch_En for any of the couch motions. See the [Digital Outputs](#) on page 121 data display for the three digital outputs for Couch_En, Collm_En, and GantryEn at port 1D, as bits b0 through b2. As each of these group-enable outputs is asserted, the column headings in [Figure 8](#) for the affected motion axes change from white characters to black.

Interpret the Control and Drive Status

[Table 23](#) lists the control and drive status elements displayed in the columns of [Figure 8](#) on page 131 for each motor axis.

Table 23 Motor Axis Control and Drive Status

Row Title	Description
Cmd (command)	Reports the type of motion command the operator initiated. Indicated as Vel (velocity), Pos (position), or P/V (position with velocity).
Auto	Displays Yes when an automatic motion is being executed.
Vcmd (velocity command)	Displays velocity data received by the communication microprocessor from the pendant, side panel controls, or console. The display format is $\pm\#$. The value ranges from 0 to 7.
Pcmd (position command)	Displays target position when a position or position with velocity command is executing. The display format is $\pm\#\#\#$.
Drv (Drive)	Displays the motor velocity DAC output from the controller. The display format is $\pm\#\#\#$. The DAC output range is $\pm 10V$.

Row Title	Description
Sts (Status)	Reports the type of motion commanded by the operator. Indicated as: ■ Ovel (open-loop velocity command) ■ Cvel (closed-loop velocity command) ■ Pos (external position command) ■ P/V (position with velocity command) ■ Int (internal command)
(No label)	Displays limit if the axis velocity or position limit has been reached, or the target if the target velocity or position has been achieved.

Hex Window Option: Common RAM

The **Hex Window** option in the **DISPLAY** menu allows you to display the real-time hex values of 80 bytes in common RAM starting at a location that you specify. This option is usually used to display CTRL and HWFA interlock error codes.

Selecting the **Hex Window** option opens the **MEMORY DISPLAY OFFSET** data box, where you specify a RAM *starting* location.

MEMORY DISPLAY OFFSET

Enter Offset as Hex (0-7FFF): 0

After you specify a RAM starting location, the system opens the **COMMON RAM MEMORY** data display, which shows the 2-digit hex value of each of the subsequent 80 bytes in common RAM, starting at the specified location.

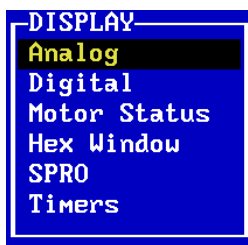
COMMON RAM MEMORY																
0	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
00	00	00	00	56	37	2E	30	2E	30	33	2D	30	30	00	00	01
10	36	30	30	43	00	00	00	00	69	00	02	36	30	30	43	2F
20	44	00	00	56	37	2E	30	2E	30	33	2D	30	30	00	00	56
30	37	2E	30	2E	30	33	2D	30	30	00	00	01	36	30	30	43
40	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00

View Common RAM

To display the contents of common RAM:

1. Select the **DISPLAY** menu heading and press **Enter**.

The DISPLAY menu appears.



2. Select **Hex Window** and press **Enter**.

The MEMORY DISPLAY OFFSET menu appears.



3. Type the offset as a four-digit hex number and press **Enter**.

The COMMON RAM MEMORY display appears.

COMMON RAM MEMORY																
0	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
00	00	00	00	56	37	2E	30	2E	30	33	2D	30	30	00	00	01
10	36	30	30	43	00	00	00	00	69	00	02	36	30	30	43	2F
20	44	00	00	56	37	2E	30	2E	30	33	2D	30	30	00	00	56
30	37	2E	30	2E	30	33	2D	30	30	00	00	01	36	30	30	43
40	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00

To close the display, press **Enter** twice, or repeat these steps.

SPRO Option: Secondary Position Readouts

Secondary Position Readout (SPRO) values use the same scale as the primary PRO values displayed at the bottom of the Service mode screen. The HWFA interlock is asserted if the PRO and SPRO values for the same motor axis differ.

The **SPRO** option in the **DISPLAY** menu allows you to display the real-time values of all enabled secondary PRO positions. Selecting the **SPRO** option opens the **SPRO DISPLAY**.

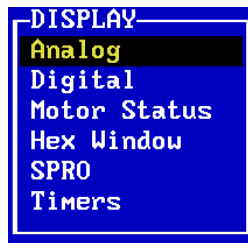
SPRO DISPLAY					
COLL RTN	180.0	deg	COLL Y1	5.0	cm
FIELD Y	9.9	cm	COLL Y2	4.9	cm
FIELD X	9.9	cm	COLL X1	5.0	cm
GANTRY RTN	180.1	deg	COLL X2	4.9	cm
			COUCH VRT	101.1	cm
			COUCH LNG	49.9	cm
			COUCH LAT	100.0	cm
			COUCH RTN	180.1	deg

View Secondary Position Readouts (SPROs)

To display SPRO values:

1. Select the **DISPLAY** menu heading and press **Enter**.

The DISPLAY menu appears.



2. Select **SPRO** and press **Enter**.

The SPRO DISPLAY appears.

SPRO DISPLAY					
COLL RTN	180.0	deg	COLL Y1	5.0	cm
FIELD Y	9.9	cm	COLL Y2	4.9	cm
FIELD X	9.9	cm	COLL X1	5.0	cm
GANTRY RTN	180.1	deg	COLL X2	4.9	cm
			COUCH VRT	101.1	cm
			COUCH LNG	49.9	cm
			COUCH LAT	100.0	cm
			COUCH RTN	180.1	deg

To close the display, press **Enter** twice, or repeat these steps.

Timers Option: Hardware Timers Output Status

The **Timers** option in the **DISPLAY** menu displays the output status of each control system timer. Selecting the **Timers** option opens the **H/W TIMER OUTPUTS** display.

To display the status of the hardware timers:

1. Select the **DISPLAY** menu heading and press **Enter**.

The DISPLAY menu appears.



Note: You must hold down a motion enable control to move any axis (except collimator jaw axes in automatic mode) in both manual and automatic modes.



Selecting the **MOTOR** menu heading on the Service mode screen opens the **MOTORS** menu, where you select what you want to move and the type of motion.

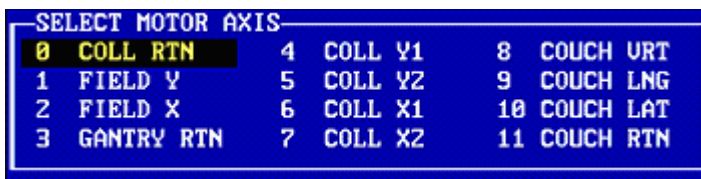


Manual Motion Option: Gantry, Collimator, or Couch



CAUTION: Controlling machine axis motions from the dedicated keyboard can be hazardous to the equipment because of the possibility of collision. Always monitor motions on a closed-circuit TV monitor to avoid collisions between the couch, gantry, and other equipment.

The **Manual motion** option in the **MOTORS** menu allows you to select an axis for manual motion in Service mode. Selecting the **Manual motion** option opens the **SELECT MOTOR AXIS** menu, allowing you to select the motor axis that you want to move manually.



Select a Motor Axis for Manual Control

You can select only one axis at a time for manual motion. When you select a different axis, your selection replaces the prior selection.

When you select an asymmetric motion (COLL Y1, Y2, X1, or X2) and press the **MOTION ENABLE** bar on the dedicated keyboard, the system is placed in asymmetric mode. To return to symmetric mode, select **FIELD Y** or **FIELD X** and press the **MOTION ENABLE** bar.

When a motor axis has been selected for manual motion, its position readout at the bottom of the Service mode screen is selected on a light blue background. As motion occurs, the readout changes in real-time.



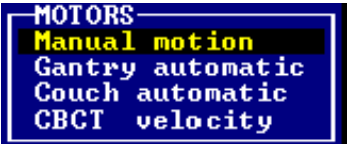
Note: Neither manual nor automatic motion is allowed when the HWFA, STPS, or CTRL interlock is active.

Move a Motor Axis Manually

To move a motor axis manually:

1. Select the **MOTOR** menu heading and press **Enter**.

The MOTORS menu appears.



2. Select **Manual motion** and press **Enter**.

The SELECT MOTOR AXIS menu appears.



3. Select the motor axis to move and press **Enter**.



Note: This action corresponds to pressing a motion selection key on the dedicated keyboard as described in appropriate accelerator Instructions for Use.

4. To start motion, hold down a directional motion control key and the **MOTION ENABLE** bar on the dedicated keyboard.

The motion begins.

5. To stop motion, release the motion control key, **MOTION ENABLE** bar, or both.

The motion ceases.

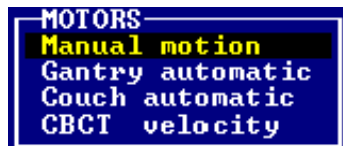
To close the menu, press **Enter** or **Esc**.

Cancel Motion Selections

To cancel all motion selections made using the MOTORS menu:

1. Select the **MOTOR** menu heading and press **Enter**.

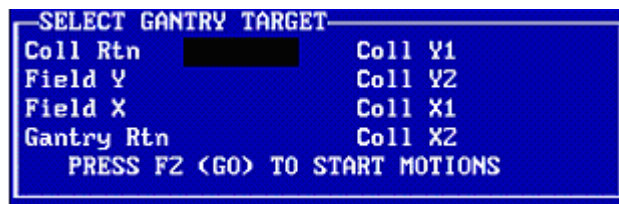
The MOTORS menu appears.



2. Press **Esc**.

Gantry Automatic Option

The **Gantry automatic** option in the **MOTORS** menu allows you to select a motor axis and move it automatically. Selecting the **Gantry automatic** option opens the **SELECT GANTRY TARGET** data box, which allows you to specify the gantry or collimator axis for automatic motion.



Each axis remains selected for automatic motion until you return to the **SELECT GANTRY TARGET** data box or you select manual motion for the same axis. You can select multiple axes for simultaneous motion.



Note: Automatic motion is not allowed when the PRO is set for raw scale or if a CTRL, HWFA, or STPS interlock is active.

When a motor axis is selected for automatic motion, the system displays its position readout on a light grey background at the bottom of the Service mode screen. As motion occurs, the readout changes in real-time.

Control Automatic Motion with the Hand Pendant

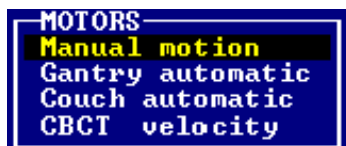
Once you specify target positions and press **F2**, you can also drive the selected motors from a hand pendant in the treatment room. To command motion, squeeze and hold both motion enable bars on the sides of the pendant, and momentarily press the **AUTO** key on the pendant paddle. To stop motion, release either or both motion enable bars.

Move the Gantry or Collimator Automatically

To move the gantry automatically:

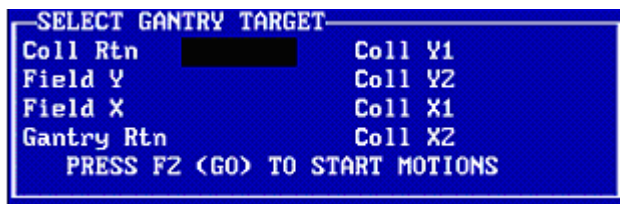
1. Select the **MOTOR** menu heading and press **Enter**.

The MOTORS menu appears.



2. Select **Gantry automatic** and press **Enter**.

The SELECT GANTRY TARGET data box appears.



3. Use the **Enter** or arrow keys to select the axis you want.
4. Type the target position in the black text box beside the axis you have selected. You may enter the position using either an integer or a decimal expression.
5. After you have selected each desired axis and entered its corresponding target position, press **F2**.

Collimator jaw motion starts immediately, and the menu closes. To continue with manual operation of rotational axes:

- To rotate the gantry or collimator, hold down the **MOTION ENABLE** bar on the dedicated keyboard to start and maintain motion until the target position is reached.
- To stop rotation of the gantry or collimator, release the **MOTION ENABLE** bar.

To stop automatic motion of a collimator jaw before it reaches its target position:

- Display the **SELECT GANTRY TARGET** data box again.
- Select the same jaw axis in manual motion mode (see [Manual Motion Option: Gantry, Collimator, or Couch](#) on page 136).

Couch Automatic Option

The **Couch automatic** option in the **MOTORS** menu allows you to move the couch automatically. Selecting the **Couch automatic** option opens the **SELECT COUCH TARGET** data box, which allows you to specify the type of motion and target position.

While a couch motion is selected for automatic motion, its position readout is displayed on a light grey background at the bottom of the Service mode screen. As motion occurs, the selected readout changes in real-time. In addition, each couch position remains selected for automatic motion until you again display the **SELECT COUCH TARGET** data box or select the same couch motion for manual motion. You can select multiple couch axes for simultaneous automatic motion.



Note: Automatic motion is not allowed when the PRO is set for raw scale or if a CTRL, HWFA, or STPS interlock is active.



CAUTION: Controlling machine axis motions from the dedicated keyboard can be hazardous to the equipment because of the possibility of collision. Always monitor motions on a closed-circuit TV monitor to avoid collisions between the couch, gantry, and other equipment.

To move the couch automatically:

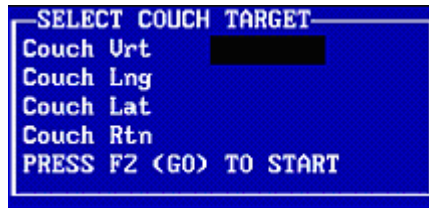
1. Select the **MOTOR** menu heading and press **Enter**.

The **MOTORS** menu appears.



2. Select **Couch automatic** and press **Enter**.

The SELECT COUCH TARGET data box appears.



3. Use the **Enter** or arrow keys to select the couch motion you want.
4. Type the target position in the black text box beside the axis you have selected. You may enter the position using either an integer or a decimal expression.
5. After you have selected each desired axis and entered its corresponding target position, press **F2**.

The menu closes.

6. To start motion, press and hold a directional motion control key and the **MOTION ENABLE** bar on the dedicated keyboard.

The motion begins.

7. To stop motion, release the motion control key, **MOTION ENABLE** bar, or both.

The motion ceases.

To stop motion of the couch before it reaches its target position:

- Release the **MOTION ENABLE** bar.
- Display the **SELECT COUCH TARGET** data box again.
- Select the same couch motion in manual motion mode (see [Manual Motion Option: Gantry, Collimator, or Couch](#) on page 136).

Cone Beam CT (CBCT) Velocity Option

The Cone Beam CT (CBCT **velocity**) option in the **MOTORS** menu allows you to set the gantry rotation speed for cone beam CT acquisition. This option must be purchased from Varian.



WARNING: Different gantry velocity settings for CBCT acquisition result in varying degrees of image quality. Be sure that the correct velocity option is selected to support the intended imaging.



Note: This menu item works only in Clinical mode when moving to CT setup field #2. It does not set speed or velocity for any other gantry motions.

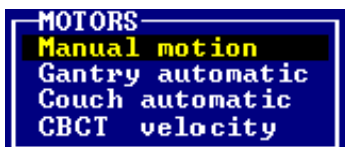
You can choose one of three velocity settings:

- High
- Medium
- Low

To set the Cone Beam CT velocity:

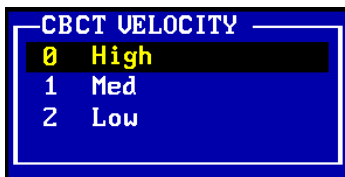
1. Select the **MOTOR** menu heading and press **Enter**.

The **MOTORS** menu appears.



2. Select **CBCT velocity** and press **Enter**.

The **CBCT VELOCITY** menu appears.



3. Select one of the three gantry **CBCT VELOCITY** settings and press **Enter**.

Selection	Setting	Velocity of Gantry Rotation
0	High	6 degrees per second
1	Medium	3 degrees per second
2	Low	1 degree per second

The gantry velocity for CBCT acquisition is now set.

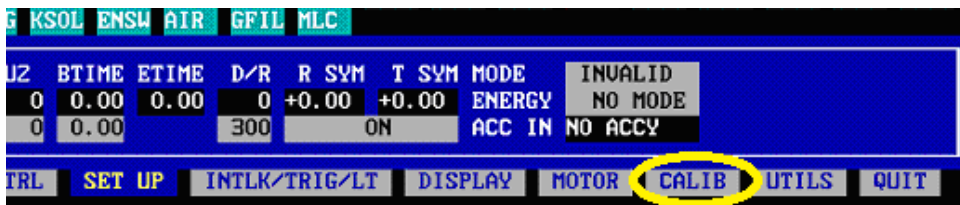
Initiate Motion While the Beam is On

To initiate motion while the beam is on:

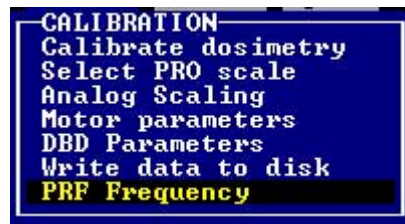
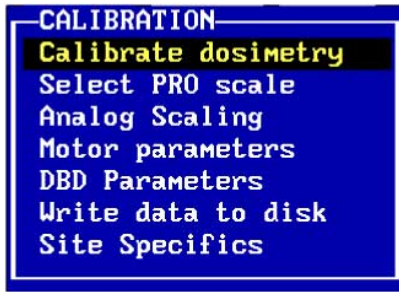
1. In the **SET UP** menu, use the **Set all** option to specify the beam-on parameters.
See [Set All](#) on page 86.
2. In the **INTLK/TRIG/LT** menu, use the **Override intlks** option to override the KEY and MOTN interlocks.
See [Override Intlks Option: Override Interlocks](#) on page 109.
3. Use the **MOTORS** menu to choose the motions desired during beam-on.
The motions can be automatic or manual (see [MOTOR Menu: Set the Type of Control](#) on page 135).
4. Turn the **DISABLE/ENABLE** keyswitch on the dedicated keyboard to the **ENABLE** position.
5. In the Service mode screen, select the **BEAM CTRL** menu heading and confirm beam-on.
6. Press the **MOTION ENABLE** bar.

CALIB Menu: Calibrate Process Values

Selecting the **CALIB** menu heading opens the **CALIBRATION** menu.



The High Energy model **CALIBRATION** menu is on the left and the **UNIQUE CALIBRATION** menu is on the right.



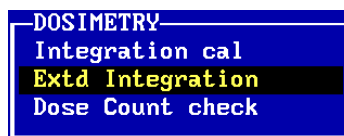
The **CALIBRATION** menu options allow you to:

- Test the dosimetry integration system and verify the dose counters and timers
- Register the software (requires the Service personnel password)
- Select a PRO scale to use for viewing position readouts
- Work with readout, analog, and PRO meter calibration
- Check the limits of motor axes
- View control parameters for a selected dynamic beam delivery (DBD) motor axis
- Save configuration changes
- Check the pulse repetition frequency (PRF) on the UNIQUE accelerators

Calibration is the process of determining the relative accuracy of a measuring instrument by comparing it to a standard, calculating the necessary corrections, and applying the corrections. Each **CALIBRATION** option is described in its own section within this guide.

Calibrate Dosimetry Option

Selecting the **Calibrate dosimetry** option from the **CALIBRATION** menu opens the **DOSIMETRY** menu.



The **DOSIMETRY** menu allows you to:

- Test the dosimetry integration system with a time constraint of 0.10 minutes.
- Test the dosimetry integration system without a time constraint; this option allows you to stop the test when you wish.

- Verify the dose counters and timers.

Test the Dosimetry Integration System

In the dosimetry integration test, the system passes a test current through the dosimetry integrator system to simulate a radiation exposure. The injected current should result in MU1 and MU2 readings between 100 and 109 (between 195 and 205 for C/D machines) within a time interval of 0.1 minute.

Running the Dosimetry Integration Test automatically sets ION1 and ION2 parameters.

The **DOSIMETRY** menu provides two options for testing:

- **Integration cal:** Allows you to test dosimetry integration using a time limit of 0.10 minutes.

To test the dosimetry integration system using the 0.10 minute time limit, select **Integration cal** in the **DOSIMETRY** menu and press **Enter**.

- **Extd Integration:** For the purposes of calibration, an extended integration capability is provided which runs integration calibration without the time constraint of 0.10 minutes. This option allows you to stop the test when you wish.

To test the dosimetry integration system for an unlimited amount of time, select **Extd Integration** in the **DOSIMETRY** menu and press **Enter**.

To stop the test, press **Enter** again to select **Dosimetry > Extd Integration**.

Test Dose Counters and Timers

When you select the **Dose Count check** option, the system verifies that the dose counter/timers can count up to the set MU and time, and verifies operation of the dose rate interlocks. To test the dose counters and timers, select the **Dose Count check** option in the **DOSIMETRY** menu and press **Enter**.

Select PRO Scale Option

The **Select PRO scale** option in the **CALIBRATION** menu allows you to select a scale in which to display position readouts (PROs) in Service mode. Selecting the **Select PRO scale** option displays the **SELECT PRO SCALE** menu.

The **SELECT PRO SCALE** menu offers the following options, including three PRO scales for use in clinical modes (your PRO scale is configured by Varian at the factory):

- **RAW SCALE:** Specifies that the actual voltage levels appear for axis position readouts. Software motion limits are disabled and automatic motion and symmetric jaw motion are not allowed.
- **INTERNAL:** This option is like **VARIAN SCALE**, except that gantry axis rotation angles range from -5 to 365.
- **VARIAN SCALE (PRO):** When the gantry is rotated to 180°, the head is at the top position. It is at the bottom for 0°. Gantry angles increase counterclockwise.
- **IEC SCALE (PRO):** Conforms to IEC Standard 60601-2-1. When the gantry is rotated to 0°, the head is at the top position. It is at the bottom for 180°. Gantry angles increase clockwise.
- **IEC 1217 SCALE (PRO)** (accelerator software version 6.3 and up): Conforms to IEC Standard 61217. When the gantry is rotated to 0°, the head is at the top. It is at the bottom for 180°. Gantry angles increase clockwise. Field labels and couch direction labels are different from IEC 60601-2-1 scale.

Figure 9 illustrates the three PRO scales.

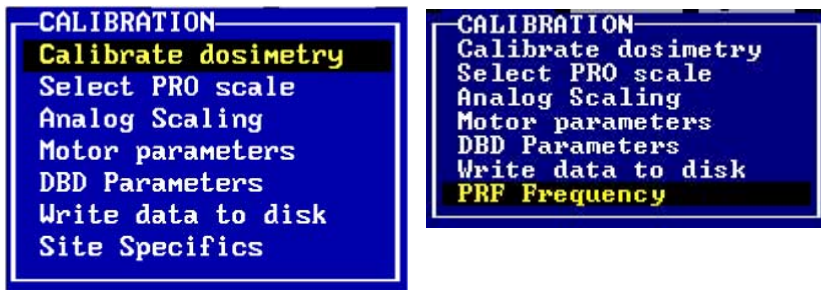
Scale	IEC 61217	IEC 60601-2-1	Varian
Gantry Rotation			
Collimator Rotation Observer position: standing by couch, looking up at collimator			
Jaws: X1/X2, Y1/Y2	 X1,X2 : -1 0 +1 Y1,Y2 : -1 0 +1	 X2,Y2: -1 0 +1 X1,Y1: +1 0 -1	 X1,Y1: +1 0 -1 X2,Y2: -1 0 +1
Couch Vertical	 +1 0 -1	 999 0 1	 99 100 101
Couch Longitudinal seen from above	 101 100 99	 101 100 99	 101 100 99
Couch Lateral seen from above	 -1 0 +1	 999 0 1	 99 100 101
Couch Rotation seen from above			

Figure 9 Three PRO Scales

To select the PRO scale:

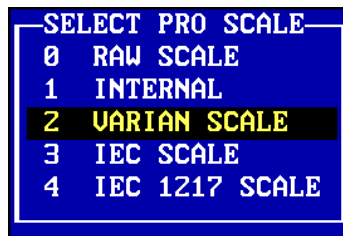
1. Select the **CALIB** menu heading and press **Enter**.

The CALIBRATION menu appears. The High Energy model CALIBRATION menu is on the left and the UNIQUE CALIBRATION menu is on the right.



2. Select the **Select PRO scale** option and press **Enter**.

The SELECT PRO SCALE menu appears.

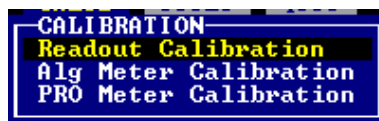


3. To select a scale:
 - Select the scale you want and press **Enter**.
 - Type the index number that precedes the scale.

The menu closes.

Perform Analog Scaling

When you select **Analog Scaling** from the **CALIBRATION** menu, the system displays the **CALIBRATION** menu. Use this menu to select the analog scaling items that you want to calibrate.

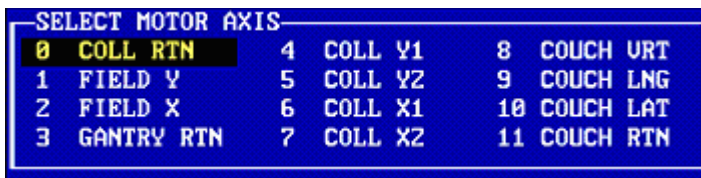


This **CALIBRATION** menu offers these analog scaling calibration options:

- **Readout Calibration** calibrates primary and secondary PROs (for details, see [Readout Calibration: Calibrate PROs](#) on page 149).
- **Alg Meter Calibration** displays the gain and offset factors applied to the raw value for a selected machine meter (for details, see [Alg Meter Calibration: Analog Meter Gain and Offset](#) on page 154).
- **PRO Meter Calibration** displays the gain and offset factors applied to the raw value for a selected PRO meter (for details, see [PRO Meter Calibration Option: PRO Meter Gain and Offset](#) on page 156).

Readout Calibration: Calibrate PROs

The **Readout Calibration** option allows you to calibrate the primary and secondary PROs. Selecting the **Readout Calibration** option opens the **SELECT MOTOR AXIS** menu, where you select the motor axis you want to calibrate.



SELECT MOTOR AXIS					
0	COLL RTN	4	COLL Y1	8	COUCH URT
1	FIELD Y	5	COLL Y2	9	COUCH LNG
2	FIELD X	6	COLL X1	10	COUCH LAT
3	GANTRY RTN	7	COLL X2	11	COUCH RTN

Cancel Calibration

To cancel calibration without saving the new values, even during the current session:

- Press the **Clear** key on the pendant.
- Press the **Esc** key on the keyboard.

Save Calibration Values

The system discards new readout calibration values at the end of the current Service mode session unless you save the new data to the console hard disk before you exit Service mode. For instructions on saving the data, see [Write Data to Media Option: Save Service Mode Data](#) on page 163.

Calibrate Readouts

You must calibrate independent jaws separately. If you attempt to select **FIELD Y** or **FIELD X** on an accelerator with independent jaws, the following message appears:

Must Calibrate Jaws Independently

Conversely, on an accelerator with jaws limited to symmetric operation, you must calibrate the jaws simultaneously. If you try to select an individual jaw, the following message appears:

Lower Independent Jaws Not Enabled

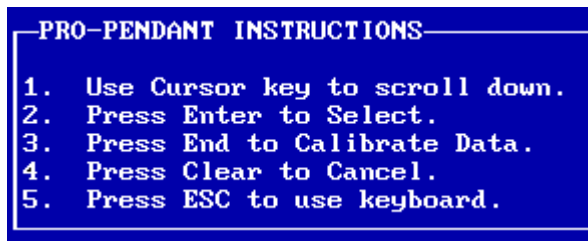
When you select a motor axis, the position readouts at the bottom of the screen change to raw scale, and the voltages are displayed.

Use the Hand Pendant

Selecting a motor axis for calibration activates several keys on the hand pendant. This allows you to calibrate the readouts without leaving the treatment room. You can also use the pendant redirection function in the **UTILS** menu to manually activate the pendant keys. For more information, see [Pendant Redirection Option: Redirect Pendant Keys](#) on page 179.

To use the pendant for readout calibration, you should have a monitor in the treatment room. If there is no monitor, you must use the alphanumeric service keyboard.

Once you have selected a motor axis for calibration, the **PRO-PENDANT INSTRUCTIONS** message appears, which lists the pendant keys to use for readout calibration.



Switch Back to the Keyboard



Note: You can switch from pendant to keyboard calibration at any time by pressing **Esc** on the keyboard.

When you have switched to keyboard calibration, the keyboard instructions appear on the screen in place of the pendant instructions. All key names are the same, except that the **Esc** on the keyboard is the **Clear** key on the pendant.

Capture Position Readouts

You can view the captured position readouts and saved values for the motor axis in a **CALIBRATION** data display. The captured position readouts (**Capture Posn**) are the measured locations at which you should obtain calibration values for the selected axis.

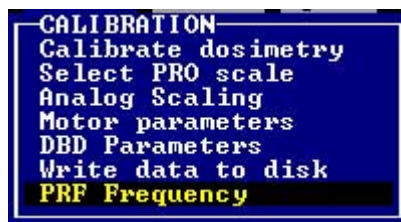
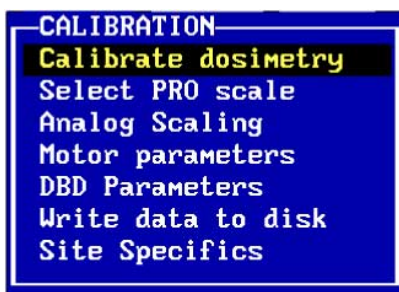
CALIBRATION			
GANTRY RTN		Value	
Capture Posn		PRO	SPRO
0.0		0	0
90.0		0	0
180.0		0	0
270.0		0	0
0.0		0	0

Calibrate a Position Readout

To calibrate a position readout:

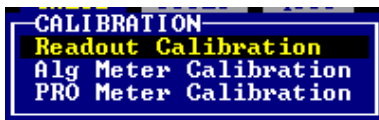
1. Select the **CALIB** menu heading and press **Enter**.

The **CALIBRATION** menu appears. The High Energy model **CALIBRATION** menu is on the left and the **UNIQUE CALIBRATION** menu is on the right.



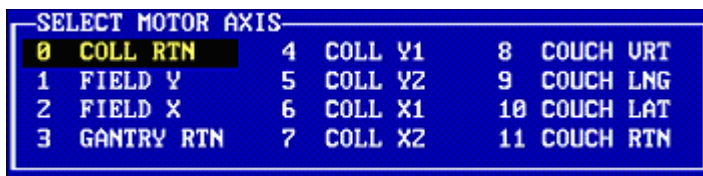
2. Select **Analog Scaling** and press **Enter**.

A second **CALIBRATION** menu appears.



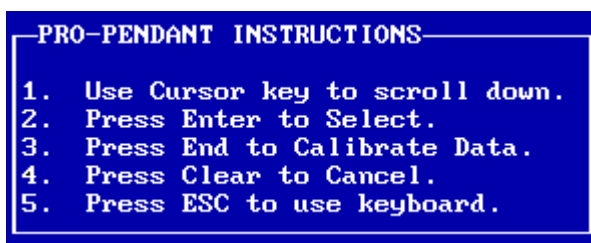
3. Select **Readout Calibration** and press **Enter**.

The SELECT MOTOR AXIS menu appears.



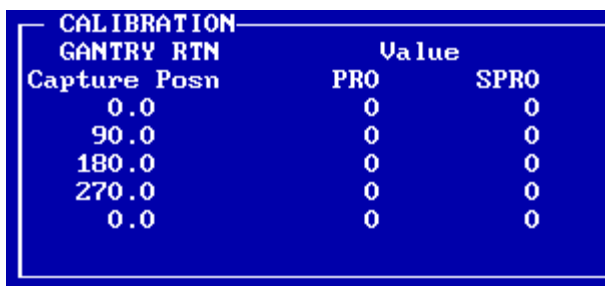
4. Select the motor axis you want to calibrate and press **Enter**.

Pendant or keyboard key instructions appear on the screen, with the captured and saved values for the motor axis.



5. Move the motor axis to each capture position listed and verify its alignment. The motor axes are listed and illustrated in [Figure 9](#).

The GANTRY RTN CALIBRATION dialog box opens.



6. When the motor axis is aligned at the true capture position, use the down arrow key to select the same position on the screen.
7. Press **Enter** to capture the value.

The value appears as an unscaled number.

8. Repeat these steps for all positions.
9. After you capture values for all the positions, press the **End** key (**Shift + Enter** on the pendant).

The system calculates the linear deviation among all the capture positions and displays the relative deviation. The calibration is tested and saved as described in [Write Data to Media Option: Save Service Mode Data](#) on page 163.



Note: A *Potentiometer Check* utility checks the direction of angle rotation. If you are calibrated backwards, the utility prevents you from saving the calibration. For more information, see [Potentiometer Check](#) on page 237.



CAUTION: Ideally, the relative deviation number should be less than 1. If it is greater than 2, you should repeat the calibration after carefully measuring any or all of the capture positions.

To close the menu, press **Enter** or **Esc**.



WARNING: Varian strongly recommends that you verify the displayed digital position readouts against the actual positions. A mismatch could indicate that the digital position readouts are miscalibrated, which could cause any rotation motor axis to rotate in the reverse direction and mistreat the patient. In particular, the collimator rotation digital position readouts can be calibrated incorrectly, causing the collimator to rotate in the reverse direction (see [Figure 10](#)). To prevent possible mistreatment, every time you calibrate the rotation digital position readouts, use the Auto GoTo test to verify the readouts. Then save the data as part of the permanent machine configuration.

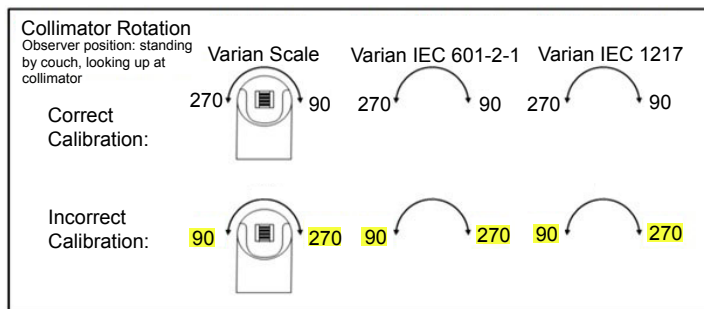


Figure 10 Correct and Incorrect Collimator Readout Calibration

Perform the Auto GoTo Test

Perform the Auto GoTo test each time you calibrate any of the rotation digital position readouts and save the results as part the machine configuration file. It is important that you perform this test to ensure correct treatment of the patient. Developed by Varian, this test verifies the rotation readout by driving the rotational motor axis to the central target position, as shown in [Figure 9](#) on page 147.

To perform the Auto GoTo Test:

1. Set the rotational motor axis to the central target position.
The central target position is indicated by the second of the three numbers shown in [Figure 9](#) for each rotational axis.
2. Rotate the rotational motor axis 10° counterclockwise.
3. Using the Auto GoTo feature, drive the rotational motor axis to the central target position:
 - a. Select the **MOTORS** menu and press **Enter**.
 - b. Choose an option:
 - Gantry automatic (see [Gantry Automatic Option](#) on page 138)
 - Couch automatic (see [Couch Automatic Option](#) on page 140)
 - c. Press **Enter**.
 - d. Select the same rotational motor axis that you rotated in Step 2.
 - e. Type in the central target position and press **Enter**.
 - f. Press **F2**.
 - g. To start motion, press and hold the **MOTION ENABLE** bar on the dedicated keyboard.

If the rotational motor axis automatically drives to the limit instead of to the central target position, the calibration is reversed and must be corrected.

4. When the calibration is correct, save the data by following the instructions in [Write Data to Media Option: Save Service Mode Data](#) on page 163.

You must save the calibrations to make them part of the permanent machine configuration. If you do not save the configuration changes to the computer hard disk, the system discards them when you exit Service mode.

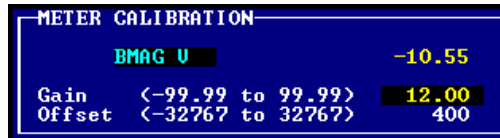
Alg Meter Calibration: Analog Meter Gain and Offset

The **Analog Scaling** option in the **CALIBRATION** menu allows you to display the gain and offset factors that are applied to the raw value detected by the software for a selected machine meter. (To display the resulting scaled meter value, use the **Analog** option in the **DISPLAY** menu.)

Selecting the **Analog Scaling** option opens the **CALIBRATION** menu, allowing you to select the **Alg Meter Calibration** option.

Selecting the **Alg Meter Calibration** option opens the **SELECT ANALOG METER** menu, where you select an analog meter.

Once you have selected the desired analog meter, the corresponding gain and offset values set previously appear in a **METER CALIBRATION** data display.

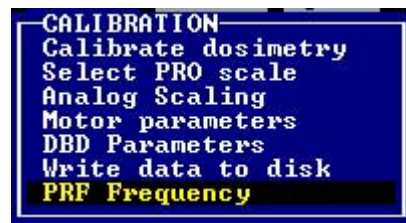
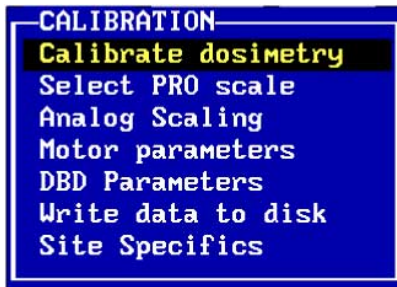


Display Analog Meter Offsets

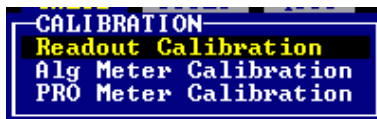
To display analog meter offset values:

1. Select the **CALIB** menu heading and press **Enter**.

The **CALIBRATION** menu appears. The High Energy model **CALIBRATION** menu is on the left and the **UNIQUE CALIBRATION** menu is on the right.



2. Select **Analog Scaling** and press **Enter**.

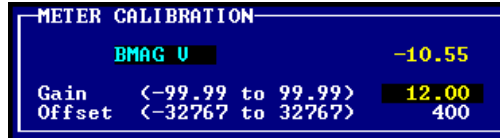


3. Select **Alg Meter Calibration** and press **Enter**.

The **SELECT ANALOG METER** menu appears.

4. Select the desired analog meter and press **Enter**.

The METER CALIBRATION data display appears.



To close the menu, press **Enter** or **Esc**.

PRO Meter Calibration Option: PRO Meter Gain and Offset

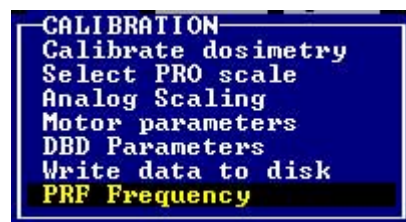
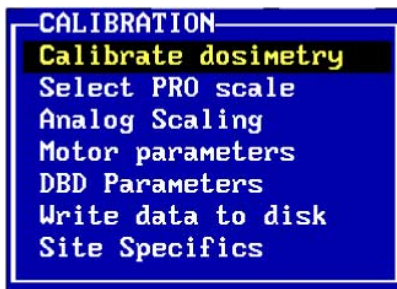
The **PRO Meter Calibration** option in the **CALIBRATION** menu allows you to display the gain and offset factors that are applied to the raw value detected by the software for a selected PRO meter. When you select the **PRO Meter Calibration** option, the system displays the **PRO METER CALIBRATION** menu, where you select the PRO meter you want to check.

Once you have specified the desired PRO meter, the system displays the corresponding gain and offset values set previously in a **METER CALIBRATION** data display.

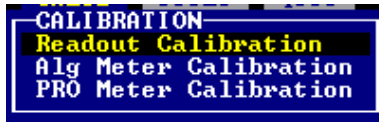
To check PRO meter offsets:

1. Select the **CALIB** menu and press **Enter**.

The **CALIBRATION** menu appears. The High Energy model **CALIBRATION** menu is on the left and the **UNIQUE CALIBRATION** menu is on the right.

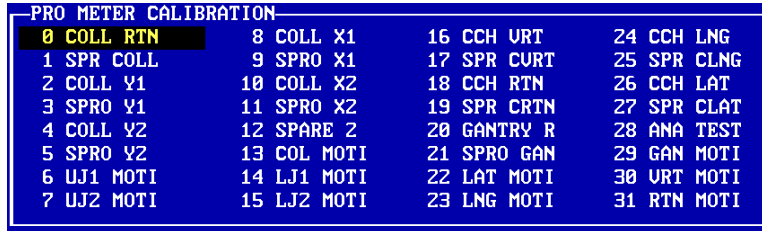


2. Select **Analog Scaling** and press **Enter**.



3. Select **PRO Meter Calibration** and press **Enter**.

The PRO METER CALIBRATION menu appears.



4. Select the PRO meter you want and press **Enter**.

The METER CALIBRATION menu appears.

To close the menu, press **Enter** or **Esc**.

Motor Parameters Option: Parameters for a Motor Axis

The **Motor parameters** option in the **CALIBRATION** menu displays the control parameters for a selected motor axis. Control parameters are the absolute minimum and maximum values for position and velocity. The **Motor parameters** option allows you to display the following parameters that are set at the factory:

- Position limits set for a selected motor axis
- Number of capture positions used to calibrate the PRO readout for that axis
- Location of capture positions

Selecting the **Motor parameters** option displays the **MOTOR PARAMETERS** menu.



This menu offers two options:

- Parameters
- Limits (to skip to this section, see [Check the Limits of a Given Axis](#) on page 160).

Select the Axis with the Parameters Option

Selecting the **Parameters** option opens the **SELECT MOTOR AXIS** menu, which allows you to select a collimator, gantry, or couch axis.

SELECT MOTOR AXIS					
0	COLL RTN	4	COLL Y1	8	COUCH URT
1	FIELD Y	5	COLL Y2	9	COUCH LNG
2	FIELD X	6	COLL X1	10	COUCH LAT
3	GANTRY RTN	7	COLL X2	11	COUCH RTN

In response to your motor axis selection, the system displays the **ADJUST MOTORS** menu with the control parameters for that motor axis.

ADJUST MOTORS		
		HEX
	COLL RTN	<8001-7FFF>
Position	Param 1	150
Position	Param 2	3000
Velocity	Param 1	FFCC
Position	Param 3	0

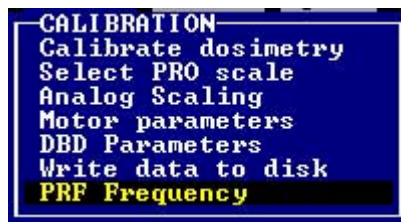
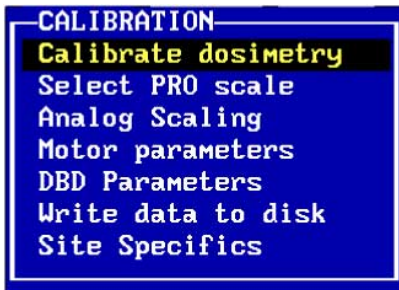
The parameters are displayed in hexadecimal format. The numbers in parentheses are the range within which the parameter values can fall (8000h is understood to be signed negative).

Check the Motor Parameters for a Given Axis

To display the control parameters of a motor axis:

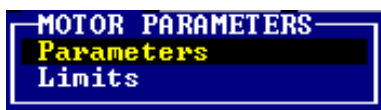
1. Select the **CALIB** menu heading and press **Enter**.

The **CALIBRATION** menu appears. The High Energy model **CALIBRATION** menu is on the left and the **UNIQUE CALIBRATION** menu is on the right.



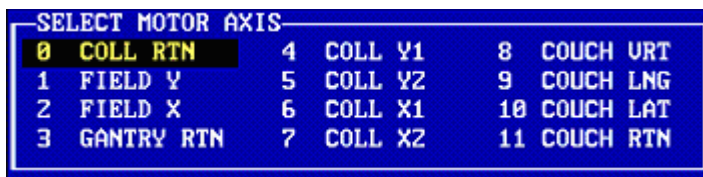
2. Select **Motor parameters** and press **Enter**.

The MOTOR PARAMETERS menu appears.



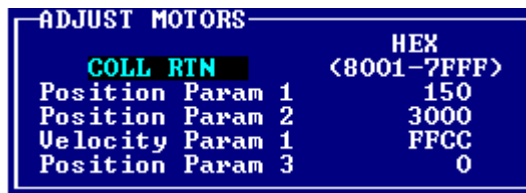
3. Select **Parameters** and press **Enter**.

The SELECT MOTOR AXIS menu appears.



4. Select the motor axis you want and press **Enter**.

The ADJUST MOTORS data display appears.



To close the display, press **Enter** or **Esc**.

Check the Limits of a Given Axis

Selecting the **Motor parameters** option opens the **MOTOR PARAMETERS** menu, where you can select the **Limits** option.



Note: You can select the **Limits** option only when the PRO scale is set to INTERNAL. To reset the scale, see [Select PRO Scale Option](#) on page 146.

Selecting the **Limits** option opens the **SELECT MOTOR AXIS** menu, which allows you to specify the desired motor axis. Once you have specified the motor axis, the system displays its motor control limits in the **MOTORS LIMITS** data display.

MOTORS LIMITS		
COLL RTN	< -10 - 370>	
Motor Position Min		15.0
Motor Position Max		345.0

The limits of the specified axis are expressed in decimal format. The numbers in parentheses indicate the input limits of motion for the axis. The software limits are always less than the full mechanical range. For example, the input range for gantry rotation is -10 to 370, but the software limits of travel must be set within the mechanical limits, which are -5.0 and 365.0.

Check the Limits of a Motor Axis

To check the limits of a motor axis:

1. Select the **CALIB** menu heading and press **Enter**.

The CALIBRATION menu appears. The High Energy model CALIBRATION menu is on the left and the UNIQUE CALIBRATION menu is on the right.

CALIBRATION
Calibrate dosimetry
Select PRO scale
Analog Scaling
Motor parameters
DBD Parameters
Write data to disk
Site Specifics

CALIBRATION
Calibrate dosimetry
Select PRO scale
Analog Scaling
Motor parameters
DBD Parameters
Write data to disk
PRF Frequency

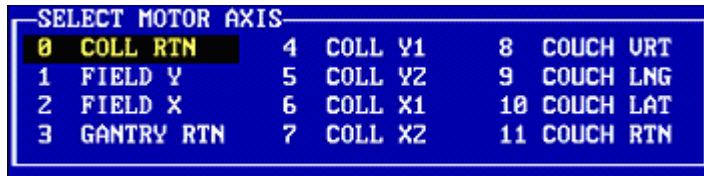
2. Select **Motor parameters** and press **Enter**.

The MOTOR PARAMETERS menu appears.



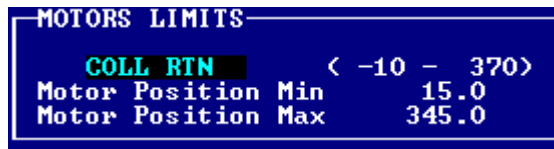
3. Select **Limits** and press **Enter**.

The SELECT MOTOR AXIS menu appears.



4. Select the motor axis you want and press **Enter**.

The MOTORS LIMITS data display appears.



5. To display the number of position captures for the selected motor axis:
 - Press **Enter**.
 - Press **Esc**.

The number of position captures appears.

6. To display the values of the capture positions for the motor axis:
 - Press **Enter**.
 - Press **Esc**.

The values of the capture positions appear.

To close the menu, press **Enter** or **Esc**.

DBD Parameters Option: Dynamic Beam Delivery Parameters

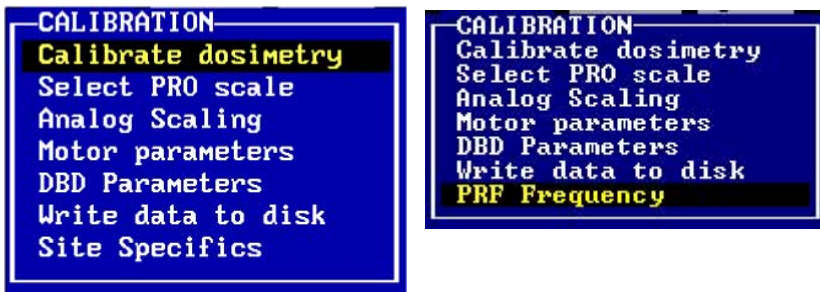
The **DBD Parameters** option in the **CALIBRATION** menu displays the control parameters for a selected dynamic beam delivery (DBD) motor axis. The control parameters are set at the factory.

Selecting the **DBD Parameters** option opens the **SELECT MOTOR AXIS** menu, allowing you to select a motor axis. After you select a motor axis, the system displays the **ADJUST DYNAMIC PARAMS** data display with the DBD control parameters for the motor axis. The DBD parameters are expressed in hexadecimal format; the numbers enclosed in parentheses indicate their valid range for the selected parameter.

To check the DBD control parameters:

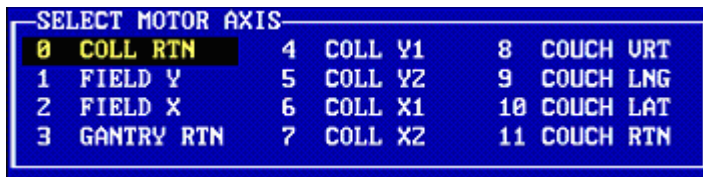
1. Select the **CALIB** menu heading and press **Enter**.

The **CALIBRATION** menu appears. The High Energy model **CALIBRATION** menu is on the left and the **UNIQUE CALIBRATION** menu is on the right.



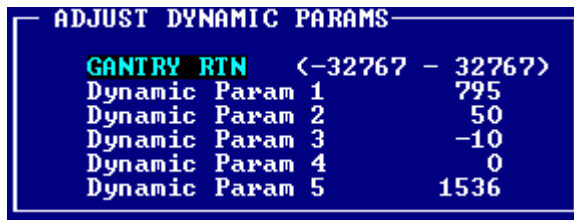
2. Select **DBD Parameters** and press **Enter**.

The **SELECT MOTOR AXIS** menu appears.



3. Select the motor axis you want and press **Enter**.

The **ADJUST DYNAMIC PARAMS** data display appears.



To close the menu, press **Enter** or **Esc**.

Write Data to Media Option: Save Service Mode Data

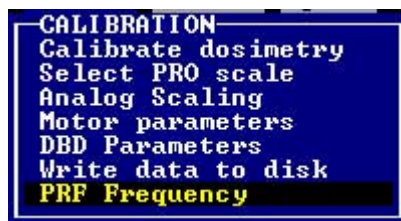
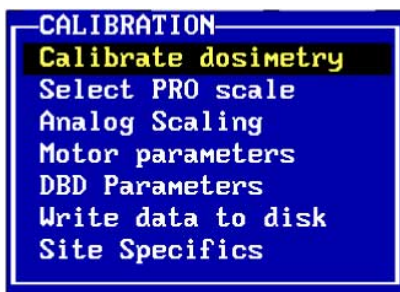
The **Write data to Media** option in the **CALIBRATION** menu allows you to save the configuration changes you made during the current Service mode session. The changes become part of the permanent machine configuration. If you do not save the configuration changes to the computer hard disk, the system discards them when you exit Service mode.

If you select the **Write data to Media** option, a message appears where you can confirm that you wish to save your configuration changes.

To save the Service mode configuration data:

1. Select the **CALIB** menu heading and press **Enter**.

The **CALIBRATION** menu appears. The High Energy model **CALIBRATION** menu is on the left and the **UNIQUE CALIBRATION** menu is on the right.



2. Select **Write data to Media** and press **Enter**.

The **SAVE CONFIG DATA CHANGES** menu appears.

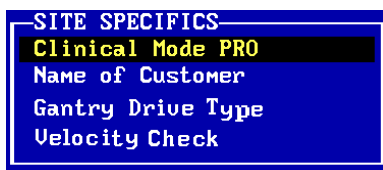
3. To save the configuration data to the hard disk, select **YES** and press **Enter**.

A message appears confirming that the calibration changes are saved to media, and the menu closes.

Site Specifics Option

The **Site Specifics** menu option in the **CALIBRATION** menu allows an authorized user to:

- Select a PRO scale for Clinical mode
- Register accelerator software
- Select a Gantry drive type: Clutch or clutchless drive
- Enable/disable Velocity Check



Note: The *Site Specifics* menu option is not available to users logging on with the Customer password. To select *Site Specifics* from the **CALIBRATION** menu, you must log on using the Service password.



WARNING: Incorrect gantry drive selection (for example, selection of the *Clutch Drive* option where a clutchless drive is installed) can result in hazardous conditions, including damage to equipment and potential injury to personnel, due to premature failure of the gantry chain.

Clinical Mode PRO: Select a PRO Scale for Treatments

The **Clinical Mode PRO** menu option in the **SITE SPECIFICS** menu is only available to users with the Service password. This option allows Varian personnel to change the PRO scale used for Morning Checkout and Clinical treatment modes. Only the PRO scales that should be used for patient treatment are offered. This PRO selection becomes the default PRO setting, regardless of which PRO scale is used for calibration and testing.

For a description of PRO scales, see [Select PRO Scale Option](#) on page 146.

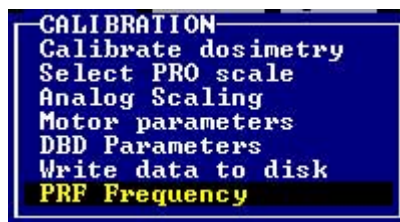
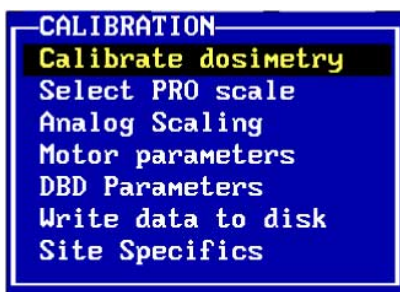


Note: The **PRO Scale** option available with the Customer password at the top level of the **CALIB** menu allows a user to use any scale (clinical, raw, or PRO) for calibration and testing, but that selection is not saved for clinical modes.

To change the Morning Checkout and Clinical modes PRO scale:

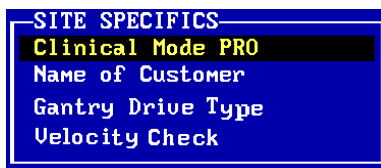
1. Select the **CALIB** menu heading and press **Enter**.

The CALIBRATION menu appears. The High Energy model CALIBRATION menu is on the left and the UNIQUE CALIBRATION menu is on the right.



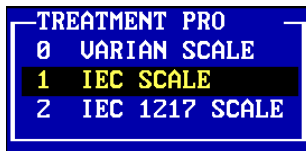
2. Select **Site Specifics** and press **Enter**.

The SITE SPECIFICS menu appears.



3. In the **SITE SPECIFICS** menu, select **Clinical Mode PRO** and press **Enter**.

The TREATMENT PRO menu appears.



4. Select a PRO scale and press **Enter**.

After you save your configuration changes, the PRO scale you have selected is used for morning checkout and patient treatments.

Name of Customer: Register Software

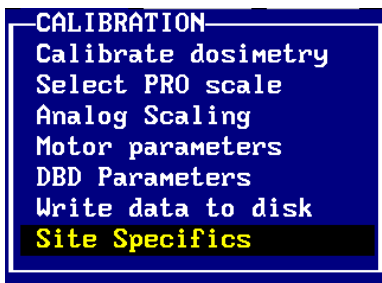
The **Name of Customer** option allows an authorized user (logged in with the Service password) to register the software. Software registration is done by Varian Field Service personnel at the time of installation.

The software can only be registered once. After a name is entered, the field becomes read-only. A subsequent modification to the name requires you to re-install the software using the upgrade process. The customer name input string is displayed at startup of Clinical mode, along with the versions of the installed software components. If the customer's name has not been properly entered and saved, then the following message appears at startup:

WARNING: UNLICENSED SOFTWARE

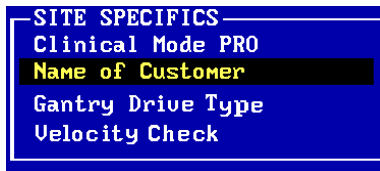
To register the software:

1. Enter Service mode using the Service password (the Customer password prevents you from opening the SITE SPECIFICS menu).



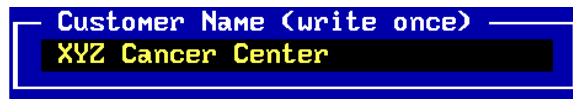
2. From the **CALIBRATION** menu, select **Site Specifics**.

The SITE SPECIFICS menu appears.



3. Select **Name of Customer** and press **Enter**.

A Customer Name data box appears where you can type your registration information.



4. Type your customer name, department name, or other registration data and press **Enter**.

The registration information appears in the data box. When you save your configurations, the registration is saved.

Gantry Drive Type: Clutch or Clutchless Drive

Authorized Varian Service personnel can select the type of gantry drive motion control using the **Gantry Drive Type** option in the **SITE SPECIFICS** menu.



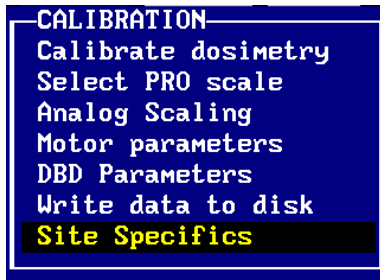
Note: The PRO Scale option available with the Customer password at the top level of the **CALIB** menu allows a user to use any scale (clinical, raw, or PRO) for calibration and testing, but that selection is not saved for clinical modes.

The gantry drive setting enhances smooth motion of the gantry. Regardless of the type of gantry drive installed on your accelerator, you must set the gantry drive to **Clutchless Drive**.



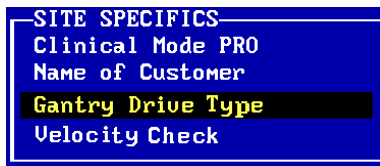
WARNING: Incorrect gantry drive selection (for example, selection of the *Clutch Drive* option where a clutchless drive is installed) can result in hazardous conditions, including damage to equipment and potential injury to personnel, due to premature failure of the gantry chain.

1. Enter Service mode using the Service password (the Customer password prevents you from opening the **SITE SPECIFICS** menu).



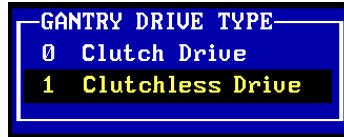
2. From the **CALIBRATION** menu, select **Site Specifics** and press **Enter**.

The **SITE SPECIFICS** menu appears.



3. Select **Gantry Drive Type** and press **Enter**.

The GANTRY DRIVE TYPE menu appears.



4. Select **Clutchless Drive** and press **Enter**.

When you save your configurations, the gantry drive setting is saved.

Velocity Check: Enable or Disable Check

Velocity check is a feature that monitors manual and automated/remote gantry motions. When **Velocity Check** is set to Enable, any excessive, reverse, or unexpected motions trigger the HWFA interlock and all motions stop.

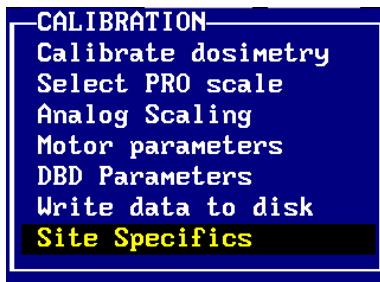
This section describes how to enable **Velocity Check** in case it is disabled.



WARNING: To prevent injury or damage due to uncontrolled gantry motion, enable Velocity Check. Disabling Velocity Check can result in hazardous conditions if motion component failure occurs.

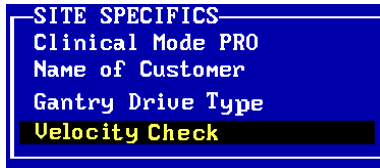
To enable or disable velocity check:

1. Enter Service mode using the Service password (the Customer password prevents you from opening the SITE SPECIFICS menu).



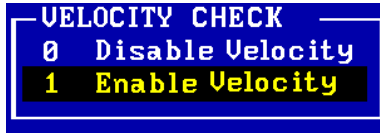
2. From the **CALIBRATION** menu, select **Site Specifics** and press **Enter**.

The SITE SPECIFICS menu appears.



3. Select **Velocity Check** and press **Enter**.

The VELOCITY CHECK menu appears.



4. Make a selection:
 - Select **0 Disable Velocity**.
 - Select **1 Enable Velocity**.

When you save your configuration settings, the velocity check setting is saved.

PRF Frequency Option (UNIQUE Accelerators Only)

The **PRF Frequency** option in the **CALIBRATION** menu displays the current pulse repetition frequency (PRF). The PRF is the discrete on/off cycle of the beam pulse that determines the intensity of the radiation beam.

Selecting **PRF Frequency** opens the **PRF FREQUENCY** menu. The PRF FREQUENCY menu allows you to check the current PRF or write it to a file.



Selecting **Set PRF frequency** displays the **ENTER PRF FREQUENCY** data box, showing the current PRF value.



Selecting **Write PRF to file** displays the **WRITE PRF TO FILE** data box

```
WRITE PRF TO FILE
PRF in file           0
Measured PRF         0
Operating PRF        0

Press Enter to save Measured PRF to file.
Press ESC to exit.
```

View the Current PRF

To view the current PRF:

1. Select the **CALIB** menu heading and press **Enter**.

The CALIBRATION menu appears.

```
CALIBRATION
Calibrate dosimetry
Select PRO scale
Analog Scaling
Motor parameters
DBD Parameters
Write data to disk
PRF Frequency
```

2. Select **PRF Frequency** and press **Enter**.

The PRF FREQUENCY menu appears.

```
PRF FREQUENCY
Set PRF frequency
Write PRF to file
```

3. Select **Set PRF frequency** and press **Enter**.

The ENTER PRF FREQUENCY data box appears, showing the current PRF value.

```
ENTER PRF FREQUENCY
PRF Frequency (0, 120 - 380): 0
```

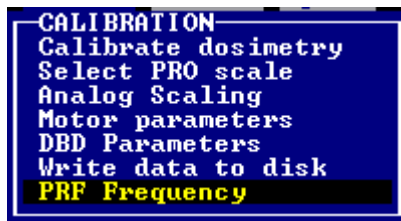
To close the data box, press **Enter** or **Esc**.

Write the PRF to a File

To write the PRF to a file:

1. Select the **CALIB** menu heading and press **Enter**.

The CALIBRATION menu appears.



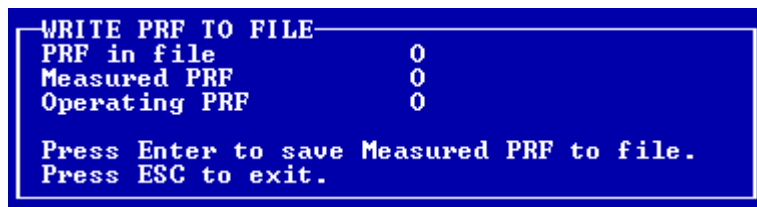
2. Select **PRF Frequency** and press **Enter**.

The PRF FREQUENCY menu appears.



3. Select **Write PRF to file** and press **Enter**.

The WRITE PRF TO FILE data display appears, showing the current PRF values.



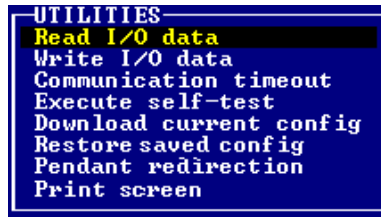
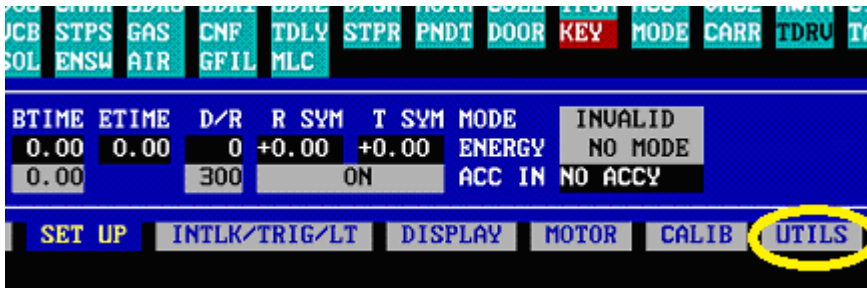
4. To save to a file, press **Enter**.

The calibrated data is saved, and the display closes.

UTILS Menu: Service Mode Utilities

Selecting the **UTILS** menu heading in Service mode opens the **UTILITIES** menu, which allows you to:

- Read and write I/O data
- Set a communications time-out
- Run a self-test
- Download and restore the current configuration
- Activate certain pendant keys
- Print a hard copy of the screen



Read I/O Option: Read the Status of the Input/Output Ports

The system displays the current data at the specified port, as a two-digit number in hexadecimal format. The hexadecimal number representing the data at the port is converted to binary format, in order to display an eight-digit array of 1s and 0s indicating the on/off status of each bit at the port. For example, the indication for a port might be 23h (0010 0011) with bits 3, 7, and 8 active.

The **Read I/O data** option in the **UTILITIES** menu allows you to display the status of a selected input/output (I/O) port. Selecting the **Read I/O data** option displays the **READING I/O DATA** display, which allows you to specify an I/O port address (in hexadecimal).



```
READING I/O DATA
Enter Port Number < 0-FF>: 1b
Data Read from Port <HEX>: 7D
```

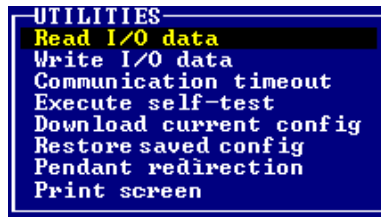
The default address is the last port read. If you have not used the **Read I/O data** option during the current Service mode session, the default is 0.

Display the Status of an I/O Port

To display the status of an I/O port:

1. Select the **UTILS** menu heading and press **Enter**.

The UTILITIES menu appears.



```
UTILITIES
Read I/O data
Write I/O data
Communication timeout
Execute self-test
Download current config
Restore saved config
Pendant redirection
Print screen
```

2. Select **Read I/O data** and press **Enter**.

The READING I/O DATA display appears.



```
READING I/O DATA
Enter Port Number < 0-FF>: 1b
Data Read from Port <HEX>: 7D
```

3. Type the two-digit hexadecimal number of the port whose data you want to read and press **Enter**.

The current data appears in the display.

To close the display, press **Enter** or **Esc**.

Write I/O Data Option: Write Data to an I/O Port

Certain functions such as writing data to an I/O port require special authorization and entry of an appropriate password. If you attempt to write data to an I/O port without having entered an authorized password, the system displays a message letting you know that this option is not available.



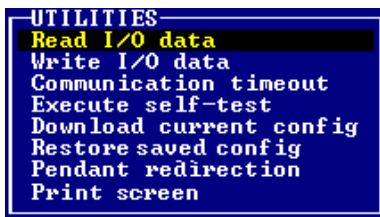
CAUTION: Incorrect use of the Write I/O Data function can damage the accelerator. Press **Esc** to exit this function.

Selecting the **Write I/O data** option displays the **WRITE I/O DATA** box, which allows you to specify an I/O port address (in hexadecimal format) and the data to be written to that port.

To write to an I/O port:

1. Select the **UTILS** menu heading and press **Enter**.

The UTILITIES menu appears.



2. Select **Write I/O data** and press **Enter**.

The WRITING I/O DATA data box appears.



3. Type the two-digit hexadecimal number of the port where you want to write the data and press **Enter**.

The data entry field is selected.

4. Type the data you want to write (using hexadecimal format) and press **Enter**.

The system writes the specified data to the indicated port and closes the data box.

Communication Timeout Option

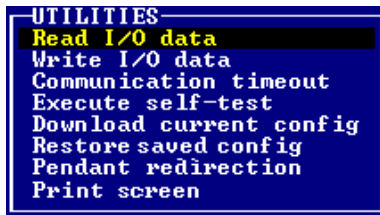
The communication timeout value sets the maximum time allowed between control commands exchanged by the console computer and the controller. The default communication timeout is set at 60000 units, where each unit has a value of 0.6 seconds.

The **Communication timeout** option in the **UTILITIES** menu allows you to change the communication timeout value for Service mode. This value applies only during the current Service mode session and is discarded when you exit Service mode. Selecting the **Communication timeout** option displays the **COMMUNICATIONS** dialog box.

To set the communication timeout:

1. Select the **UTILS** menu heading and press **Enter**.

The UTILITIES menu appears.



2. Select **Communication timeout** and press **Enter**.

The COMMUNICATIONS data box appears.



3. Type in the new timeout value and press **Enter**.

The menu closes.

If the Communications Timeout Runs Out

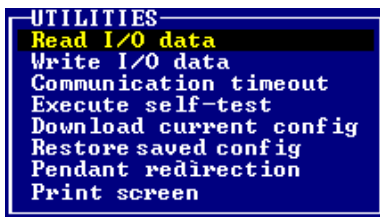
If the communications timeout runs out, a communications failure occurs and the system activates a CTRL interlock. To recover from a CTRL interlock, perform the diagnostic self-test.

Execute Self-Test Option: Perform a Diagnostic Self-Test

To perform a self-test:

1. Select the **UTILS** menu heading and press **Enter**.

The UTILITIES menu appears.



2. Select the **Execute self-test** option and press **Enter**.

The control software performs the self-test. During the self-test, the system displays:

```
Executing  
Controller Self-Test  
Please Wait...
```

If the power keyswitch in the console electronics cabinet is in the on position (extreme clockwise), the system displays:

```
Self-Test Not Executed  
Power Key was ON.
```

If the test finishes successfully, the system displays:

```
Self-Test Completed Successfully.
```

When a self-test fails, the system displays a **DIAGNOSTIC FAILURE** data display with the reason the test failed, an integer address, and test data. In addition, the system creates a file containing a record of the system status, including treatment settings, interlocks, position readouts, analog meters, digital I/O, and accessory data. This file is stored as an event log and can be used to help diagnose a system failure.

Download Current Config Option

The **Download current config** option in the **UTILITIES** menu allows you to download configuration parameters.

If you select **Download current config**, the console computer downloads the following parameters from the computer memory to the controller:

- Beamstopper configuration
- Console gun delay (high energy accelerators only)
- Klystron delay (high energy accelerators only)
- PRF frequency (UNIQUE accelerators only)
- Motor control parameters
- Dynamic Beam Delivery (DBD) parameters
- Customer minor interlock configuration
- Position readout (PRO) scale configuration
- Machine configuration
- Energy table
- Motor configuration parameters



CAUTION: Be sure to save this configuration, because the system discards it when you exit Service mode.

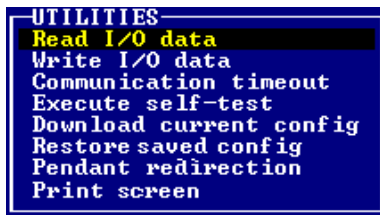
See [Write Data to Media Option: Save Service Mode Data](#) on page 163.

Download the Current Configuration Parameters

To download the current configuration parameters:

1. Select the **UTILS** menu heading and press **Enter**.

The UTILITIES menu appears.



2. Select **Download current config** and press **Enter**.

The system downloads the current configuration data to the controller, and then closes the menu.

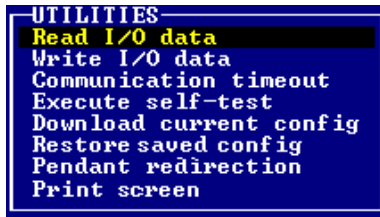
Restore Saved Config Option: Restore Original Paramters

The **Restore saved config** option in the **UTILITIES** menu allows you to restore the original (or last saved) values for configuration parameters. Selecting the **Restore saved config** option downloads the original (or last saved) configuration files from the console computer hard disk to the controller. Any data you modified during the current session is discarded. Use the **Restore saved config** function if you changed configuration parameters during a Service mode session, and now want restore the original parameters without saving the new configuration parameters.

To restore the original configuration:

1. Select the **UTILS** menu heading and press **Enter**.

The UTILITIES menu appears.



2. Select **Restore saved config** and press **Enter**.

The previously saved configuration is restored and the menu closes.

Pendant Redirection Option: Redirect Pendant Keys

The **Pendant redirection** function in the **UTILITIES** menu allows you to activate certain keys on the pendant so they perform keyboard functions. Then, you can run some Service mode features from the treatment room, such as overriding various interlocks.

When pendant redirection is enabled, you cannot run features that require numeric entries nor turn on the beam.

To turn off the pendant redirection function, press any key on the alphanumeric service keyboard.

[Table 24](#) lists each keyboard key (on the alphanumeric keyboard) that activates a corresponding pendant key.

Table 24 Keyboard and Pendant Corollary Keys

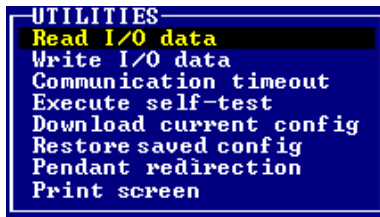
Keyboard Key	Pendant Keys
Right arrow	▶ ◀
Left arrow	Shift + ▶ ◀
Down arrow	▼ ▲
Up arrow	Shift + ▼ ▲
Enter	Enter
End	Shift + Enter
Esc	Clear

Turn On Pendant Redirection

To turn on pendant control:

1. Select the **UTILS** menu heading and press **Enter**.

The UTILITIES menu appears.



2. Select **Pendant redirection** and press **Enter**.

The screen closes and the pendant keys are activated.

3. On the pendant, use the arrow keys (or **Shift** + arrow) to navigate to desired selections.
4. Perform the desired function using the pendant.

Pendant or keyboard instructions appear.

(See [Example of Pendant Operation](#) on page 181.)

5. Follow the instructions that appear.

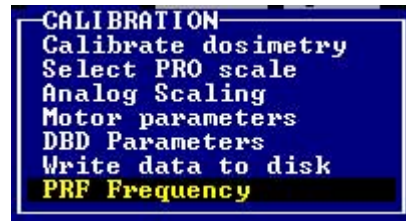
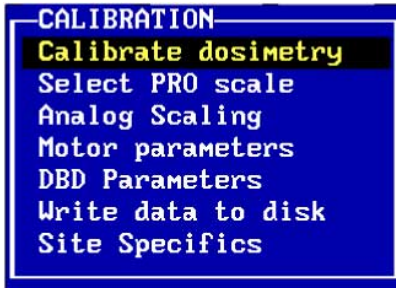
6. To close the menu, press **Enter** or **Esc**.

Example of Pendant Operation

This is an example of using pendant operation to check the calibration of a motor axis. Turn on the pendant control and follow these steps using the pendant:

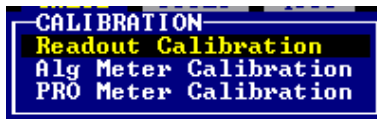
1. Select the **CALIB** menu heading and press **Enter**.

The CALIBRATION menu appears. The High Energy model CALIBRATION menu is on the left and the UNIQUE CALIBRATION menu is on the right.



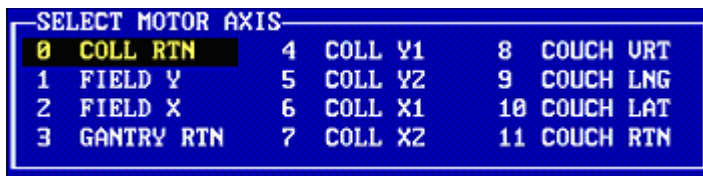
2. Select **Analog Scaling** and press **Enter**.

The second CALIBRATION appears.



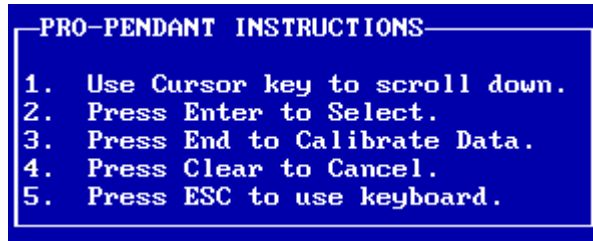
3. Select **Readout Calibration** and press **Enter**.

The SELECT MOTOR AXIS menu appears.



4. Select the motor axis you want to check and press **Enter**.

Pendant instructions appear, with the captured and saved values for the motor axis.



5. Move the motor axis to each capture position listed and measure its alignment.



Note: When the selected motor axis physically reaches the true capture position, use the down arrow key on the pendant to select the corresponding position on the screen.

6. Press **Enter** to capture the value.

The value appears as an unscaled number.

7. Repeat these steps for all positions.
8. After you capture values for all the positions, press the **Shift + Enter** on the pendant.

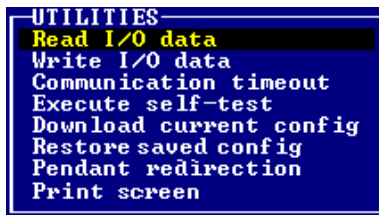
Print Screen Option

The **Print screen** option in the **UTILITIES** menu allows you to create a text version of the active screen. This file goes to print directory in Windows.

To turn on printing:

1. Select the **UTILS** menu heading and press **Enter**.

The UTILITIES menu appears.



2. Select **Print screen** and press **Enter**.

The Print screen menu closes and the system sends a text version of the screen to the print directory.

Chapter 8 COMMUNICATIONS Mode

COMMUNICATIONS Mode

Functions formerly in COMMUNICATIONS mode have been moved to the LogViewer application. These functions include:

- Viewing and printing Clinac logs
- Viewing and printing screenshots
- Backing up and restoring logs and machine configuration files



Note: *All functions except printing are available the 4DITC workstation running LogViewer. Printing is available on a customer workstation running LogViewer and connected to a printer.*

Chapter 9 LogViewer: Logs, Screenshots, and Backup/Restore

About LogViewer

LogViewer is an application that allows you to:

- View, sort, and back up Clinac logs and screenshots.

Logs and screenshots are saved in the 4DITC workstation. If LogViewer is installed on a networked workstation, then logs and screenshots can also be backed up and viewed there.

- Back up and restore configuration data.

this can only be done on the 4DITC workstation.

- Print logs and screenshots.

This can only be done on a customer-provided workstation with LogViewer and a printer installed. Log files and screenshots are mirrored from the 4DITC workstation.

Clinac logs can be used for diagnostic purposes and record machine parameters. Log types include:

- Morning Checkout

Records parameters and activity for each treatment during Morning Checkout.

- Clinac Dynalogs

Records parameters and statistics for dynamic treatments.

- Clinac Event Logs

Provides information on errors, interlocks, and operating statistics that can be used for diagnostic purposes.

Other types of logs are available for diagnostic purposes, and are accessible only by Varian service personnel.

The Backup and Restore tab allows you to back up log files, and back up and restore the machine configuration files. You can only do this from the 4DITC workstation.

LogViewer Configuration

LogViewer is configured by installing it with associated log folders on the 4D Integrated Treatment Console (4DITC) workstation. LogViewer and LogViewer folders can also be installed and viewed on a customer workstation on the network.

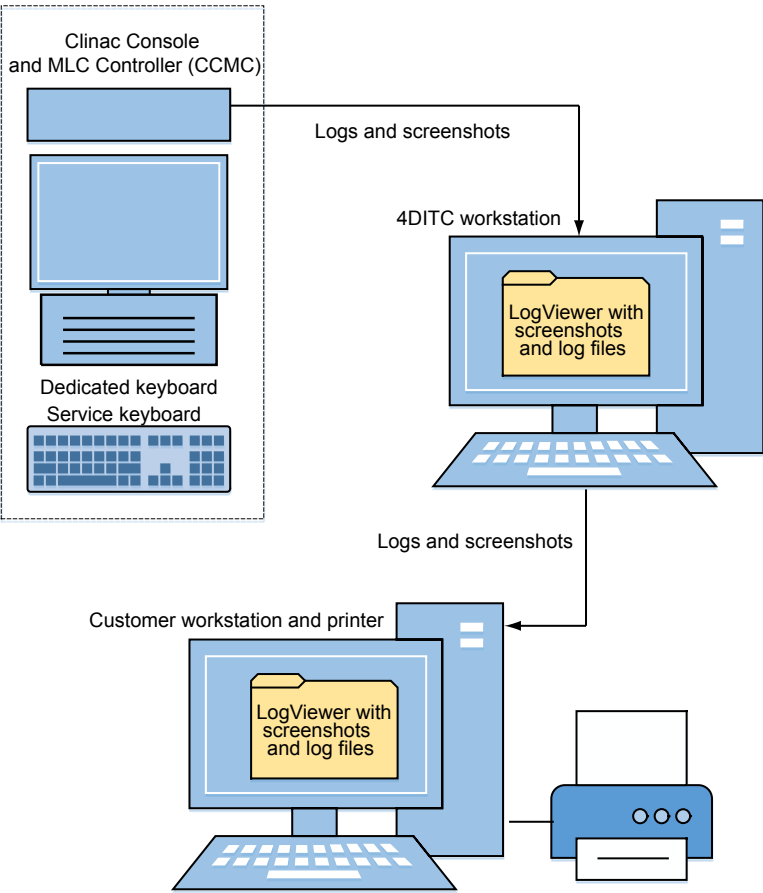


Figure 11 LogViewer System Diagram

Not all LogViewer capabilities are available on all workstations.

To do this...	Use 4DITC workstation	Use Customer workstation
View logs and screenshots	Yes	Yes
Print logs and screenshots	No	Yes, if connected to a printer

To do this...	Use 4DITC workstation	Use Customer workstation
Back up logs	Yes	Yes
Back up and restore machine configuration	Yes	No

LogViewer Quick Start

Date/Time and Type List

LogViewer log tabs display a Date/Time and Type list that lists the logs for that log type. This list allows you to select logs for viewing and right-click individual logs to filter, refresh, or print them.

- Sort logs by date and time (click **Date/Time**)
- Sort logs according to type (click **Type**)
- Filter, refresh, or print logs (right-click the log and choose an option).

The Filter option opens the **Select the Filter** dialog box where you can select the type and time range for logs you want to view.



Note: If LogViewer is open, new logs will not be displayed until the list is refreshed.

Quick Start

Table 25 LogViewer Quick Start

Task	Procedure
View and sort a log	On either the 4DITC workstation, or an optional networked customer workstation with LogViewer: 1. To view, open the appropriate log tab. For Event logs you can select sub tabs to view specific logs. 2. Sort the list according to date and time by clicking Date/Time, or sort by Type by clicking Type.
Refresh the log list	If LogViewer is open, new logs will not be displayed until the list is refreshed. To refresh the list, do one of the following: ■ At the workstation, press the F5 function key ■ Select View > Refresh ■ Close and reopen LogViewer.

Task	Procedure
Filter a log	On either the 4DITC or optional networked customer workstation with LogViewer: 1. Right-click the log in the Date/Time list and select Filter, or select View > Filter <log type>. 2. Select filter parameters for the log you are viewing: ■ Select individual items from the Select Event Types to filter to view logs according to event type. If you do not select or deselect individual event types, all logs are selected by default. If you want to change your selections you can click Select All to select all types of logs, or click Select None to clear all selections. ■ Select a date range to limit your search to a time period you specify. In the Start Date and End Date fields, you can select dates by clicking the calendar icon, or use the up and down buttons to select the Start Date and End Date. You can navigate by months as well as years.
Print a log or report	Note: Logs and screenshots can only be printed on a workstation with LogViewer and a printer installed. To print, do one of the following: ■ Right-click the log or screenshot in LogViewer and select Print. ■ Select the log or screenshot and select File > Print. ■ Open the Print Files tab, right-click the log or screenshot, and select Print. Note: You can print reports for Morning Checkout logs.
Capture screenshots of the Clinac console screen and save them for printing in LogViewer	To take a screenshot: 1. At the console workstation, in the console screen you want to capture, click the print screen button on the Service keyboard. 2. Save the screenshot in the Print folder for printing in LogViewer: Right-click it in the list and select Print. 3. To view the screenshot: In a workstation with LogViewer installed and configured (such as the customer workstation), open the Print Files tab, sort the list according to Type, and select the screenshot from the list.

Task	Procedure
Back up event logs	Logs can be backed up from either the 4DITC or customer workstation with LogViewer installed. 1. Open the Backup and Restore tab. 2. Under Log Files, select the date range. The file size for the files you selected, and date of last backup, are displayed. 3. Click Backup Log Files. 4. Navigate to the location where you want to store the backup files (either file location or external media). 5. Click Save. Files are saved in a .ZIP file.
Backup or restore configuration files	Configuration files can be backed up and restored on the 4DITC workstation only. 1. Open the Backup and Restore tab. Below the Backup and Restore options, the date of the last backup is displayed. 2. Under Machine Configuration, choose an option: ■ To back up the machine configuration files, click Backup Configuration. Navigate to the location where you want to keep the backup files, and click Save. Files are saved in a .ZIP file. ■ To restore machine configuration files, click Restore Configuration. Navigate to the location of the backup configuration files (.ZIP file) and initiate restore by clicking Open. The Restore function uses the .ZIP file to restore machine configuration.
Change the root directory/locations for the LogViewer program files or log files	■ To change the root directory for the LogViewer program files: 1. Click File > Change Root Directory. 2. Navigate to the location where you want to keep the program files, and click OK. ■ To change the root directory for the LogViewer logs: 1. Click File > Change (name of log tab) Tab View Directory. 2. Click the Browse button to navigate to the location where you want to keep the program files, and click OK.

Task	Procedure
Set the startup path LogViewer will use to access files on the customer workstation	Upon installation, LogViewer is configured with the path to access files on the 4DITC workstation. For the LogViewer installation on the customer workstation, you must set up this path. This is a one-time procedure. To set the path for LogViewer on the customer workstation: On the customer workstation: 1. Select File > Set Startup Path. 2. Deselect the Use default Directories option. (This option is selected only for LogViewer running on the 4DITC workstation.) 3. Enter the startup directory address, or click Browse and navigate to the desired location. 4. Click OK.

Clinac Morning Checkout Logs

Morning Checkout logs record the results of treatments performed daily during Morning Checkout. After each Morning Checkout, a Morning Checkout log is saved automatically in a Morning Checkout folder on the 4DITC workstation.

In the LogViewer Clinac Morning Checkout tab, you can:

- View Morning Checkout and Dynamic Treatment (DT) Summary logs
- Sort the logs according to date and time or type
- Filter the logs
- Print logs and log reports.

When you click Print in LogViewer in the customer workstation, or select the Print option in MORNING CHECKOUT mode, the log is also copied to the Print folder.

View and Filter Clinac Morning Checkout Logs

The Clinac Morning Checkout tab opens by default in the MC log tab. To view Dynamic Treatment Summary statistics for Morning Checkout, click the DT Summary tab.

1. To sort the list, click **Date/Time** at the top of the log list.

The list toggles between ascending or descending date and time.

2. To view a log, click to select it from the **Date/Time** list.

3. To filter the logs, do one of the following:
 - Right-click in the list, and select **Filter**.
 - In the **View** menu, click **Filter 'Clinac Morning Checkout'**.

The **Select the filter for Morning Checkout** dialog box appears. In **Select the Event Types to filter**, Morning Checkout is the only choice and is selected by default. Changing this selection invalidates the filter.

4. Select a date range to limit your search to a time period you specify. In the **Start Date** and **End Date** fields, you can select dates by clicking the calendar icon, or use the up and down arrows.
5. Click **OK**.

Print Morning Checkout Logs and Reports

You can print individual Morning Checkout logs, or a report of multiple logs.



Note: To print files, you must use a customer-provided workstation connected to a printer.

1. To print an individual log, do one of the following:
 - Right-click the log in the **Date/Time** list and select **Print**.
 - Select the log from the **Date/Time** list, and select **File > Print**.
 - Open the **Print Files** tab, right-click the log in the **Date/Time** list, and select **Print**.

The Print dialog box appears.

2. To filter the logs you want to print, choose one of the following: In the **View** menu, click **Filter 'Clinac Morning Checkout'**.

The **Select the filter for Morning Checkout** dialog box appears. In **Select the Event Types to filter**, Morning Checkout is the only choice and is selected by default. Changing this selection invalidates the filter.

3. Select a date range to limit your search to a time period you specify. In the **Start Date** and **End Date** fields, you can select dates by clicking the calendar icon, or use the up and down arrows.
4. Click **OK**.

The filter is applied to the list.

5. Open the **Print Files** tab, right-click the log in the **Date/Time** list, and select **Print**.

Clinac Dynalog Files

Dynamic beam delivery (dynamic treatment) logs displayed in the Clinac Dynalogs tab are records of the machine settings and actual values after an Arc or Enhanced Dynamic Wedge (EDW) treatment. They are created after delivering the treatment in CLINICAL mode or while performing the Morning Checkout procedure.

Each Dynalog contains machine identification, machine serial number, treatment setup information, and dynamic beam statistics; each log is dated and time stamped when it was created.

TREATMENT SETUP

The TREATMENT SETUP part of the log identifies:

- Treatment type
- Energy
- Monitor
- Units delivered
- Slots status

It also lists information related to EDW or Dynamic Arc treatments, depending on the type of treatment.

DYNAMIC BEAM STATISTICS

The DYNAMIC BEAM STATISTICS part of the log lists:

- Total dose delivered
- Standard deviation for the dose
- Dose-position standard deviation
- Number of samples for the treatment

The STT (Segmented treatment table) and SEGMENT BOUNDARY SAMPLES (ACTUAL) sections display dose and gantry angle for each instance, or segment, of the treatment.

View, Filter, and Print Clinac Dynalogs

To view Clinac Dynalogs, click the Clinac Dynalogs tab.

1. To sort the list, do one of the following:
 - To sort according to date and time, click **Date/Time** at the top of the list.
The list toggles between ascending or descending date and time.
 - To sort according to treatment type, click **Treatment Type**.
The list sorts into blocks of treatments, and can be toggled in alphabetical ascending or descending order.
2. To view a log, click to select it from the list.
3. To filter the logs, do one of the following:
 - Right-click the log in the list, and select **Filter**.
 - In the **View** menu, click **Filter 'Clinac Dynalogs'**.
4. Choose the filter method:
 - Select **Filter by Log Type** and select an event type. You can select all event types by clicking **Select All**, or deselect your selections by clicking **Select None**.
Click **OK**. The list of logs is filtered.
 - To view logs by a date range, select **Filter By Date Range** and enter the **Start Date** and **End Date**, then click **OK**. The Date/Time list is narrowed to the date range you selected.
5. To print a Clinac Dynalog, do one of the following:
 - Right-click the log in the Date/Time list and select **Print**.
 - Select **File > Print**.
 - Open the Print tab, right-click the log in the Date/Time list, and select **Print**.



Note: Logs can only be printed using a customer-provided workstation connected to a printer.

6. In the Print window, make your print selections and click **OK**.

Clinac Event Logs

Event logs displayed in the Clinac Event Logs tab contain details about the status of the accelerator at the time when an event, such as an interlock, occurred. The recorded data can help determine the cause of the problem.

Settings and Status Tab

The Settings and Status tab shows actual data at the time of the event, such as monitor units 1 and 2, time, dose rate (calculated from time and dose), and accessory details.

- Event Status

The Event Status section displays technical details generated by the accelerator including MU, Time, and accessory status. A second set of machine status entries also appears if the event was generated at the console.

- Settings

The Settings section shows the set values the operator entered for the treatment during which the event occurred. These settings include mode, energy, and MU.

- PRO and SRO

The PRO and SRO columns show the position of each axis at the time of the event. The PRO column lists the primary position readouts. The SPRO column lists the secondary position readouts.

- STT Summary

An STT, or dynamic beam summary, tab appears only if a dynamic treatment was being performed when an event occurred. The information appears in the same format as the morning checkout dynamic summaries, showing set, actual doses, and positions over the course of the treatment.

Interlocks Tab

The interlock status tab shows the state of the interlocks at the time of an event. Asterisks appear in columns to indicate the state of associated interlocks at the time of an event.

When a CTRL, HWFA, or MOTN interlock causes an event, the interlock and an abbreviated status message appear at the lower left.

Table 26 Columns on the Interlock Status Tab

Column	Description
DS (Disabled)	Prevents the hardware/software (HS) interlock from becoming active. Can be set by the control system to disable interlocks as a function of mode, such as to disable the IPSN interlock for a fixed treatment. Can also be set manually in Service mode.
HS	Interlock input from an outside source. Can be hardware-processed input, such as the GAS interlock, or software-processed input, such as the CNF interlock.
AC (Active)	Interlock that is processed to inhibit beam. Results from HS interlocks that are not disabled.
LA (Latched)	Record of an interlock that was active during a certain event, such as beam-on. Used to see transient condition interlocks that are otherwise self-clearing, or those that could be reset by subsequent events.

Analog Meter Tab

The Analog Meter tab shows the values of the analog machine and position meters at the time of an event. The first two columns on the page display the scaled meter values for low-voltage power supply voltage, for steering coil current, and for beam symmetry. The next two columns display meter values for process parameters from a variety of sources in the accelerator.

The position meters section in the lower section of the window displays:

- PRO potentiometer wiper voltage
- Motor current readings from all ten motor axes
- Secondary PRO values for the gantry and the upper collimator jaws
- Readings for the negative and positive power supply values for the gantry, collimator, and couch PROs

Digital I/O Tab

The Digital I/O tab of the event log is organized into two main areas: Digital input in the upper section of the window, and digital output in the lower section of the window. This window displays the status of the digital ports. The bits that were active at the time of an event are selected in black. An asterisk appears after the active bits to identify them on a printout. Each column displays the 8 bits of digital I/O data at a port (one byte). The eight bits are arranged from bottom (b0) to top (b7).

Each code or number in a column represents a bit of the port data byte. For example: BO_Enable on the input board is bit 4 of I/O port 10 and BP_LASER is bit 7 of I/O port 18 on the output board. The bit number itself appears in the column if the bit has not been assigned to a function. For example, the unused bit 7 of I/O port 1C on the output board is labeled 7.

The port on the far left of the input display handles output bits. The port on the far right of the output display handles input bits.

Accessory Tab

The Accessory tab displays the bit status associated with the four accessory slots and the accessory controller status. This tab also shows certain errors associated with the slots and accessory controller.

The tab is organized into two main areas: Accessory Data, and Accessory Error Data. Each area lists five columns with the following headings:

- IM = Interface Mount
- AM = Accessory Mount
- EA = Electron Applicator
- CM = Compensator Mount
- CTRL = Accessory Controller

Each column shows specific information about the status or error condition of an accessory.

Various Errors Tab

The Various Errors tab displays more information about the status of the CTRL, HWFA, or MOTN interlocks that cause an event.

STT Summary Tab

A, STT, or dynamic beam, summary appears only if a dynamic treatment was being performed when an event occurred. The information appears in the same format as the morning checkout dynamic summaries, showing set, actual doses, and positions over the course of the treatment.

View, Filter, and Print Clinac Event Logs

To view Clinac Event logs, click the Clinac Event Logs tab, then click the appropriate log sub tab.

1. To sort the list, do one of the following:
 - To sort according to date and time, click **Date/Time** at the top of the list.
The list toggles between ascending or descending date and time.
 - To sort according to treatment type, click **Type**.
The list sorts into blocks of treatments, and can be toggled in alphabetical ascending or descending order.
2. To view a log, click to select it from the list.
3. To filter the logs, do one of the following:
 - Right-click the log in the list, and select **Filter**.
 - In the **View** menu, click **Filter 'Clinac Event Logs'**.
4. Choose the filter method:
 - Select **Filter by Log Type** and select an event type. You can select all event types by clicking **Select All**, or deselect your selections by clicking **Select None**.
Click **OK**. The list of logs is filtered.
 - To view logs by a date range, select **Filter By Date Range** and enter the **Start Date** and **End Date**, then click **OK**. The Date/Time list is narrowed to the date range you selected.
5. To print a Clinac Event Log, do one of the following:
 - Right-click the log in the Date/Time list and select **Print**.
 - Select **File > Print**.
6. In the Print window, make your print selections and click **OK**.

View and Print Clinac Screenshots

You can save screenshots of Clinac screens by clicking the **Prtsc** key in any screen on your Varian Service keyboard. This action saves the screenshot in the Print folder.



Note: Logs can only be printed using a customer-provided workstation connected to a printer.

The following steps describe how to view and print screenshots in LogViewer.

1. To sort the list, do one of the following:
 - To sort according to date and time, click **Date/Time** at the top of the list.
The list toggles between ascending or descending date and time.
 - To isolate the list of screenshots, click **Type**.
The list sorts into blocks of treatments and screenshots, and can be toggled in alphabetical ascending or descending order.
2. To view a screenshot, click to select it from the list.
3. To filter the logs, do one of the following:
 - Right-click the log in the list, and select **Filter**.
 - In the **View** menu, click **Filter**.
4. Choose the filter method:
 - Select **Filter by Log Type** and select an event type. You can select all event types by clicking **Select All**, or deselect your selections by clicking **Select None**.
Click **OK**. The list of logs is filtered.
 - To view screenshots by a date range, select **Filter By Date Range** and enter the **Start Date** and **End Date**, then click **OK**. The Date/Time list is narrowed to the date range you selected.
5. To print, in LogViewer on the customer workstation, do one of the following:
 - Right-click the screenshot in the Date/Time list and select **Print**.
 - Select **File > Print**.
6. In the Print window, make your print selections and click **OK**.

Print Physics Mode Configuration Data

You can print a summary report of Physics mode configuration data in LogViewer. The report includes the morning checkout treatments and a list of commissioned Enhanced Dynamic Wedge treatments.



Note: *The report shows the data in memory at the time you requested the printout. If you did not save the data that was printed before you exit Physics mode, the printout will be inconsistent with current data.*

1. In the Physics mode **SELECT OPTION** menu, use the dedicated keyboard to select **CONFIGURATION** and press **ENTER**.
The **SELECT CONFIGURATION** menu appears.
2. To send the information to the print directory, press **F6**.
A message appears asking you whether passwords should be printed with the configuration data.
3. Choose whether or not to print the password with the configuration data.
4. Press **ENTER**.

The **SELECT CONFIGURATION** menu reappears.

The physics configuration information is saved as a report in the Print directory.

5. On the customer workstation, open **LogViewer** and open the **Print Files** tab.
(Printing can only be done in LogViewer on the customer workstation.)
6. Select the physics report in the list of files.
7. To print, do one of the following:
 - Right-click the report in the list and select **Print**.
 - Select **File > Print**.

The printer window opens.

8. Make your selections and click **OK**.

Back Up Logs

The Backup and Restore tab allows you to back up log files on the 4DITC workstation, customer workstation, or external media.

1. Open the **Backup and Restore** tab.
Under **Log Files**, the date and file location of the last backup is displayed at the bottom of the Log Files section.
2. Select a date range for the backup.
The size of the files you have selected is displayed.

3. Click **Backup Log Files**.
4. Navigate to the location where you want to store the backup files. You can use a location of the workstation or external media.
5. Click **Save**.

Back Up or Restore Machine Configuration Files

The Backup and Restore tab allows you to back up or restore the machine configuration from the 4DITC workstation only. Backup configuration files can be created in the root directory on the 4DITC workstation hard drive, or on external media.

1. Open the **Backup and Restore** tab.
2. Choose an option:
 - To back up machine configuration files:
 1. In **Machine Configuration**, click **Backup Configuration**.
 2. Use the Browse button to navigate to the location where you want to keep the backup files.
 3. Click **Save**.
 - To restore machine configuration files:
 1. In **Machine Configuration**, click **Restore Configuration**.
 2. Navigate to the location of the backup configuration files.
 3. Initiate restore by clicking **Open**.

Appendix A Hand Pendant Diagnostic Tests and Error Messages

Hand Pendant Overview

During normal operation, the hand pendant performs self-diagnostic tests in the background and displays an error message if it is not operating correctly. You can run diagnostic tests manually to check pendant operation, for example, if the pendant is dropped.

This section describes:

- Three types of hand pendant diagnostic tests
- How to perform hand pendant diagnostic tests
- How to resolve hand pendant error messages

Warning Regarding Hand Pendant Safety



WARNING: To prevent unplanned motion of equipment that could cause collisions and injuries, observe the following hand pendant safety guidelines:

- If a pendant fails any test and you are unable to quickly clear the fault, discontinue using the pendant until it is repaired and passes all tests. Use the other pendant or the couch side panel controls.
- It is important to run the power-up diagnostic tests whenever the pendant is dropped, to make sure that it has not suffered serious damage. For example, if the motion enable bars are damaged, releasing them may not stop accelerator or couch motions.
- When picking up a dropped pendant, do not press any buttons, and do not press the motion enable bars. Test the pendant to make sure it is working properly before using the pendant to operate the equipment.

For more information about safety precautions, refer to the *Clinac Safety Guide*.

Hand Pendant Diagnostic

The hand pendant runs three types of self-diagnostic tests: Power-up diagnostic tests, runtime diagnostic tests, and self-tests. The pendant does not allow motions during these tests.

If at any time the hand pendant does not pass a diagnostic test:

- Pendant LED displays an error code
- All pendant keys are disabled except for the **CLEAR** key
- Pendant does not function until the issue is resolved

After the issue is resolved and the user presses the **CLEAR** key, the pendant clears the LED display and repeats the self-diagnostic tests.

After the self-diagnostics are completed successfully, the LED displays the message *Varian4* or *Varian6* (depending on the pendant configuration). Then the LED clears the message, and normal operation can proceed.

Power-Up Diagnostic Tests

The pendant automatically runs a series of power-up diagnostic tests on itself each time the pendant receives power. You can also initiate the power-up diagnostic tests by pressing the **CLEAR** or **CLEAR ALL** keys.

During the tests (which last less than 30 seconds and may take less than a second), the pendant prevents itself from sending control signals to the accelerator. The tests check pendant memory, motion enable bars, and thumbwheels.

When power-up diagnostic testing begins, the display clears. If an error occurs, the pendant displays an error message and all control labels turn off. The **CLEAR** key remains active. In this case, the only available options are to:

- Press **CLEAR** to restart the power-up diagnostics. If these finish with no error detected, the pendant displays *Varian4* or *Varian6* for 5 to 10 seconds. The display then clears and the pendant is ready for normal operation.
- Run the pendant self-tests (see [Pendant Self-Tests](#) on page 203). When you exit the self-tests, the power-up diagnostic tests begin.



Note: To exit a test, press **CLEAR**.

[Table 27](#) lists the power-up diagnostic tests, and what happens if each test fails or succeeds.

Table 27 Hand Pendant Power-Up Diagnostic Tests

Test	If this test fails	If this test succeeds
Memory	The pendant displays ERROR M.	The next test runs.
Motion enable bars	The pendant displays ERROR E.	The next test runs.

Test	If this test fails	If this test succeeds
Thumbwheel	The pendant displays ERROR nT, where n is a number from 1 to 6 that indicates the position of the first failed thumbwheel relative to the top of the pendant. The number 1 represents the position closest to the display, and the number 6 represents the position farthest from the display. For example, if the third and fifth thumbwheels from the top fail, the pendant displays ERROR 3T.	The pendant displays Varian4 or Varian6 for 5 to 10 seconds. Then the display clears and the pendant functions normally.

Runtime Diagnostic Tests

While you are using the hand pendant, it is running diagnostic tests in the background. [Table 28](#) lists the tests, what can cause a test to fail, and what happens if a test does fail.

Table 28 Hand Pendant Runtime Diagnostic Tests

Test	What causes this test to fail	If this test fails
Single thumbwheel	Any thumbwheel remaining uncentered for more than about 2 seconds without both enable bars being pressed.	All motion commands to the accelerator cease and both labels associated with the uncentered thumbwheel flash. You must center the thumbwheel to re-enable the pendant.
Multiple thumbwheels	Impaired electrical conduction to the thumbwheels, possibly involving the pendant itself.	All motion commands to the accelerator cease and the hand pendant LED displays the message ERROR 7T. If electrical failure has occurred, the error may appear as ERRO or n.
Motion enable bars	A failure in the enable bar mechanism.	All motion commands to the accelerator cease and power-up diagnostic tests begin (see Power-Up Diagnostic Tests on page 202).

Pendant Self-Tests

The pendant self-tests allow you to manually check various pendant functions. [Table 29](#) describes how to start each test and what the result should be.



Note: Pendant self-testing stops and power-up diagnostic tests begin if:

- You do not press a key for two to four minutes.
- You press and release the **CLEAR** key.

Table 29 Hand Pendant Runtime Diagnostic Tests

Function	Press these keys	Expected result
Key test	Enter + SSD	The pendant displays the name of any key you press. This test terminates if you do not press a key for 5 to 10 seconds.
LED test	Enter + EDW	All display LEDs and key names light.
Motion enable bars test	Enter + Field	The pendant displays: E : - - - - - Pressing the left enable bar should cause the pendant to display: E : - - L - - - Pressing the right enable bar should cause the pendant to display: E : - - - - - Pressing both enable bars should cause the pendant to display: E : - - L - R - -
Single Thumbwheel test	Enter + Room	The pendant should display: A : - - - - - when all the thumbwheels are exactly centered. (Each dash from left to right in the display corresponds to a thumbwheel from top to bottom of the pendant.) A, -, or a are all acceptable indicators of a centered thumbwheel. Moving a single thumbwheel to the right should display consecutive uppercase letters in the corresponding position in the display. Moving the thumbwheel to the left should display consecutive lowercase letters.

Function	Press these keys	Expected result
Multiple thumbwheels test	Move five thumbwheels out of center position (more than 20% deflection from home position)	The pendant should display: ERROR 7T
Software version display	Enter + TBS	The pendant displays the version number of the pendant microprocessor program.

Troubleshoot Hand Pendant Error Messages

Table 30 describes hand pendant error messages and actions you can take to resolve them.

Table 30 Pendant LED Messages

Message	Interpretation	Resolution
ERROR M	Problem with pendant memory.	Press the CLEAR key to re-initialize the pendant. The pendant will retest itself. If the pendant does not pass, contact your Varian service representative.
ERROR E	Motion Enable is selected.	Make sure no Motion Enable bars are pressed. Then press the CLEAR key to re-initialize the pendant. The pendant will retest itself. If the pendant does not pass, contact your Varian service representative.
ERROR nT (where n = 1 to 6)	A thumbwheel error has occurred. n indicates which thumbwheel is involved.	Press the CLEAR key to re-initialize the pendant. The pendant will retest itself. If the pendant does not pass, call your Varian service representative.
ERROR 7T	- Five or more thumbwheels are deflected - There is an electrical error involving the pendant.	The pendant cable may need to be replaced.

Message	Interpretation	Resolution
ERRO n	If the electrical error involves the pendant display, the error code may be cut off, or appear as a n.	This error condition will not clear until the cause is resolved and the user presses CLEAR on the pendant.  Note: When any form of display problem occurs on the pendant, the pendant should be serviced by a Varian service representative.
ERROR 9T	Appears if the thumbwheel voltage reference line or floating ground line are malfunctioning while the motion bars are enabled.	Press the CLEAR key to clear the pendant error code. If the error does not clear, contact your Varian service representative.
ERROR AT	Appears if any thumbwheel produces voltage outside an allowable range. This error will only be displayed if the thumbwheel is moved while the motion enable bars are pressed.	Press the CLEAR key to clear the pendant error code. If the error does not clear, contact your Varian service representative.

Appendix B Troubleshooting Interlocks

Interlocks



WARNING: Injury, death, patient mistreatment, or equipment damage may result from clearing an interlock before taking appropriate corrective action.

Follow these precautions:

- Before clearing an interlock, be sure to understand the possible causes of the interlock and consequences of clearing it. See [Clinac Interlock Codes](#) on page 208.
- If an interlock repeats or persists, contact Varian Service.
- Never use Service mode to override an interlock for clinical use of the machine.

An interlock alerts you to conditions that can cause minor or moderate injury or damage to equipment. The alert appears as a three- or four-character code in a yellow box in the center of the screen.



There are three types of interlocks:

- *Major*—Most major interlocks indicate conditions that, if not corrected, can seriously damage the linear accelerator. However, the KFIL or MFIL interlock is a major interlock that appears as part of the normal Standby state. (KFIL appears on High Energy accelerators and MFIL appears on UNIQUE accelerators.)
- *Dosimetry*—Prior to beam-on, the beam passes through the ion chamber, where it is measured for flatness, symmetry, and other qualities. Dosimetry interlocks are usually caused by transient variations in the beam, and a second attempt to deliver the treatment should succeed. If a dosimetry interlock occurs, you must type a password to continue.
- *Minor*—Minor interlocks involve conditions that can be easily corrected; for example, DOOR (treatment room door is open) or ACC (the accessory programmed for the treatment does not match the installed accessory).

Clear Interlocks

Varian Medical Systems recommends that you keep a daily log dedicated to each linear accelerator. If an interlock occurs:

1. Record:

- Date and time
- Type of interlock
- State of the system
- Any other observations

2. Record the monitor units delivered.

If the interlock appears during beam-on, you may need this information to perform a partial treatment.

3. Look up the interlock code to determine the cause of the interlock.

See [Clinac Interlock Codes](#) on page 208.

The interlock message area is limited to eight interlocks; if more than eight interlocks are active, as you clear one interlock, another active interlock appears at the end of the list. In some cases, a message appears telling you how to clear the interlock.

4. If appropriate, remove the patient from the treatment room.

Resolve an Interlock Activated During Beam On

If one or more interlocks are activated during beam-on, the beam shuts off, and motions stop. The interlock responsible for beam-off appears in the list of active interlocks.

To resolve an interlock during beam-on:

1. Note the amount of MU remaining for the treatment.
2. Follow the onscreen prompts to reprogram the treatment.
3. Perform the remainder of the treatment as a partial treatment if the treatment is a dynamic wedge or arc treatment.

If the treatment is not a dynamic treatment, program a new treatment using the remaining MU only.

Clinac Interlock Codes

Clinac interlocks fall into the general categories of minor, dosimetry, or major interlocks. [Table 31 Interlocks, Conditions to Correct, and Actions](#) on page 209 lists interlock codes alphabetically, with causes and the actions that may correct them.

Some corrections must be performed by qualified personnel.

In the case of a Clinac running with an RV system, a list of active interlocks appears on the in-room monitor, in addition to the console monitor. EXCI supports the verification and recording of a treatment, and also supports the features of Auto Mode-up. This includes sending a list of active interlocks to the in-room monitor.

Interlocks that apply only to High Energy models follow the code with a suffix of (HE); interlocks that apply only to UNIQUE models follow the code with a suffix code of (LE).

Table 31 Interlocks, Conditions to Correct, and Actions

Code	Type	Condition Requiring Correction	Action
ACC	Minor	There is a problem with the installed accessory combination.	
 WARNING: An improperly mounted accessory could fall on a patient and cause serious injury or death. To prevent harm to the patient when using accessories, follow safety guidelines and precautions and your department's protocols.			
		Selected accessory does not match the mounted accessory.	Select the correct accessory.
		Accessory is improperly positioned. A loose accessory can fall out of the accessory slot.	Reposition the accessory.
		Red Reset button on the collimator lights up and machine movements are locked; no interlock message appears. On High Energy accelerators only: The electron applicator touch guard may have been tripped.	Clear the path of the electron applicator and press the red Reset button.
AIR (HE)	Minor	Air pressure is below 30 psig limit. (This condition may be temporary.)	Try the treatment again: 1. Press Setup. 2. Reprogram the treatment and attempt beam-on.

Code	Type	Condition Requiring Correction	Action
BMAG (HE)	Minor	■ Voltage for bending magnet power supply exceeds the set value limit. The allowable range is $\pm 5\%$ ■ Current is too high or low for the energy mode.	Check the water temperature. Try the treatment again: 1. Press Setup. 2. Reprogram the treatment and attempt beam-on.
CARR (HE)	Minor	■ Carousel is not in position for the selected mode. ■ Communication failed between the console computer and carousel controller.  Note: <i>The CARR interlock cannot be cleared in NO_MODE.</i>	Try the treatment again: 1. Press Setup. 2. Reprogram the treatment and attempt beam-on.
CDOS	Minor	External customer-defined dosimetry interlock has tripped (The interlock was not asserted during the check cycle.) The CDOS interlock is connected to PortalVision, Yield Monitor, and third-party systems.	Try the treatment again: 1. Press Setup. 2. Reprogram the treatment and attempt beam-on.

Code	Type	Condition Requiring Correction	Action
CKFA	Dosimetry	Dosimetry check cycle failure. The Controller activated the following interlocks during the calibration cycle, but the console did not detect one or more of them: XDP1, XDP2, XDRS, XDR1, XDR2, EXQ1/SYM1, EXQ2/SYM2, EXQT/YLD, ION1, ION2, LVPS, DS12. DOS1, DOS2, and TIME. The actual values of DOS1, DOS2, and TIME must match the set values. This interlock is activated in Clinical mode only.	Try the treatment again: 1. Enter the dosimetry password. 2. Press Setup. 3. Reprogram the treatment and attempt beam-on.
CKTO	Dosimetry	Check cycle did not complete before timeout. The Console initiates a Dose Count Check cycle and asserts the interlock if the cycle does not complete within 1.8 to 2.4 seconds after the command is received.	
CLFA	Dosimetry	Dose calibration cycle failure. Calibration failed during calibration check. At the end of the Dose Calibration cycle (in Clinical mode only), the Console Processor reads MU1, MU2, and TIME values, and checks the condition of the DS12 interlock.	
CLTO	Dosimetry	The calibration cycle took too long, and did not finish before timeout (Clinical mode only).	
CMNR	Minor	An external customer-defined minor interlock is activated.	1. Correct the condition that activated the interlock. (For example, close the extra door). 2. Proceed with the treatment.

Code	Type	Condition Requiring Correction	Action
CNF	Minor	Configuration data is missing due to one of the following conditions: ■ The console controller did not receive the required configuration data from the console computer. ■ PRO readouts are in RAW units. ■ (UNIQUE) A dynamic treatment type was selected but the Magnetron Pulse Rate Frequency was not determined.	Try the treatment again: 1. Press Setup. 2. Reprogram the treatment and attempt beam-on. 3. Choose from the following options: ■ If this interlock recurs, measure the pulse repetition frequency (PRF). ■ If measuring the PRF does not resolve the problem, have an authorized user check the position readout (PRO) settings in Physics mode.
COLL	Minor	■ The accessory combination is invalid. ■ The accessory combination is invalid for the set Treatment type or Energy. ■ The collimator jaws are positioned incorrectly (that is., not within the prescribed 2mm tolerance of the specified positions or range of positions). ■ Either set of jaws is in Symmetric mode and is not tracking within 2mm.	1. Reset the collimator jaws. 2. Reattempt the treatment.
CTRL	Major	The console control computer detected an abnormal operating condition.	Press F2 and then Enter to run the computer's power-up self-test cycle. Before the test starts, press Clear to cancel. Once the test begins, it cannot be canceled.
DOOR	Minor	Treatment room door is open.	1. Close the treatment room door. 2. Proceed with treatment.
DOS1	Minor	The number of accumulated monitor units in the primary dosimetry channel equals the value that was set for MU1.	No action. (This message indicates normal beam termination and does not usually appear in Clinical mode.)

Code	Type	Condition Requiring Correction	Action
DOS2	Dosimetry	 WARNING: If DOS2 terminates the beam, do not perform further treatments until the linear accelerator has been checked by qualified service personnel.	
			Indicates DOS1 failed. Issue with main circuit prevents completion of the dose. Number of accumulated monitor units in secondary dosimetry channel exceeds the preset dose by 10% (10% or 25MU, whichever is less, for all Clinical mode and electron arc treatments [available on High Energy accelerators only]). Do not perform further treatments. Contact authorized service personnel. (You can put the accelerator in Standby mode. It is unnecessary to power off.)
DPSN	Minor	Dose vs. Position mismatch.	
		In dynamic mode, an incorrectly positioned physical element (MLC leaf or gantry) caused the beam to terminate.	Check physical elements, and correct those that are out of position.
		For respiratory gating, gating is out-of-tolerance because the beam was held too long (more than 30 seconds for an external beam hold, or more than 15 seconds for clinical conditions).	Correct the gating condition that is holding the beam.
DS12	Dosimetry	For dynamic treatments, a discrepancy was detected when comparing: ■ dose delivered and one or more axis positions ■ primary and secondary dosimetry channels ■ two or more counters/timers that are counting the same signal	Reprogram the treatment and attempt beam-on. If the treatment was interrupted: ■ Record the gantry location and Start angle, to determine the new Start angle. ■ Reprogram the treatment and enter a partial MU value to continue from where the interlock occurred.

Code	Type	Condition Requiring Correction	Action
DSFA	Dosimetry	 WARNING: If DSFA terminates the beam, do not perform further treatments until the linear accelerator has been checked by qualified service personnel.	
		Actual dose exceeds set dose.	Do not perform further treatments. Contact authorized service personnel. (You can put the accelerator in Standby mode. It is unnecessary to power off.)
ENSW (HE)	Minor	The Accelerator energy switch is not in the correct position for the selected mode. If an AIR interlock appears at the same time, there may be a problem with the hydraulic air system that moves the energy switch.	Check the air pressure. Try the treatment again: 1. Press Setup. 2. Reprogram the treatment and attempt beam-on.
EXBH	Minor	An external beam hold occurred during an electron mode or X-ray arc mode treatment. For example, Respiratory Gating or a third party connected to the linear accelerator can produce an external beam hold.	1. Press Enter to clear the EXBH interlock. 2. Discontinue use of the external gating system during arc treatments or electron treatments (available only on High Energy accelerators). 3. Continue treatment. If the problem persists, contact authorized service personnel.

Code	Type	Condition Requiring Correction	Action
EXQ1 (HE)	Dosimetry	Radial beam symmetry is out of tolerance or instantaneous dose rate exceeds 800 MU/min per pulse. EXQ does not apply to 4MeV treatments. For 4MeV, asymmetry of < 5% terminates beam with a UDR interlock.	Try the treatment again: 1. Enter the dosimetry password. 2. Press Setup . 3. Reprogram the treatment and attempt beam-on.
EXQ2 (HE)	Dosimetry	Transverse beam symmetry is out of tolerance or instantaneous dose rate exceeds 800 MU/min per pulse. EXQ does not apply to 4MeV treatments. For 4MeV, asymmetry of < 5% terminates beam with a UDR interlock.	
EXQT (HE)	Dosimetry	Total asymmetry, total dosimetry channel charge per pulse, or average dose rate exceeds limits. EXQ does not apply to 4MeV treatments. For 4MeV, asymmetry of < 5% terminates beam with a UDR interlock.	
EXT	Minor	External record/verify. Treatment delivery software has not yet accepted the actual parameters as being within tolerance of the plan. The linear accelerator software has detected a problem in communication with the RV system.	1. Check the patient setup for discrepancies. 2. Edit or acquire the RV parameters to match the actual parameters on the Varian linear accelerator, if appropriate. 3. Check the serial connection between the console computer and the RV system.

Code	Type	Condition Requiring Correction	Action
FLOW	Major	 CAUTION: The FLOW interlock can result in overheating of the linear accelerator. If the FLOW interlock does not clear or recurs after you have attempted to correct the problem, do not continue. Power off the linear accelerator and contact authorized service personnel.	
			The cooling water flow is too slow. Insufficient water flow opened a flow switch in the stand or gantry unit. 1. Check the water tank level. 2. If it is low, fill the tank (distilled water only) to the optimum level. 3. If this does not correct the problem and the FLOW interlock recurs, turn off the linear accelerator and contact qualified service personnel.
FOIL	Minor	Carousel locking pin is not in the home position or the plunger for the energy selected is not pushed in.	1. Check the carousel locking pin and fix it if it is jammed or stuck. 2. Try the treatment again: a. Press Setup. b. Reprogram the treatment and attempt beam-on.
GAS	Minor	 WARNING: Do not adjust the main regulator knob that controls the flow of gas from the tank. The position of this knob is set by service personnel, and adjusting it can seriously damage the system.	
			■ High Energy: Waveguide dielectric gas pressure is < 30 psig. Adjust the gas or change the gas tank. ■ UNIQUE: Waveguide dielectric gas pressure is < 17 psig.

Code	Type	Condition Requiring Correction	Action
GFIL (HE)	Minor	The filament and bias voltages are below proper operating level for the linear accelerator gun. In digital gun drivers, the gun controller detected a faulty condition.	Try the treatment again: 1. Press Setup. 2. Reprogram the treatment and attempt beam-on (you may need to turn the linear accelerator power switch to Standby and then on again).
HVCB	Minor	The high voltage circuit breaker is open.	1. Look at the breaker handle to detect a circuit breaker trip. 2. Check the doors, screen switches, and S1 switch in the main thyratron chassis. 3. Reset the circuit breaker.
HVOC	Minor	High voltage power supply over-current. Peak or average charging current in the high-voltage power supply may have exceeded its limits.	Try the treatment again: 1. Press Setup. 2. Reprogram the treatment and attempt beam-on.
HWFA	Minor	The control processor detected a minor hardware failure during runtime integrity checks.	If the problem does not resolve, contact authorized service personnel.
ION1	Dosimetry	Power supply voltage for radial ion chamber is below 350 V DC.	Try the treatment again: 1. Enter the dosimetry password.
ION2	Dosimetry	Power supply voltage for transverse ion chamber is below 350 V DC.	2. Press Setup. 3. Reprogram the treatment and attempt beam-on.
IPSN	Minor	Dynamic treatment actual position parameters are out of tolerance with plan position parameters.	Check the actual position parameters on system with the plan position parameters and make appropriate changes.
KEY	Minor	The Disable/Enable keyswitch on the dedicated keyboard is in the Disable position.	Turn the Disable/Enable keyswitch to the Enable position.

Code	Type	Condition Requiring Correction	Action
KFIL (HE)	Major	The klystron filament current is at or below half nominal value. KFIL is disabled for 5 seconds after the power key is turned on.	1. Turn the power keyswitch in the console electronics cabinet counterclockwise to the Standby position 2. Then turn clockwise to the On position.
 Note: The KFIL interlock normally appears when the key is turned to the Standby position at the end of the day.			
KSOL (HE)	Minor	Klystron power is low.	Try the treatment again: 1. Press Setup. 2. Reprogram the treatment and attempt beam-on.
LVPS	Dosimetry	Low voltage power supply (+12, ±15, or +5 V) failed.	Try the treatment again: 1. Enter the dosimetry pass-word. 2. Press Setup. 3. Reprogram the treatment and attempt beam-on.
MFIL (LE)	Major	The magnetron filament current is below nominal value. If the Clinac is in Standby mode, MFIL is disabled for 15 seconds during the transition from Standby to On while the filament warms up.	1. Turn the power keyswitch in the console electronics cabinet counterclockwise to the Standby position 2. Then turn clockwise to the On position.
MGNT (LE)	Minor	■ Magnetron electromagnet voltage is ±7% of nominal. ■ Magnetron electromagnet current is ±2% of nominal.	Try the treatment again: 1. Press Setup. 2. Reprogram the treatment and attempt beam-on.
MLC	Minor	The multileaf collimator is preventing linear accelerator operation because: ■ MLC leaves are not in the planned treatment position. ■ MLC is disabled in Physics mode. ■ MLC controller asserted an interlock.	Correct the problem with the MLC and re-attempt treatment.

Code	Type	Condition Requiring Correction	Action
MOD	Minor	Negative voltage buildup (exceeding 9.5 V) on the modulator pulse-forming network occurs at the input to the sensing circuit. This is a transient but significant interlock. For UNIQUE machines, reverse voltage is caused by an arcing magnetron. For high energy machines, negative voltage is caused by a modulator, klystron, or pulse transformer overcurrent.	Try the treatment again: 1. Press Setup. 2. Reprogram the treatment and attempt beam-on.
MODE (HE)	Minor	The selected energy mode does not match the check code returned from the console controller.	
MOTN	Minor	Unplanned motions were detected on an axis not selected for dynamic therapy, or before or during treatment. The patient may have moved on the treatment couch.	1. Correct for the motions, and retry the treatment. 2. If the interlock recurs repeatedly or does not appear to have an obvious cause, contact qualified service personnel.
ORNT	Minor	Enhanced Dynamic Wedge orientation selected at the console does not match the in-room orientation selection. No Enhanced Dynamic Wedge orientation has yet been selected from the hand pendant.	1. Correct discrepancies between the console and in-room orientation selection. 2. Select an EDW orientation from the hand pendant.
PNDT	Minor	Hand pendant is not in the pendant cradle on the treatment couch. Treatment couch longitudinal and/or lateral brakes are not engaged at the treatment couch side panel (couch top is unlocked and can be moved manually).	Hang the pendant in the holder at the end of the treatment couch. Lock the treatment couch longitudinal and/or lateral brakes at the treatment couch side panel (so the couch top is locked).

Code	Type	Condition Requiring Correction	Action
PUMP	Major	 WARNING: The PUMP interlock indicate conditions that may result in overheating of the linear accelerator. If the PUMP interlock does not clear or recurs after you have attempted to correct the problem, do not continue. Power off the linear accelerator and contact authorized service personnel.	
			The water tank level is low, or the pump was shut off manually, or by a fault condition. Fault examples include: water temperature too high, or controller malfunction. Read the water and temperature gauges. Based on the readings, do the following to ensure normal temperature and pressure: ■ Fill water tank to optimum level. ■ Maintain recommended operating water temperature range: ■ For High Energy: between 15°–26° C (59°–79° F). ■ For UNIQUE: between 38°–42° C (100°–107° F). ■ Ensure that the water temperature does not exceed 48° C (118° F). ■ Ensure that the water pressure does not exceed 100 psi. The pump reset switch on the primary power distribution pane clears the PUMP interlock after the water level and temperature are within limits.
RVBP	Minor	One or more beam parameters downloaded by the treatment delivery software differ from the actual beam setting at the linear accelerator.	1. Correct discrepancies between the plan downloaded by the treatment delivery (RV) system and the linear accelerator settings. 2. Check patient setup. 3. Reattempt treatment.

Code	Type	Condition Requiring Correction	Action
STPR (HE)	Minor	The beamstopper is not fully extended or retracted. Gantry positions that require the beamstopper to be extended for the specified energies may be programmed into the system.	Correct the beamstopper position.
STPS	Minor	Stand power supply failure. Motion is disabled as long as the interlock is asserted. For UNIQUE machines this interlock is asserted for motion control power supply only (± 27 V). For high energy machines the interlock is asserted for motion control or beam steering power supply (± 27 V or ± 16 V).	Try the treatment again: 1. Press Setup. 2. Reprogram the treatment and attempt beam-on.
SYM1 (LE)	Dosimetry	Radial symmetry exceeds 2% of symmetrical adjustment.	Try the treatment again: 1. Enter the dosimetry password.
SYM2 (LE)	Dosimetry	Transverse symmetry exceeds 2% of symmetrical adjustment.	2. Press Setup. 3. Reprogram the treatment and attempt beam-on.
TARG (HE)	Minor	Target is not in the correct position for the selected mode. The target drive mechanism position is jammed or mechanism proximity switches are either misaligned or defective.	Try the treatment again: 1. Press Setup. 2. Reprogram the treatment and attempt beam-on. 3. If this does not clear the interlock, try to program another energy.

Code	Type	Condition Requiring Correction	Action
TDLY	Minor	A time delay of 12 minutes for component warm-up allows equipment filaments to reach their operating temperatures. The delay starts when the power key is turned clockwise from Off to On. The length of the delay varies with how long the machine was off: ■ 2-minute delay if the machine was off less than 1 minute ■ 4-minute delay if the machine was off more than 2 minutes ■ 12-minute delay if off more than 5 minutes. A delay longer than 12 minutes indicates that a major interlock must be cleared to proceed through the time delay.	If the TDLY interlock does not clear after 12 minutes, check for other interlocks. It is not possible to adjust or override the standard 12-minute time delay.
TDRV (HE)	Minor	The RF system shunt-tee power-attenuator turntable is not in the correct position for the selected mode and the power attenuator system setting disagrees with the energy set at the console.	Reboot the linear accelerator. If this does not resolve the problem, contact authorized service personnel.
 Note: This interlock applies only to Clinac models, manufactured prior to and including 1996.			
TIME	Minor	The displayed beam-on time is greater than or equal to the value set for time.	■ Check the MU1 and MU2 readouts for discrepancies to determine whether this interlock resulted from a failure of both MU1 and MU2 readouts. ■ If a failure of MU1 or MU2 readouts caused the TIME interlock or the interlock cause is unclear, do not perform further treatments. Contact authorized service personnel. (You can put the accelerator in Standby mode. It is unnecessary to power off.)

Code	Type	Condition Requiring Correction	Action
UDR1	Minor	Dose rate measured by the radial ion chambers is too low. For 4MeV treatments, UDR may be triggered by asymmetry < 5%. Alternatively, the radial dose rate is less than 80% of the set value, possibly from: ■ Too few MU per servo period (1/20 of a second). ■ AFC may be out of tune.	Try the treatment again: 1. Press Setup. 2. Reprogram the machine with the remaining MU and attempt beam-on.
UDR2	Minor	Dose rate measure by the transverse ion chambers is too low. For 4MeV treatments, UDR may be triggered by asymmetry < 5%. Alternatively, the transverse dose rate is less than 80% of the set value, possibly from: ■ Too few MU servo period (1/20 of a second). ■ AFC may be out of tune.	
UDRS	Minor	For 4MeV treatments, UDR may be triggered by asymmetry < 5%. Alternatively, the servo-ed dose rate is less than 80% of the set value, possibly from: ■ Too few MU per servo period (1/20 of a second). ■ AFC may be out of tune.	Try the treatment again: 1. Press Setup. 2. Reprogram the machine with the remaining MU and attempt beam-on.
VAC1	Major	 WARNING: VAC1 can result in major damage to the linear accelerator. If VAC1 does not clear, push an EMERGENCY OFF button and contact qualified service personnel. There is a major vacuum leak somewhere in an area controlled by the power supply circuits that monitor pump current. Push an EMERGENCY OFF button, and call qualified service personnel.	

Code	Type	Condition Requiring Correction	Action
VAC2	Minor	There may be an impurity, minor vacuum leak, or transient release of gas somewhere in the system. Impurities are usually self-clearing; leaks require repair by trained personnel.	Try the treatment again: 1. Press Setup . 2. Reprogram the treatment and attempt beam-on.
VSWR (HE)	Minor	Voltage standing wave ratio at the klystron directional coupler exceeds 1.5 to 1.	Try the treatment again: 1. Press Setup . 2. Reprogram the treatment and attempt beam-on.
XDP1	Dosimetry	Radial dose per pulse exceeded normal mode limits.	Try the treatment again: 1. Enter the dosimetry pass-word. 2. Press Setup . 3. Reprogram the treatment and attempt beam-on.
XDP2	Dosimetry	Transverse dose per pulse exceeded normal mode limits.	
XDR1	Dosimetry	Radial dose rate exceeded set dose rate.	
XDR2	Dosimetry	Transverse dose rate exceeded set dose rate.	
XDRS	Dosimetry	Servoed dose rate exceeded intended dose rate.	
YLD (LE)	Dosimetry	Yield ratio. The ratio of peak target current to peak ion chamber current deviated $\pm 3.5\%$ from its preset point.	Try the treatment again: 1. Enter the dosimetry pass-word. 2. Press Setup . Reprogram the treatment and attempt beam-on.

Appendix C Clinac System Support Application

Clinac System Support Application Overview

The Clinac System Support application includes utilities that run in the background of the 4D Integrated Treatment Console (4DITC) application to monitor accelerator status.



Note: For systems without 4DITC, an alternate workstation is provided.

Utilities in the System Support application include:

- Clinac Yield Monitor (see [Clinac Yield Monitor](#) on page 226)
- Potentiometer check (see [Potentiometer Check](#) on page 237)

Tabs in the application include:

- Messages

General instructions and status for the application.

- Yield Monitor

Measures dose rate, target current, and the predicted dose rate for all 6 MV beams. It also shows the deviation between the predicted dose rate and the measured dose rate along with upper and lower control limits. Software tools are provided to establish the Yield Monitor baseline and to test Clinac System Support application behavior.

- Settings

Reference information for the application, such as SSIB (System Support Interface Board) IP address. These settings can only be changed by service personnel with appropriate user rights.

- Potentiometer

Monitors electrical polarity within the Primary and Secondary potentiometers (PRO-SPRO) that provide position information for all axes, and displays a message if there is a polarity mismatch.

Clinac Yield Monitor

The Clinac System Support application monitors accelerator photon yield for every 6 MV beam and compares it against a baseline established for the machine. The Yield Monitor algorithm uses the target current to predict a dose rate. It then compares this prediction with the dose rate measured by the system monitor chambers. The prediction algorithm parameters are set during the baselining process, and are machine-specific.

The Clinac System Support application starts monitoring automatically when the 4D Integrated Treatment Console (4DITC) workstation is powered on, and operates in the background. A user can view the application by clicking the Clinac System Support application in the system tray.



Note: For systems without 4DITC, an alternate workstation is provided.

If the application detects a deviation that exceeds the upper or lower control limits, it automatically displays a message with information about the deviation. Any significant deviation between the predicted dose rate and the measured dose rate can indicate problems in the beam line, such as damage to the target, ion chamber drift, or energy change.

Clinac System Support Application Yield Monitor Messages

The Yield Monitor notifies the user of beam deviations using three types of messages.

Table 32 Yield Monitor Notification Messages

Message	Notes	User Action
 NOTICE: Yield deviation has exceeded Notification control limit during last treatment field. After completion of treatment, review QA for beam output, symmetry, and flatness.	■ Message appears after beam-off. ■ May indicate a problem in the beam line.	1. Click Acknowledge to dismiss the message. 2. Notify physicist that this message appeared. 3. After treatment is completed, review machine QA trends for output, symmetry, and flatness. 4. Continue treatment if appropriate.
 WARNING: Yield deviation has exceeded Notification control limit 4 times within 4 hours. Continued use could result in patient mistreatment and damage to the machine. Check beam output, symmetry and flatness. Proceed to treat if machine passes QA.	■ CDOS interlock is asserted when the Notification control limit is exceeded a specified number of times within a specified number of hours. ■ May indicate a problem in the beam line.	1. Physicist or other authorized user must sign off to dismiss this message and release the interlock. 2. Review machine QA trends for output, symmetry and flatness. 3. Physicist should run QA tests for flatness, symmetry, and dose output. 4. If machine passes QA, continue treatment. Otherwise, contact Varian Customer Service.
 WARNING: Yield deviation has exceeded the Interlock control limit. Continued use could result in patient mistreatment and damage to the machine. Check beam output, symmetry and flatness. Proceed to treat if machine passes QA.	■ CDOS interlock interrupts beam as soon as Yield deviation exceeds the Interlock control limit. ■ May indicate a problem in the beam line.	If yield deviation triggers a CDOS interlock, contact Varian Customer Service.

Troubleshooting Yield Monitor Messages

If the yield deviation exceeds the upper or lower control limits for Notices or Interlocks, Use the tips below to help identify potential causes.

- If the error message is intermittent and goes away for other dose rates for a given 6 MV beam, then it is likely to be a false positive. Consider checking the energy for proper tuning.
- If the error message occurs repeatedly for different dose rates or treatment modalities, this can indicate a problem in the beam line.

The following table describes the three most likely issues with the beam line.

Table 33 Troubleshooting the Beam Line

Cause	Description	Action
Ion chamber drift	Yield deviations in the positive direction indicate that the measured dose rate is lower than the predicted dose rate. Generally, ion chamber drift occurs slowly over several months. In cases of ion chamber drift, the yield deviation is similar across 6 MV, 6 SRS, and 6 FFF beams.	Check machine output, symmetry, flatness, and if appropriate, recalibrate accelerator output.
Target damage	■ Yield deviations in the negative direction indicate that the measured dose rate is higher than the predicted dose rate. ■ Target damage can start slowly, eventually leading to sudden target failure. In cases of target failure, the yield deviation is different across different beams: least for 6 MV, medium for 6 SRS, and maximum for 6 FFF.	■ Check machine output, symmetry, and flatness. ■ Contact Varian Customer Service if machine is outside beam specifications.
Energy change	Small changes in beam energy can lead to large changes in photon yield. This can occur due to issues with RF power or the bend magnet.	

Yield Monitor Display

The Clinac Yield Monitor displays the measured dose rate, target current, and the predicted dose rate for all 6 MV class beams. It also shows the deviation between the predicted rate and the measured dose rate, along with upper and lower control limits. Software tools are provided to establish the yield monitor baseline and to test the Clinac System Support application. These tools are available only after login by an authorized user.

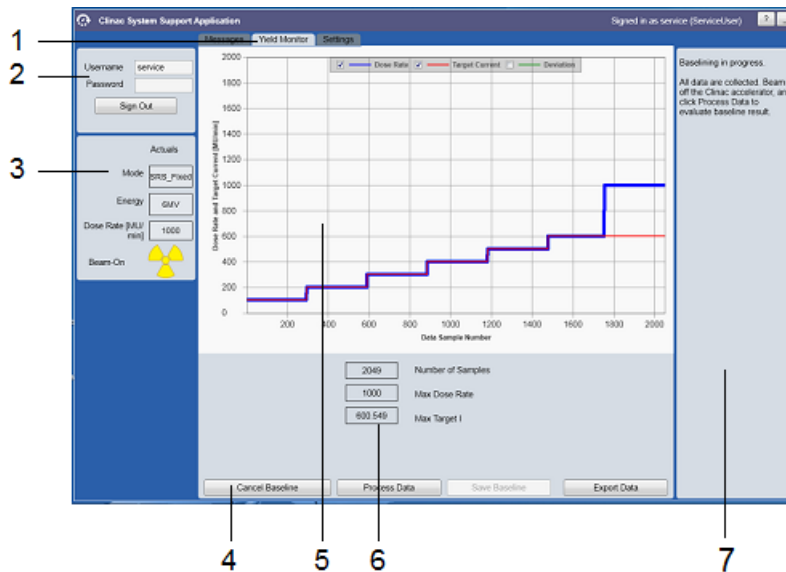


Figure 12 Yield Monitor Display

- 1 Clinac System Support application tabs—View messages and settings for the application, and open the Yield Monitor.
- 2 Log in—Log in with Service or Physicist credentials. Only an authorized user can view and use the tools.
- 3 Beam parameters—View energy, mode, dose rate, and beam-on status for the current beam.
- 4 Baselining tools—Perform baselining. Process, save, and export the data.
- 5 Graph—View the baselining procedure and baseline deviations. For more information, see [Interpreting the Yield Monitor Graph](#) on page 229.
- 6 Baseline details—Detailed baseline values reflecting parameters for the Yield Monitor algorithm.
- 7 Message area—View baselining information, status, and next steps.

Interpreting the Yield Monitor Graph

The Yield Monitor graph displays beam yield metrics and data processing.



If trace lines overlap, you can use the boxes at the top of the graph to select or deselect the tracings you want to view.



Note: The predetermined control limits for Notifications and Interlocks may differ for positive and negative deviations.

Table 34 Yield Monitor Graph

	Line		Description
1	Interlock control limit	Red broken lines	If the yield deviation exceeds the upper or lower control limit for Interlock, an Interlock message appears and the CDOS interlock interrupts the beam in the middle of the treatment field.
2	Notification control limit	Gold broken lines	If the yield deviation exceeds the upper or lower control limit for notification, a Notification message appears after beam-off.

	Line		Description
3	Relative Deviation (%)	Green line—Values shown along right Y axis	Deviation between the predicted dose rate and the measured dose rate. Gaps that occur in the green deviation line are expected behavior (as in the figure above). Approximately 200 samples are required before the first deviation value is computed. The gap at the beginning of data collection indicates the time it takes for the yield monitor to collect the samples. A gap during baselining reflects a period of dose rate below 90. If the dose rate is 90 MU/min or less, the data is not used by the algorithm.
4	Dose Rate	Blue line—values shown along left Y axis	Dose rate in real time during baselining.
5	Target Current	Red solid line	Target current in real time during baselining.

Baseline Details

During the baseline process, you can view detailed baseline values for the previous baseline and the new baseline by clicking Detailed Baseline Values. These values reflect parameters for the prediction model used by the Yield Monitor algorithm.

The screenshot shows a window titled "Clinac System Support Application". Inside, there are two columns: "Previous Values" and "New Values". Each column has six input fields. To the right of these fields are labels for each parameter. At the bottom right is an "OK" button.

Previous Values	New Values	Label
0.999086	0.999086	Gain Fixed Mode
4.35989E-11	2.71569E-11	Offset
1.66514	1.66514	Gain SRS/FFF Mode
8.76016E-14	6.27534E-14	Residual Error RMS [MU/min]
1.66575E-15	1.05363E-15	Second Order Coefficient
-1.54292E-18	-9.80415E-19	Third Order Coefficient

- Gain Fixed Mode

First parameter for the fit to the prediction model, 6X

- Offset

Second parameter for the fit to the prediction model, both 6X and 6 FFF or SRS

- Gain FFF Mode

First parameter for the fit to the prediction model for 6 FFF or SRS

- Residual Error RMS (% of max Dose Rate)

Measure of the quality of the fit to the prediction model. Value above a configured threshold results in baseline failure

- Second Order Coefficient

High-order corrections to the fit model

- Third Order Coefficient

High-order corrections to the fit model

Baseline Process for High-Dose Beams

High-dose beams display a wide divergence between target current and dose rate.

In the following figure, the first set of samples reflect standard fixed 6X beam.

For FFF, the target current stays the same, but the dose rate increases to FFF levels. The wide divergence between target current and dose rate, and the target current line, are normal for FFF and SRS beams.

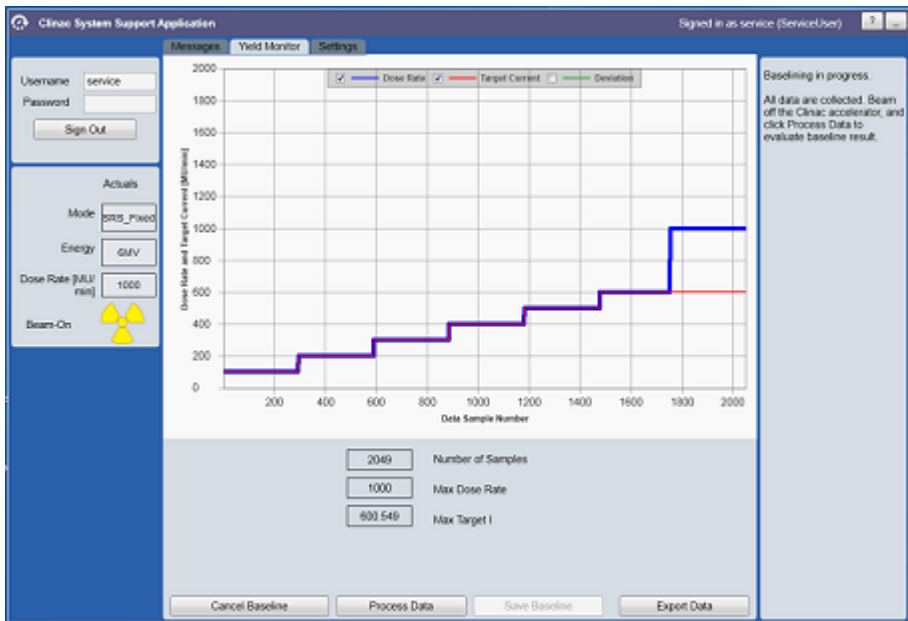


Figure 13 Normal Baseline Data Collection for 6X and FFF Beams

Overview of the Baseline



WARNING: Re-establishing the Yield Monitor baseline resets the parameters of the Yield Monitor algorithm. Perform machine QA and ensure that beam output, symmetry and flatness are within specifications before saving new baseline data using the Yield Monitor utility.

The Yield Monitor utility collects dose rate and target current data from the accelerator and uses that data to establish baseline values for yield monitoring. This process is also known as *baselining*. The process includes three overall steps.

1. Gather baseline data while beaming on with each 6 MV beam in sequence while running the Yield Monitor.
2. Process and verify the data by beaming on using the new baseline values.
3. Save baseline data. The application replaces the previous Yield Monitor baseline. You can also export data to external media.



WARNING: Repeated baselining can mask underlying problems in the accelerator. Do not re-establish the baseline for a machine that is repeatedly showing beam deviations.

When to Reestablish the Baseline

A new baseline must be reestablished after:

- New Clinac was installed; you have confirmed symmetry and flatness; and you have completed initial beam commissioning and final dose calibration.
- The dose output was calibrated at a different depth from the original calibration depth.
- Major repair work was performed with the beam line, such as replacing the ion chamber, target, energy switch, or bend magnet.

A new baseline may be reestablished:

- Anytime beam tuning work is performed that results in a yield notification or yield interlock.

Before baselining, you must confirm that flatness and symmetry as well as dose output are correct.

- If a target current monitor cable was replaced. This should only be required if the previous baseline was established with a coax cable that had higher impedance than normal (for example, if the cable was in the process of failing but had not yet completely failed).

- After replacing the B15 Symmetry Monitor and Dose Rate Integrator and adjusting the linearity for 6MV, 6MV SRS or 6MV FFF.

When Not to Reestablish the Baseline

Do not reestablish the baseline:

- After GUN I and RFDR adjustments for basic beam tuning, unless the tuning results in a Yield notification or interlock.
- After replacing an SSIB PCB or its CF card (the baseline data is stored within the CSSA application and transferred to the SSIB each time CSSA connects to SSIB).
- After you recalibrate the dose output during daily or weekly beam QA checks.

For baselining after recalibrating the dose output during QA checks, the following reasoning applies:

1. If a target was failing, reestablishing a new baseline after a customer dose output calibration would mask the target failure symptoms. But if the baseline is not reestablished, the yield monitor algorithm would eventually detect the dose increase and produce a Yield Notification or Interlock warning, which would indicate a possible failing target.
2. If an ion chamber was drifting (or seasoning), the dose output calibration correction should return the dose back to the original dose output calibration value and return the yield monitor deviation back towards zero.

There is one exception to this condition. If you did not calibrate the dose output to an accuracy of $\pm 0.5\%$ of the desired value when the original baseline was established, then the baseline would need to be re-established after you correct the dose output calibration.

Establish the Baseline

During this process, you beam on with each 6 MV beam in sequence while running the Yield Monitor. The Yield Monitor collects data on dose rate and target current.



WARNING: Re-establishing the Yield Monitor baseline resets the parameters of the Yield Monitor algorithm. Perform machine QA and ensure that beam output, symmetry and flatness are within specifications before saving new baseline data using the Yield Monitor utility.

1. Set up the accelerator to use Morning Checkout positions.
2. Warm up the machine by beaming on for at least two minutes using the maximum 6 MV dose rate available on the machine.
3. Sign into the Clinac System Support application.

4. Mode up the accelerator in Service mode.



Note: *Baselining may cause beam deviations that trigger a CDOS interlock. You may be required to override the CDOS interlock to complete this process.*

5. Set the mode and dose rate to **6MV FIXED, 100MU/min**.

6. Click **Establish Baseline**.

7. Beam on for at least 15 seconds.

The baselining process starts. The message area displays status, and prompts you when to change to the next 6MV dose rate. You can change dose rates without beaming off.



Note: *During baselining, the Relative deviation line and values (right Y axis) and the Interlock and Notification control limits are not displayed on the graph.*

- If you beam off between dose rates, the CDOS interlock may be asserted and must be overridden to proceed.
- If you turn the beam off while baseline data is being collected, the Yield Monitor prompts you to restart that beam.
- If you skip a dose rate, the Yield Monitor prompts you to beam on with the missing dose rate after the current beam type is completed.
- If you repeat a beam, the data will not be included in data processing. To repeat a beam you must restart the baselining procedure.



Note: *If you have a 6 SRS or 6 FFF beam, the System Support application will prompt you to beam on with appropriate dose rates.*

8. Continue baselining, following the instructions that appear on the Yield Monitor screen.

Troubleshooting Baseline Failure

If the baseline process fails, a message appears accompanied by a red X symbol.

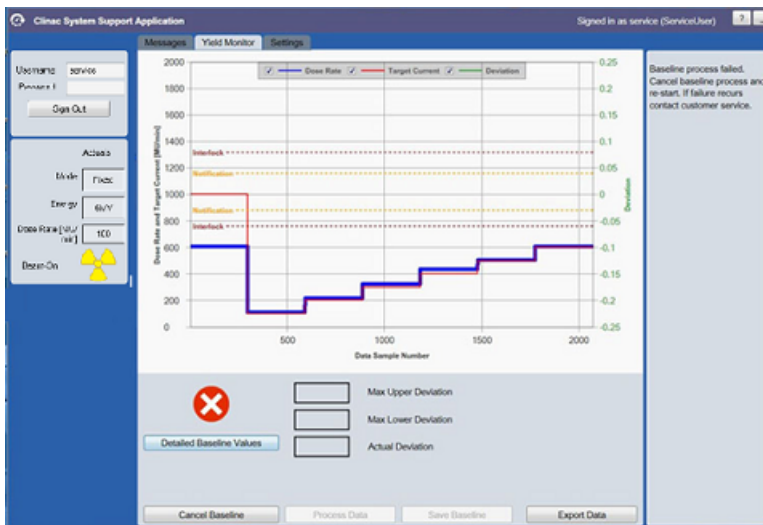


Figure 14 Failed Baseline

The baseline process may fail under the following circumstances:

- The beam was switched off.
- Not enough time was allowed to collect enough samples to establish the baseline. Yield monitor requires at least 200 samples to run the algorithm.
- A residual error occurred.

For each of these circumstances, the baseline process can be resumed at the point at which the failure occurred.

Process and Save Baseline Data

After running the baseline, the Yield Monitor application captures the parameters necessary to set the prediction algorithm.



WARNING: Saving incorrect Yield Monitor baseline data can result in patient injury and machine damage. Ensure dose output, flatness, and symmetry are calibrated correctly.

1. When baselining is complete, beam off and click **Process Data**.



Note: During this step, the Relative deviation line and values (right Y axis) and the Interlock and Notification control limits are displayed on the graph.

2. Test the data by beaming on with any 6MV energy.

If the dose output, flatness, and symmetry are calibrated correctly, the deviation value will be less than the Notification level for all dose rates when verifying the baseline.

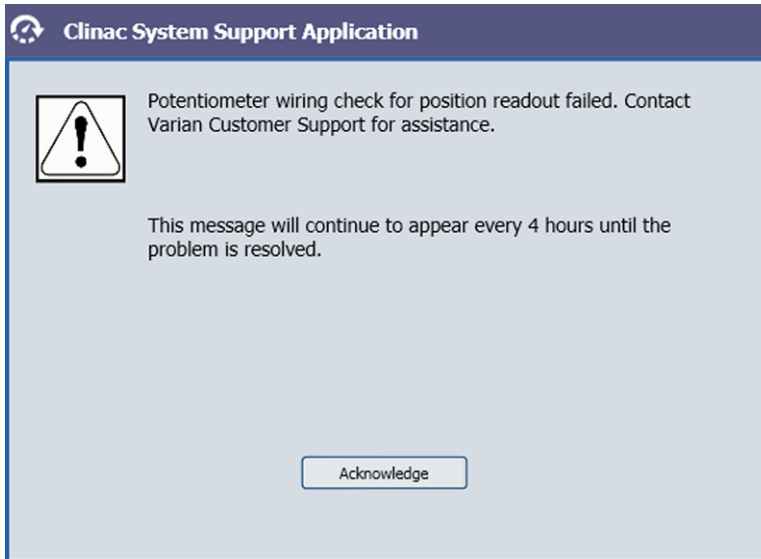
- If beam deviations appear that meet or fall outside one of the control limits, do not save the data. Call Varian Customer Service to determine the cause of the beam deviations.
- If beam deviations stay inside the control limits with the new baseline data, click **Save Data**. The data are saved as the new Yield Monitor baseline for the accelerator.

Potentiometer Check

The Clinac System Support application Potentiometer Check utility runs on the 4D Integrated Treatment Console (4DITC) workstation, and monitors the electrical polarity of the primary and secondary readout potentiometers for all motion axes of the machine.

The Primary and Secondary readout potentiometers provide position information to the Gantry display and Clinac Control Console for all motion axes (jaws, gantry, couch and collimator). The Potentiometer Check utility provides an additional check of the positions reported by the PRO-SPRO system.

If the Potentiometer Check utility detects a polarity mismatch within the PRO-SPRO system, a Potentiometer Check message appears notifying you of the anomaly.



The machine is not in an unsafe state if this message appears. You can acknowledge the message and continue to use the machine.

The message will continue to display every four hours. The potentiometer should be checked by Varian Customer Service to correct the polarity mismatch. Call Varian Customer Service to have the potentiometer checked by Varian personnel.



Note: *There are two circumstances under which this message may appear without a polarity mismatch:*

- *In cases of Emergency Off or interruption of power to the accelerator. In these cases, acknowledge the notification to dismiss it.*
- *In couches that are installed a 36-inch turntable. These couches do not have secondary readout potentiometers. This will cause the message to appear every four hours.*

Appendix D Recommendations for Dosimetry Calibration

Recommended Daily Photon Output Checks

Varian strongly recommends that all sites perform daily output constancy checks on photon beams as recommended by the AAPM, specifically those provided by *AAPM Task Group 142 Report: Quality Assurance of Medical Accelerators*, American Association of Physicists in Medicine, Medical Physics publication 36 (9) (2009), and *AAPM Report No. 46, Report of Task Group No. 40 Radiation Therapy Committee: Comprehensive QA for Radiation Oncology*, American Association of Physicists in Medicine, Medical Physics publication 21 (4) (1994). (See the end of the appendix for complete references to publications.)

Varian recommends that all sites perform these checks:

- Daily output constancy checks should be performed for all photon beams consistent with the AAPM TG-51 protocol (P. R. Almond et al., 1999), or equivalent dosimetry protocols such as IAEA TRS 398 (IAEA, 2000).
- X-ray output measurements should be made with buildup (5 cm water equivalent) that is beyond the electron range to avoid electron contamination. This extra buildup may not be necessary for array-type QA devices that measure energy/flatness/symmetry in addition to the usual output.
- Daily output constancy checks must include all 6MV beams (6SRS, 6FFF, and 6X). For more information, see *Notification and Correction Notice: Unexpected 6MV Beam Output Variations Affecting C-Series High Energy Linear Accelerators Versions Novalis Tx, Trilogy, Trilogy Tx, Clinac iX, Clinac CX, Clinac 2100 C/D, Clinac 2300 C/D, Clinac DHX, Clinac 21 EX, Clinac 23 EX*, Varian Medical Systems, PNL-CP-12459 (2014).

Calibration Depth Other Than Depth of Maximum Dose

Varian recommends that the High Energy Clinac dosimetry system be adjusted so that 1 MU displayed on the console corresponds to 1 cGy of dose at $d_{\max}$ (depth of maximum dose) in water for a 10 x 10 cm field at SSD = 100 cm.¹

¹ The calibration method described in this section can also be used for UNIQUE and other Clinac models.

The following section is intended for Clinac users who do not use $d_{\max}$ for their photon dose calibration depth, and provides guidelines for setting D'_0 (cGy/MU) for non-standard calibration depths (where D'_0 = dose per MU under calibration conditions). If the guidelines described in this section are not followed, operating at calibration depths other than $d_{\max}$ may damage the accelerator.

AAPM TG-51 is the calibration protocol used by many facilities to perform dose output calibration of photon and electron beams. TG-51 is compatible with non-isocentric dose calibration that defines the dose/MU factor D'_0 as 1 cGy/MU at 100 cm SSD and $d_{\max}$. TG-51 is also compatible with an isocentric system that defines D'_0 at 100 cm SAD and a depth of 10 cm. Using the isocentric system has the advantage that all beam energies are calibrated at the same depth.

Clinac systems are designed to allow TG-51 calibration depths other than $d_{\max}$, such as 5 cm or 10 cm. When using calibration depths greater than $d_{\max}$, it is necessary to reduce D'_0 from the standard value of 1 cGy/MU.

For the isocentric case where calibration depth is 10 cm, AAPM report TG-71 (John P. Gibbons et al., 2014) tabulated suggested values of D'_0 . These values are listed in the following table.



NOTICE: For isocentric calibration at 10 cm, the $d_{\max}$ for the Clinac accelerator must be calibrated at or below the D'_0 values suggested in AAPM report TG-71. Operating outside the D'_0 values suggested in AAPM report TG-71 may cause damage to the Clinac accelerator system.

Table 35 Values Required for Dose Output Calibration Depth 10 cm

Energy MV (BJR 17/BJR11)	D'_0 cGy/MU
4/4	0.75
6/6	0.81
8/8	0.85
10/10	0.88
16/15	0.93
23/18	0.95



WARNING: The depth used for AAPM TG-51 dose output measurements must match the depth used to create models for each treatment planning system algorithm. If the depth used to create models for each treatment planning system algorithm is not consistent with the actual measurement depth, significant dose delivery errors will occur.

Changing the dose output calibration point on a Clinac system requires all algorithms in your treatment planning system to be updated for all modified energies. For Eclipse, this includes unapproving the current models and recalculating the models' absolute dosimetry. Previously calculated plans that are no longer under treatment remain valid using the prior calibration. All plans to be treated on the machine with updated calibration of dose calculation algorithms must be recalculated to update their MUs.

For users not familiar with the process of modifying or calculating Eclipse models, contact Varian Help Desk for assistance. For users with other planning systems, contact the vendor's customer support line for assistance.

References

Peter R. Almond, Peter J. Biggs, B. M. Coursey, W. F. Hanson, M. Saif Huq, Ravinder Nath, D. W. O. Rogers. *AAPM's TG-51 Protocol for Clinical Reference Dosimetry of High-Energy Photon and Electron Beams*, American Association of Physicists in Medicine, Medical Physics 26 (1999)

John P. Gibbons, John Antolak, David S. Followill, M. Saiful Huq, Eric E. Klein, Kwok L. Lam, Jatinder R. Palta, Donald M. Roback, Mark Reid, Faiz M. Khan. *AAPM's TG-71 monitor unit calculations for external photon and electron beams: Report of the AAPM Therapy Physics Committee Task Group No. 71*, American Association of Physicists in Medicine, Medical Physics 41 (2014)

Eric E. Klein, Joseph Hanley, John Bayouth, Fang-Fang Yin, William Simon, Sean Dresser, Christopher Serago, Francisco Aguirre, Lijun Ma, Bijan Arjomandy, Chihray Liu, Carlos Sandin, Todd Holmes. *AAPM Task Group 142 Report: Quality Assurance of Medical Accelerators*, American Association of Physicists in Medicine, Medical Physics publication 36 (9) (2009)

Gerald J. Kutcher, Lawrence Coia, Michael Gillin, William F. Hanson, Steven Leibel, Robert J. Morton, Jatinder R. Palta, James A. Purdy, Lawrence E. Reinstein, Goran K. Svensson, Mona Weller, Linda Wingfield. *AAPM Report No. 46, Report of Task Group No. 40 Radiation Therapy Committee: Comprehensive QA for Radiation Oncology*, American Association of Physicists in Medicine, Medical Physics publication 21 (4) (1994)

IAEA International Atomic Energy Agency, *Absorbed Dose Determination in External Beam Radiotherapy: An International Code of Practice for Dosimetry Based on Standards of Absorbed Dose to Water*, Technical Report Series No. 398 (2000)

Notification and Correction Notice: Unexpected 6MV Beam Output Variations Affecting C-Series High Energy Linear Accelerators Versions Novalis Tx, Trilogy, Trilogy Tx, Clinac iX, Clinac CX, Clinac 2100 C/D, Clinac 2300 C/D, Clinac DX, Clinac 21 EX, Clinac 23 EX, Varian Medical Systems, PNL-CP-12459 (2014)

Customer Technical Bulletin: Dosimetry Monitoring System Calibration, Varian Medical Systems, CTB-GE-228 (2014)

Appendix E Transmission Factor and Radiation Fields

Transmission Factor at Each X-Radiation Energy

You can find the latest product information and beam and dose specifications by visiting the Varian Medical Systems web site at <http://my.varian.com>.

Collimators

The distance from the target to the lower surface of the block tray is 65.4 cm for an accelerator with MLC. The primary collimator half-angle may be converted to radians using the relationship: 2π radians = 360° . For both collimator and gantry rotation, the Varian scale position 180° corresponds to the IEC scale position 0° .

Wedges

Wedge information applies to the Type III series wedges supplied with Varian Oncology Systems' current accelerators. Current Type III series 15° and 30° wedges are steel; 45° and 60° wedges are lead. There were steel 45° Type III wedges produced. An accelerator with MLC includes an upper and lower wedge set. The following are nominal densities for the current wedge materials: Lead wedges, $p = 11.3 \text{ g/cm}^3$; steel wedges, $p = 7.8 \text{ g/cm}^3$. The 20 cm set includes 15 cm 60° wedges. The 30 cm set includes 15 cm 60° wedges and 20 cm 45° wedges.

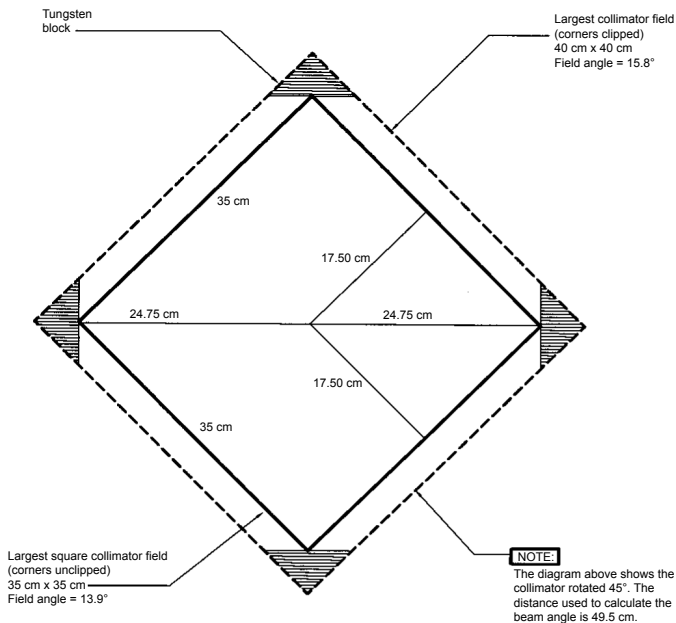
Geometrical Radiation Fields

Figure 15 provides a depiction of the geometrical radiation fields for X- and electron radiation for the accelerator at Normal Treatment Distance (NTD). For Varian systems, the calibration point is set at Source-to-Surface Distance (SSD) of 100 cm at Dmax.

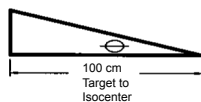


Note: Available delivery of the radiation beam on the accelerator is 360° of gantry rotation.

The corner tungsten field assemblies (blocks) are provided to produce a sharp penumbra line due to the proximity of the primary collimator circular design to the target. These corner blocks appear only for field sizes greater than 35 cm x 35 cm.



CALCULATIONS



$$\begin{aligned} \tan \theta &= \frac{24.75 \text{ cm}}{100 \text{ cm}} \\ \theta &= \left(\frac{24.75 \text{ cm}}{100 \text{ cm}} \right) \tan^{-1} \\ \theta &= 13.9^\circ \end{aligned}$$

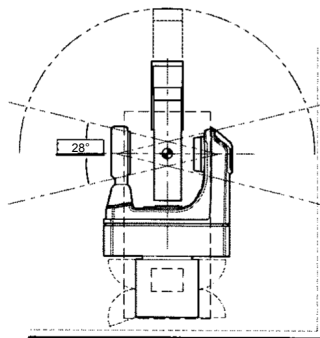


Figure 15 Geometrical Radiation Fields

Appendix F EMC Data and Guidance

EMC Data Guidance Overview

Medical electrical equipment requires special precautions regarding electromagnetic compatibility (EMC) and must be installed and put into service according to the EMC Information provided in this manual.

Portable and mobile RF communications equipment can affect medical electrical equipment.

To ensure EMC compliance, the accelerator should be used only with the cables and accessories listed in the Varian Data Book. The use of accessories, transducers, or cables other than those specified by Varian or manufactured by Varian may result in increased radiation emissions or decreased immunity of the equipment to electromagnetic interference (EMI).

The equipment should not be used adjacent to or stacked with other equipment.

If adjacent or stacked use is necessary: Prior to clinical use, place the equipment in the configuration in which it is to be used clinically, and observe it to verify normal operation.



WARNING: When an exception to permanently installed equipment has been implemented at your site, it is important that you know that Varian accelerators were not tested for radiated RF immunity over the entire frequency range of 80 MHz to 2.5 GHz. The Varian accelerators are tested with licensed transmitters only in the following frequencies: 470 MHz, 915 MHz, and 2.4 GHz. Varian recommends to test your accelerator for normal operation while the equipment generating possible RF interference is in operation. Failure to perform these tests may result in serious injury or death.

Industrial, Scientific, and Medical Equipment (ISM) Classifications

According to CISPR 11, Industrial, Scientific, and Medical (ISM) Equipment is classified under two groups and two classes.

Group 1

Contains all ISM equipment that intentionally generates and/or uses conductively coupled radio frequency (RF) energy that is necessary for the internal functioning of the equipment itself.

Group 2

Contains:

- All ISM equipment that intentionally generates or uses RF energy in the form of electromagnetic radiation for the treatment of material, and
- Spark erosion equipment.

Class A

- Equipment suitable for use in all establishments other than domestic.
- Equipment not directly connected to a low-voltage power supply network that supplies a building used for domestic purposes.

Class B

- Equipment suitable for use in domestic establishments
- Equipment suitable for use in establishments directly connected to a low-voltage power supply network that supplies a building used for domestic purposes.

According to the above classifications, the Varian medical linear accelerator is Group 1, Class A type equipment.

Varian Test Declarations and Guidance

Varian medical linear accelerators are intended for use in the electromagnetic environment specified below. The customer or user of this accelerator must ensure that it is used in such an environment.



Note: *It is recommended that the actual shielding effectiveness and filter attenuation of the shielded location be verified to ensure that they meet the minimum specifications. See [Table 36](#).*

Table 36 Electromagnetic Emission

Emission Test	Compliance	Electromagnetic Environment Guidance
RF emission CSIPR 11	Group 1	The accelerator uses RF energy only for its internal function. Therefore, its RF emissions are very low and are not likely to cause any interference in nearby electronic equipment.
RF emission CSIPR 11	Class A	The accelerator is suitable for use in all establishments other than domestic and those directly connected to the public low-voltage power supply network that supplies buildings used for domestic purposes.
Harmonic emissions IEC 61000-3-2	Not applicable	These tests do not apply because Varian systems draw more than 16 A per phase.
Voltage fluctuation/ flicker emissions IEC 61000-3-3	Not applicable	These tests do not apply because Varian systems draw more than 16 A per phase.

Varian medical linear accelerators are intended for use in the electromagnetic environment specified below. The customer or user of this accelerator should ensure that it is used in such an environment.



Note: These guidelines may not apply in all situations. Electromagnetic propagation is affected by absorption and reflection from structures, objects, and people. [Table 37](#) is Varian's guidance and manufacturer's declaration of electromagnetic immunity.

Table 37 Electromagnetic Immunity

Immunity Test	IEC 60601 Test Level	Compliance Level	Electromagnetic Environment Guidance
Electrostatic discharge (ESD) IEC 61000-4-2	±6kV contact ±8kV air	±6kV contact ±8kV air	Floors should be wood, concrete, or ceramic tile. If floors are covered with synthetic material, the relative humidity should be at least 30%.
Electrical fast transient/ burst IEC 61000-4-4	±2kV for power supply lines ±1kV for input/output lines	±2kV for power supply lines ±1kV for input/output lines	Mains power quality should be that of a typical commercial or hospital environment.
Surge IEC 61000-4-5	±1kV line(s) to line(s) ±2kV line(s) to earth	±1kV differential mode ±2kV Common mode	Mains power quality should be that of a typical commercial or hospital environment.

Immunity Test	IEC 60601 Test Level	Compliance Level	Electromagnetic Environment Guidance
Voltage dips, short interruptions, and voltage variations on power supply input lines IEC 61000-4-11	<5% Ur (>95% dip in Ur) for 0.5 cycle <40% Ur (60% dip in Ur) for 5 cycles <70% Ur (30% dip in Ur) for 25 cycles	Not applicable (>16 A/phase)	Mains power quality should be that of a typical commercial or hospital environment. If the user of the accelerator requires continued operation during power mains interruption, it is recommended that the accelerator be powered from an uninterruptible power supply or a battery.
Power frequency (50/60 Hz) Magnetic field IEC 61000-4-8	3.0 A/m	3.0 A/m	Power frequency magnetic fields should be at levels characteristic of a typical location in a typical commercial or hospital environment.
Conducted RF IEC 61000-4-6 (RF Common-mode Immunity)	3 Vrms 150 KHz to 80 MHz Swept with a 80% AM, 1 kHz sine wave	3 Vrms	Portable and mobile RF communication equipment should be used no closer to any part of the accelerator (including cables) than the recommended separation distance calculated from the equation applicable to the frequency of the transmitter.
Radiated RF IEC 61000-4-3 (RF Electromagnetic field immunity)	3 Vrms 80 MHz to 2.5 GHz	3 V/m	Recommended separation distance d from the antenna $d = \frac{3.5}{V_1} = \frac{3.5}{3} = 1.2\sqrt{P}$ for 150 kHz to 80 $d = \frac{3.5}{E_1} = \frac{3.5}{3} = 1.2\sqrt{P} \text{ for 80 MHz to 800 MHz}^2$ $d = \frac{7}{E_1} = \frac{7}{3} = 2.3\sqrt{P} \text{ for 800 MHz to 2.5 GHz}$ Calculating for test levels $V_1=3$ and $E_1=3$

² At 80 MHz and 800 MHz, the higher frequency range applies.

Immunity Test	IEC 60601 Test Level	Compliance Level	Electromagnetic Environment Guidance
			where: P = Maximum output power (ERP) Watts rating of the transmitter according to the transmitter manufacturer d = Recommended separation distance in meters (m) E = Field strength (RMS value) (V/m) Field strength from fixed RF transmitters, as determined by an electromagnetic site survey³, should be less than the compliance level in each frequency range.⁴ Interference may occur in the vicinity of equipment marked with the following symbol. 

Recommended Distance Between Communications Equipment and Accelerator

Varian medical linear accelerators are intended for use in an electromagnetic environment in which radiated RF disturbances are controlled. The customer or the user of the accelerator can help prevent electromagnetic interference by maintaining a minimum distance between portable and mobile RF communications equipment (transmitters) and the accelerator as recommended below, according to the maximum output power of the communications equipment.

³ Field strengths from fixed transmitters, such as base stations for radio (cellular or cordless) telephones and land mobile radios, amateur radio, AM and FM radio broadcast and TV broadcast cannot be predicted theoretically with accuracy. To assess the electromagnetic environment due to fixed RF transmitters, an electromagnetic site survey should be considered. If the measured field strength in the location in which the accelerator is used exceeds the applicable RF compliance level above, the accelerator should be observed to verify normal operation. If abnormal performance is observed, additional measures may be necessary, such as reorienting or relocating the accelerator.

⁴ Over the frequency range 150 kHz to 80 MHz, field strengths should be less than 3V/m.



Note: These guidelines may not apply in all situations. Electromagnetic propagation is affected by absorption and reflection from structures, objects, and people.

Table 38 Recommended Distance Between Communications Equipment and Accelerator

Radiated maximum output power of transmitter (W) ¹	Separation Distance According to Frequency of Transmitter (m)		
	150 kHz to 80 MHz $d = \left\lceil \frac{3.5}{V_1} \right\rceil \sqrt{P}$	80 MHz to 800 MHz ² $d = \left\lceil \frac{3.5}{E_1} \right\rceil \sqrt{P}$	800 MHz to 2.5 GHz $d = \left\lceil \frac{7}{E_1} \right\rceil \sqrt{P}$
0.01	0.12	0.12	0.23
0.1	0.38	0.38	0.73
1	1.2	1.2	2.3
10	3.8	3.8	7.3
100	12	12	23
Table heading notes: 1. For transmitters rated at a maximum output power not listed above, the recommended separation distance d in meters (m) can be estimated using equation applicable to the frequency of the transmitter, where P is the maximum output power rating of the transmitter in watts (W) according to the transmitter manufacturer. 2. At 80 MHz and 800 MHz, the separation distance for the higher frequency range applies.			

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